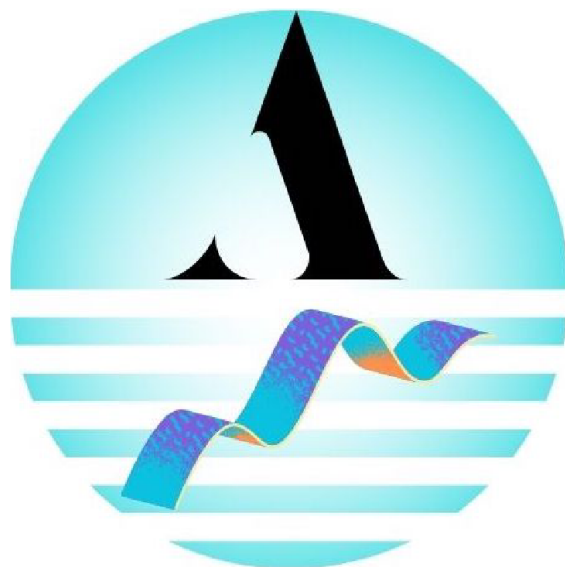




Coordinate Geometry (MMT422J2)

||||||| Written by: **DR. AIJAZ** |||||

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Bachelors with Mathematics as Major— 4th Semester
MMT422J2: Mathematics/Applied Mathematics: Geometry

Credits: 4 THEORY + 2 TUTORIAL

Theory: 60 Hours & Tutorial: 30 Hours

Course Objectives: (i) To provide idea about the basic applications of the analytical plane and solid geometry. To study the different sections of a cone (as conic section) and its properties. (ii) to familiarize the students with the illustrations of graphing standard quadric surfaces like cones, paraboloids, hyperboloids and ellipsoids.

Course Outcomes: After the completion of the paper, the student is expected to handle 2D and 3D geometrical concepts. (ii) Understand the nature of Hyperbolic functions. (iii) Trace standard curves in Cartesian coordinates and polar coordinates.

Theory: 4 Credits

Unit –I

Parabola, tangents and normals, pole and polar, pair of tangents from a point, equation of a chord of a parabola in terms of its middle point, parametric equations of a parabola.

Unit –II

Ellipse, tangents and normal, pole and polar, parametric equations of ellipse, Diameters, conjugate diameters and their properties, General second-degree equation in x and y , Conditions under which a general second degree equation represents a conic, Determination of equation of the corresponding conic.

Unit –III

Sphere, radical plane, coaxial system, simplified form of the equation of two spheres, Cone, vertex, guiding curve, generator, equation of cone with vertex as origin or a given vertex and guiding curve, Condition that the general equation of the second degree should represent a cone, Necessary and sufficient conditions for a cone to have three mutually perpendicular generators.

Unit –IV

Cylinder, equation of the cylinder whose generators intersect a given conic and are parallel to given line, enveloping cylinder of a given sphere, types of conicoids, tangent and tangent planes, Director sphere, normal to a sphere, polar and polar planes, enveloping and enveloping cylinder. The paraboloids, conjugate diameters and conjugate planes

Tutorial: 2 Credits

Unit –V

Plane: equation of plane in intercept form, normal form, reduction of general form to normal form, equation of plane through three points, angle between two planes, perpendicular distance of a point from a plane, plane through the intersection of two planes.

Unit –VI

Hyperbola, tangents and normal, equation of hyperbola referred to asymptotes as axes, rectangular and conjugate diameters and their properties, tracing of conics.

Recommended Books:

1. P. Balasubrahmanyam, K.G. Subramanian and G. R. Venkataraman, Coordinate Geometry of two and three Dimensions, Tata McGraw Hill, 1994.
2. Shanti Narayan, Analytical Solid Geometry, S. Chand and Company, 2007.
3. M. R. Puri, Co-ordinate Geometry of the Conics, New Academic Publishers, Jammu.
4. R.J.T. Bill, Elementary Treatise on Coordinate Geometry of Three Dimensions, McMillan India Ltd., 1994.
5. S.L. Loney, The Elements of Coordinate Geometry, McMillan and Company, London.
6. S.Pirzada and T.A.Chishti, Analytical Solid Geometry, Universities Press, Orient Blackswan, 2007.

Chapter 1

The Parabola

1.1 Introduction and Standard Equation

The parabola is one of the classic conic sections, historically discovered by slicing a cone with a plane parallel to one of its generating lines. In analytic geometry, however, we define it strictly via the locus of a point.

Definition 1.1.1 (Parabola). A parabola is the locus of a point which moves in a plane such that its distance from a fixed point (called the **focus**) is always equal to its perpendicular distance from a fixed straight line (called the **directrix**).

If we denote the focus by S and the closest point on the directrix to a point P as M , the defining condition for the parabola is:

$$\frac{SP}{PM} = e = 1 \quad (1.1)$$

where e is the eccentricity of the conic. For a parabola, the eccentricity is always unity.

1.1.1 Standard Equation of the Parabola

To derive the simplest (standard) equation, we choose our coordinate axes specifically to exploit the symmetry of the curve.

Let the focus S be at the point $(a, 0)$ on the x -axis. Let the directrix be the line $x = -a$. The origin $O(0, 0)$ lies exactly midway between the focus and the directrix. This point O is called the **vertex** of the parabola.

Let $P(x, y)$ be any point on the parabola. The distance from P to the focus S is:

$$SP = \sqrt{(x - a)^2 + y^2}$$

The perpendicular distance from P to the directrix $x + a = 0$ is:

$$PM = |x + a|$$

By the definition of the parabola, $SP = PM$. Squaring both sides:

$$(x - a)^2 + y^2 = (x + a)^2$$

Expanding and simplifying:

$$x^2 - 2ax + a^2 + y^2 = x^2 + 2ax + a^2$$

$$y^2 = 4ax$$

This is the standard equation of a parabola opening to the right (positive x -axis).

Definition 1.1.2 (Latus Rectum). The chord of the parabola which passes through the focus and is perpendicular to the axis of the parabola is called the **Latus Rectum**.

For the standard equation $y^2 = 4ax$:

- **Axis:** The x-axis ($y = 0$).
- **Vertex:** The origin $(0, 0)$.
- **Focus:** $(a, 0)$.
- **Directrix:** $x = -a$.
- **Length of Latus Rectum:** $4a$. The endpoints of the latus rectum are $(a, 2a)$ and $(a, -2a)$.
- **Focal Distance:** The distance of a point (x, y) from the focus is $x + a$.

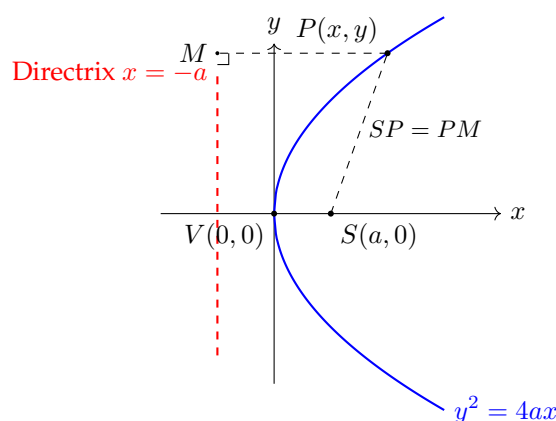


Figure 1.1: Geometry of the Standard Parabola $y^2 = 4ax$

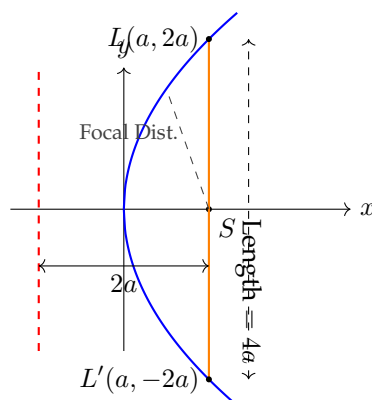


Figure 1.2: The Latus Rectum and Focal Distance

Other Standard Forms

Depending on the orientation of the focus and directrix relative to the axes, there are four standard forms:

1. $y^2 = 4ax$: Opens right. Focus $(a, 0)$, Directrix $x = -a$.
2. $y^2 = -4ax$: Opens left. Focus $(-a, 0)$, Directrix $x = a$.
3. $x^2 = 4ay$: Opens upwards. Focus $(0, a)$, Directrix $y = -a$.
4. $x^2 = -4ay$: Opens downwards. Focus $(0, -a)$, Directrix $y = a$.

Example 1.1.3. Find the coordinates of the focus, the equation of the directrix, and the length of the latus rectum for the parabola $y^2 = 12x$.

Solution. Comparing the given equation with the standard form $y^2 = 4ax$, we have:

$$4a = 12 \implies a = 3$$

Therefore, the focus is at $S(3, 0)$. The equation of the directrix is $x = -3$ (or $x + 3 = 0$). The length of the latus rectum is $4a = 12$. ■

Example 1.1.4. Find the equation of the parabola whose focus is at $(-1, -2)$ and whose directrix is the line $x - 2y + 3 = 0$.

Solution. Let $P(x, y)$ be any point on the parabola. By definition, the distance from P to the focus equals the perpendicular distance from P to the directrix.

The square of the distance from the focus is:

$$SP^2 = (x + 1)^2 + (y + 2)^2$$

The square of the perpendicular distance to the directrix is:

$$PM^2 = \left(\frac{x - 2y + 3}{\sqrt{1^2 + (-2)^2}} \right)^2 = \frac{(x - 2y + 3)^2}{5}$$

Equating $SP^2 = PM^2$:

$$5[(x^2 + 2x + 1) + (y^2 + 4y + 4)] = (x - 2y + 3)^2$$

Expanding the Right Hand Side using $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$:

$$5x^2 + 5y^2 + 10x + 20y + 25 = x^2 + 4y^2 + 9 - 4xy - 12y + 6x$$

Rearranging terms to the general form $Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0$:

$$4x^2 + 4xy + y^2 + 4x + 32y + 16 = 0$$

This is the required equation of the parabola. Note that the terms of the second degree $(2x + y)^2$ form a perfect square, which is a characteristic property of the general equation of a parabola. ■

Example 1.1.5 (Left-Handed Parabola). Consider the equation of the parabola given by $y^2 = -20x$. Determine the coordinates of the focus and the equation of the directrix.

Solution. The given equation is $y^2 = -20x$. Comparing this with the standard form of a left-handed parabola, $y^2 = -4ax$, we observe:

$$-4a = -20 \implies a = 5$$

For this standard form:

1. The focus S is located at $(-a, 0)$. Substituting $a = 5$, the focus is $(-5, 0)$.
2. The directrix is the vertical line $x = a$. Thus, the equation of the directrix is $x = 5$ (or $x - 5 = 0$). ■

Example 1.1.6 (Upward-Opening Parabola). Find the equation of the parabola with its vertex at the origin, symmetric with respect to the y-axis, and passing through the point $P(4, 2)$.

Solution. Since the vertex is at the origin and the axis of symmetry is the y-axis, the equation must be of the form $x^2 = 4ay$ or $x^2 = -4ay$. The parabola passes through the point $P(4, 2)$, which lies in the first quadrant (where $y > 0$). Therefore, the parabola must open upwards, and we select the form:

$$x^2 = 4ay$$

Since $P(4, 2)$ lies on the curve, its coordinates must satisfy the equation:

$$(4)^2 = 4a(2)$$

$$16 = 8a \implies a = 2$$

Substituting the value of $a = 2$ back into the standard equation, we get:

$$x^2 = 4(2)y \implies x^2 = 8y$$

Thus, the required equation is $x^2 = 8y$. ■

Example 1.1.7 (Downward-Opening Parabola). Calculate the length of the latus rectum and the focal distance of the point $(-2, -1)$ for the parabola $x^2 = -4y$.

Solution. The given equation is $x^2 = -4y$. This corresponds to the standard form $x^2 = -4ay$, which represents a parabola opening downwards with the vertex at the origin. Comparing coefficients:

$$4a = 4 \implies a = 1$$

Length of Latus Rectum: The length of the latus rectum is given by $4a$.

$$\text{Length} = 4(1) = 4$$

Focal Distance: For a parabola $x^2 = -4ay$, the directrix is the horizontal line $y = a$. Here, $y = 1$ (or $y - 1 = 0$). By definition, the focal distance of a point $P(x, y)$ on the parabola is its perpendicular distance from the directrix. For the point $P(-2, -1)$:

$$\text{Focal Distance} = \text{Distance from } P(-2, -1) \text{ to } y - 1 = 0$$

$$\text{Distance} = \frac{|(-1) - 1|}{\sqrt{0^2 + 1^2}} = |-2| = 2$$

Therefore, the length of the latus rectum is 4 and the focal distance is 2. ■

1.2 Parametric Equations of a Parabola

In many geometrical problems, using Cartesian coordinates (x, y) can lead to cumbersome algebraic expressions involving square roots (e.g., $y = \pm 2\sqrt{ax}$). To simplify calculations, we often employ **parametric equations**, which express both x and y in terms of a single independent variable (parameter) t .

1.2.1 Derivation of the Parametric Coordinates

Consider the standard equation of the parabola:

$$y^2 = 4ax \tag{1.2}$$

We wish to find functions $x = f(t)$ and $y = g(t)$ that identically satisfy this equation for any real value of t .

If we set $y = 2at$, substitute this into the standard equation:

$$(2at)^2 = 4ax$$

$$4a^2t^2 = 4ax$$

Dividing both sides by $4a$ (assuming $a \neq 0$):

$$x = at^2$$

Thus, the equations:

$$x = at^2, \quad y = 2at \quad (1.3)$$

satisfy the equation of the parabola. Together, they represent the **parametric equations** of the parabola $y^2 = 4ax$.

Definition 1.2.1 (Parametric Point). Any point P on the parabola $y^2 = 4ax$ is referred to as the point ' t ', and its coordinates are denoted by $(at^2, 2at)$.

1.2.2 Properties of the Parameter t

The parameter t ranges from $-\infty$ to $+\infty$:

- If $t = 0$, the point is $(0, 0)$, which is the **Vertex**.
- If $t > 0$, $y = 2at$ is positive, representing the **upper branch** of the parabola.
- If $t < 0$, $y = 2at$ is negative, representing the **lower branch** of the parabola.
- Geometrically, the parameter t is related to the slope m of the chord joining the vertex $(0, 0)$ to the point $P(at^2, 2at)$. The slope is $m = \frac{2at}{at^2} = \frac{2}{t}$. Thus, $t = \frac{2}{m}$.

1.2.3 Parametric Forms for Other Standard Parabolas

While $(at^2, 2at)$ is the most common form, other orientations have corresponding parametric equations:

Cartesian Equation	Parametric Equations
$y^2 = 4ax$	$x = at^2, \quad y = 2at$
$y^2 = -4ax$	$x = -at^2, \quad y = 2at$
$x^2 = 4ay$	$x = 2at, \quad y = at^2$
$x^2 = -4ay$	$x = 2at, \quad y = -at^2$

1.2.4 Visual Representation

The following figure illustrates a general point $P(t)$ on the parabola and its relation to the parameter.

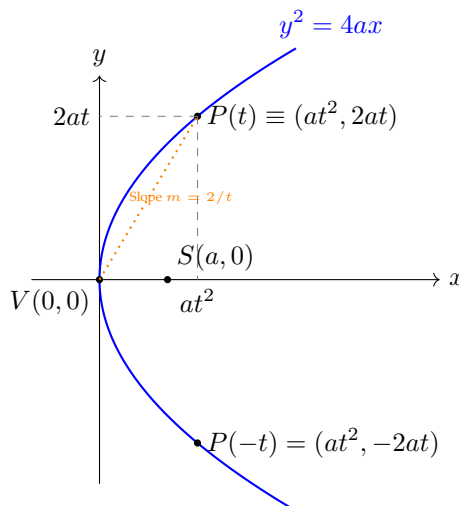


Figure 1.3: The Point $P(t)$ on the Parabola $y^2 = 4ax$

Example 1.2.2 (Basic Parameter Determination). Find the value of the parameter t for the point $(3, -6)$ lying on the parabola $y^2 = 12x$.

Solution. First, we identify the standard form of the parabola. Given $y^2 = 12x$, we compare it with $y^2 = 4ax$.

$$4a = 12 \implies a = 3$$

The parametric coordinates for $y^2 = 4ax$ are $(at^2, 2at)$. Since the point is $(3, -6)$, we equate the y -coordinate (which is linear and easier to solve) to $2at$:

$$2at = -6$$

Substituting $a = 3$:

$$2(3)t = -6$$

$$6t = -6 \implies t = -1$$

To verify, we check the x -coordinate:

$$x = at^2 = 3(-1)^2 = 3(1) = 3$$

This matches the given abscissa. Thus, the parameter is $t = -1$. ■

Example 1.2.3 (The Focal Chord Property). If the chord joining two points $P(t_1)$ and $Q(t_2)$ on the parabola $y^2 = 4ax$ passes through the focus, prove that $t_1 t_2 = -1$.

Solution. Let the coordinates of the points be $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$. The focus of the parabola is $S(a, 0)$. Since P, S , and Q are collinear (lying on the same line), the slope of the segment PS must equal the slope of the segment QS .

The slope of a line joining (x_1, y_1) and (x_2, y_2) is $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Slope of PS :

$$m_{PS} = \frac{2at_1 - 0}{at_1^2 - a} = \frac{2at_1}{a(t_1^2 - 1)} = \frac{2t_1}{t_1^2 - 1}$$

Slope of QS :

$$m_{QS} = \frac{2at_2 - 0}{at_2^2 - a} = \frac{2at_2}{a(t_2^2 - 1)} = \frac{2t_2}{t_2^2 - 1}$$

Equating the slopes $m_{PS} = m_{QS}$:

$$\frac{2t_1}{t_1^2 - 1} = \frac{2t_2}{t_2^2 - 1}$$

Dividing by 2 and cross-multiplying:

$$t_1(t_2^2 - 1) = t_2(t_1^2 - 1)$$

$$t_1 t_2^2 - t_1 = t_2 t_1^2 - t_2$$

Rearranging terms:

$$t_1 t_2^2 - t_2 t_1^2 - t_1 + t_2 = 0$$

Factor out $t_1 t_2$ from the first two terms:

$$t_1 t_2(t_2 - t_1) + (t_2 - t_1) = 0$$

$$(t_2 - t_1)(t_1 t_2 + 1) = 0$$

Since P and Q are distinct points, $t_1 \neq t_2$, so $t_2 - t_1 \neq 0$. Therefore, we must have:

$$t_1 t_2 + 1 = 0 \implies t_1 t_2 = -1$$

■

Example 1.2.4 (Area of a Triangle Inscribed in a Parabola). Show that the area of the triangle formed by three points t_1, t_2, t_3 on the parabola $y^2 = 4ax$ is $a^2|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$.

Solution. Let the three points be $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$, and $C(at_3^2, 2at_3)$. The area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is given by the determinant magnitude:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Substituting the parametric coordinates:

$$\text{Area} = \frac{1}{2} |at_1^2(2at_2 - 2at_3) + at_2^2(2at_3 - 2at_1) + at_3^2(2at_1 - 2at_2)|$$

Factor out $2a$ from the terms inside the brackets and a from the x -coordinates, giving a total factor of $2a^2$:

$$\text{Area} = \frac{1}{2} (2a^2) |t_1^2(t_2 - t_3) + t_2^2(t_3 - t_1) + t_3^2(t_1 - t_2)|$$

$$\text{Area} = a^2 |t_1^2(t_2 - t_3) + t_2^2t_3 - t_2^2t_1 + t_3^2t_1 - t_3^2t_2|$$

This expression is a cyclic polynomial. We can rearrange it to factorize:

$$\Delta = t_1^2(t_2 - t_3) - t_1(t_2^2 - t_3^2) + t_2t_3(t_2 - t_3)$$

$$\Delta = (t_2 - t_3) [t_1^2 - t_1(t_2 + t_3) + t_2t_3]$$

The term in the square brackets factors as $(t_1 - t_2)(t_1 - t_3)$.

$$\Delta = (t_2 - t_3)(t_1 - t_2)(t_1 - t_3)$$

Taking the absolute value and rearranging to cyclic order $(t_3 - t_1) = -(t_1 - t_3)$:

$$\text{Area} = a^2 |(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$$

■

Example 1.2.5 (Right Angle at the Vertex). A chord passing through points $P(t_1)$ and $Q(t_2)$ on the parabola $y^2 = 4ax$ subtends a right angle at the vertex. Prove that the chord passes through a fixed point on the axis of the parabola.

Solution. Let the vertex be the origin $V(0, 0)$. The coordinates of the points are $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$.

The chord PQ subtends a right angle at V , which means the lines VP and VQ are perpendicular. Therefore, the product of their slopes must be -1 .

Slope of VP :

$$m_1 = \frac{2at_1 - 0}{at_1^2 - 0} = \frac{2}{t_1}$$

Slope of VQ :

$$m_2 = \frac{2at_2 - 0}{at_2^2 - 0} = \frac{2}{t_2}$$

Condition for perpendicularity:

$$m_1 m_2 = -1 \implies \left(\frac{2}{t_1}\right) \left(\frac{2}{t_2}\right) = -1$$

$$\frac{4}{t_1 t_2} = -1 \implies t_1 t_2 = -4$$

Now, consider the equation of the chord passing through P and Q :

$$y - 2at_1 = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}(x - at_1^2)$$

Simplifying the slope:

$$\text{Slope} = \frac{2a(t_2 - t_1)}{a(t_2 - t_1)(t_2 + t_1)} = \frac{2}{t_1 + t_2}$$

The equation becomes:

$$y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$$

$$y(t_1 + t_2) - 2at_1(t_1 + t_2) = 2x - 2at_1^2$$

$$y(t_1 + t_2) - 2at_1^2 - 2at_1t_2 = 2x - 2at_1^2$$

Canceling $-2at_1^2$ from both sides:

$$y(t_1 + t_2) - 2at_1t_2 = 2x$$

Substitute the condition $t_1t_2 = -4$:

$$y(t_1 + t_2) - 2a(-4) = 2x$$

$$y(t_1 + t_2) + 8a = 2x$$

Rearranging to group terms involving the variable parameter $(t_1 + t_2)$:

$$2x - 8a - y(t_1 + t_2) = 0$$

For this line to pass through a fixed point for any value of t_1 and t_2 , the coefficients of the variable term and the constant term must vanish independently, or we check the intersection with the axis ($y = 0$). If $y = 0$, then $2x - 8a = 0 \implies x = 4a$. Thus, the chord always passes through the fixed point $(4a, 0)$ on the axis of the parabola. ■

1.3 Tangents and Normals

Calculus and coordinate geometry provide powerful tools to study the limiting positions of chords (tangents) and the lines perpendicular to them (normals). We analyze these lines for the standard parabola $y^2 = 4ax$.

1.3.1 Equation of the Tangent

Definition 1.3.1 (Tangent). A tangent to a curve at a point P is the limiting position of a secant line PQ as the point Q moves along the curve and approaches P .

We derive the equation of the tangent in three standard forms:

1. Point Form (Equation at a Point)

Let $P(x_1, y_1)$ be a point lying on the parabola $y^2 = 4ax$. Differentiating the equation of the parabola with respect to x :

$$2y \frac{dy}{dx} = 4a \implies \frac{dy}{dx} = \frac{2a}{y}$$

The slope of the tangent at (x_1, y_1) is $m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} = \frac{2a}{y_1}$. Using the point-slope form $y - y_1 = m(x - x_1)$:

$$y - y_1 = \frac{2a}{y_1}(x - x_1)$$

$$yy_1 - y_1^2 = 2ax - 2ax_1$$

Since (x_1, y_1) lies on the parabola, $y_1^2 = 4ax_1$. Substituting this:

$$yy_1 - 4ax_1 = 2ax - 2ax_1$$

$$yy_1 = 2ax + 2ax_1$$

Thus, the equation of the tangent is:

$$\boxed{yy_1 = 2a(x + x_1)} \quad (1.4)$$

2. Parametric Form

For a point $P(t)$ with coordinates $(at^2, 2at)$, we substitute $x_1 = at^2$ and $y_1 = 2at$ into the point form equation.

$$y(2at) = 2a(x + at^2)$$

Dividing both sides by $2a$ (assuming $a \neq 0$):

$$ty = x + at^2 \quad (1.5)$$

Therefore, the tangent at the point t is

$$\boxed{y = \frac{x}{t} + at} \quad (1.6)$$

3. Slope Form (Condition of Tangency)

Let the line $y = mx + c$ be a tangent to the parabola $y^2 = 4ax$. Substituting y into the parabola equation:

$$(mx + c)^2 = 4ax$$

$$m^2x^2 + 2mcx + c^2 - 4ax = 0$$

$$m^2x^2 + 2(mc - 2a)x + c^2 = 0$$

For the line to be a tangent, this quadratic equation must have equal roots (the points of intersection coincide). Thus, the discriminant D must be zero:

$$D = [2(mc - 2a)]^2 - 4(m^2)(c^2) = 0$$

$$4(m^2c^2 - 4amc + 4a^2) - 4m^2c^2 = 0$$

$$-16amc + 16a^2 = 0 \implies amc = a^2$$

Since $a \neq 0$, we get the condition of tangency:

$$c = \frac{a}{m}$$

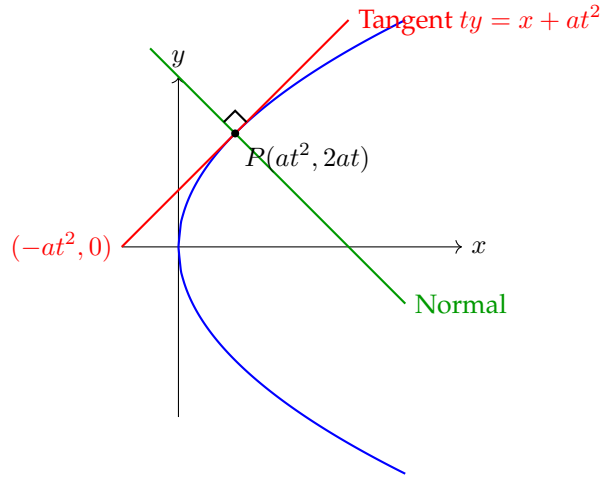
Substituting this back, the equation of the tangent with slope m is:

$$\boxed{y = mx + \frac{a}{m}} \quad (1.7)$$

The point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

1.3.2 Equation of the Normal

Definition 1.3.2 (Normal). The normal to a curve at a point P is the straight line passing through P and perpendicular to the tangent at P .

Figure 1.4: Tangent and Normal at a point P on the Parabola**1. Point Form**

Let the tangent at $P(x_1, y_1)$ have slope $m_T = \frac{2a}{y_1}$. The slope of the normal, m_N , must satisfy $m_T m_N = -1$.

$$m_N = -\frac{y_1}{2a}$$

The equation of the normal is:

$$y - y_1 = -\frac{y_1}{2a}(x - x_1) \quad (1.8)$$

2. Parametric Form

At the point $(at^2, 2at)$, the slope of the tangent is $\frac{1}{t}$. Therefore, the slope of the normal is $-t$. Equation of the normal:

$$y - 2at = -t(x - at^2)$$

Rearranging: the equation of normal becomes

$$y + tx = 2at + at^3 \quad (1.9)$$

This is a cubic equation in t , implying that from a given point (x, y) , up to three normals can be drawn to a parabola.

3. Slope Form

If we set the slope of the normal to be m , then from the parametric form, $m = -t$, which implies $t = -m$. Substituting $t = -m$ into the parametric normal equation:

$$y + (-m)x = 2a(-m) + a(-m)^3$$

$$y = mx - 2am - am^3 \quad (1.10)$$

This describes the normal with slope m . The foot of the normal (the point of incidence) is $(am^2, -2am)$.

1.3.3 Important Results regarding Tangents and Normals

- 1. Point of Intersection of Tangents:** The point of intersection of two tangents at $P(t_1)$ and $Q(t_2)$ is $(at_1 t_2, a(t_1 + t_2))$.

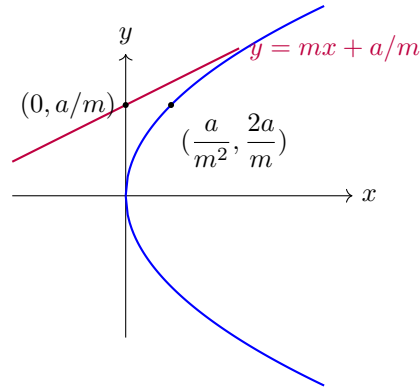


Figure 1.5: Tangent in Slope Form

2. Point of Intersection of Normals: The point of intersection of two normals at $P(t_1)$ and $Q(t_2)$ is given by:

$$x = 2a + a(t_1^2 + t_2^2 + t_1 t_2), \quad y = -at_1 t_2 (t_1 + t_2)$$

3. Subtangent and Subnormal: Let the tangent and normal at $P(x, y)$ meet the axis of the parabola at T and N respectively. Let M be the projection of P on the axis.

- Length of Subtangent $TM = 2x$ (Bisected by vertex).
- Length of Subnormal $MN = 2a$ (Constant).

Example 1.3.3 (Tangent in Slope Form). Find the equation of the tangent to the parabola $y^2 = 12x$ which is parallel to the line $2x - y + 5 = 0$.

Solution. We identify the parameters of the parabola $y^2 = 4ax$:

$$4a = 12 \implies a = 3$$

The given line can be rewritten as $y = 2x + 5$, so the slope is $m = 2$. Since the required tangent is parallel to this line, it must also have slope $m = 2$. Using the slope form of the tangent equation $y = mx + \frac{a}{m}$:

$$y = 2x + \frac{3}{2}$$

Multiplying by 2 to clear the fraction:

$$2y = 4x + 3 \quad \text{or} \quad 4x - 2y + 3 = 0$$

■

Example 1.3.4 (Intersection of Tangents (Parametric Form)). Tangents are drawn to the parabola $y^2 = 8x$ at points P and Q corresponding to parameter values $t = 1$ and $t = 3$ respectively. Find the coordinates of their point of intersection.

Solution. For the parabola $y^2 = 8x$, we have $a = 2$. We utilize the standard result that the point of intersection of tangents at t_1 and t_2 is:

$$R(at_1 t_2, a(t_1 + t_2))$$

Here, $t_1 = 1$ and $t_2 = 3$. Substituting these values:

- x-coordinate: $2(1)(3) = 6$
- y-coordinate: $2(1 + 3) = 8$

Thus, the tangents intersect at the point $(6, 8)$.

■

Example 1.3.5 (Normal in Point Form and Subnormal Length). Find the equation of the normal to the parabola $y^2 = 4x$ at the point $(1, 2)$ and verify that the length of the subnormal is constant.

Solution. Here $a = 1$. The point $P(1, 2)$ lies on the parabola. The equation of the tangent at (x_1, y_1) is $yy_1 = 2a(x + x_1)$. Slope of tangent: $m_T = \frac{2a}{y_1} = \frac{2(1)}{2} = 1$. Therefore, the slope of the normal is $m_N = -1$.

Using the point-slope form for the normal:

$$y - 2 = -1(x - 1) \implies x + y - 3 = 0$$

Verification of Subnormal: The normal $x + y = 3$ meets the x-axis (where $y = 0$) at $N(3, 0)$. The projection of $P(1, 2)$ on the x-axis is $M(1, 0)$.

$$\text{Length of Subnormal} = MN = |3 - 1| = 2$$

The theoretical constant length is $2a = 2(1) = 2$. The values match perfectly. ■

Example 1.3.6 (Normal in Slope Form). Find the equation of the normal to the parabola $y^2 = 6x$ which makes an angle of 135° with the positive x-axis.

Solution. The parabola is $y^2 = 4ax$ with $a = \frac{6}{4} = \frac{3}{2}$. The slope of the normal is given by $m = \tan(135^\circ) = -1$. We use the standard slope form of the normal equation:

$$y = mx - 2am - am^3$$

Substituting $a = 1.5$ and $m = -1$:

$$y = (-1)x - 2(1.5)(-1) - (1.5)(-1)^3$$

$$y = -x + 3 + 1.5$$

$$y = -x + 4.5$$

Multiplying by 2 yields the standard integer form:

$$2x + 2y - 9 = 0$$

Example 1.3.7 (Locus of Perpendicular Tangents). Prove that the locus of the point of intersection of two perpendicular tangents to the parabola $y^2 = 4ax$ is the directrix $x + a = 0$.

Solution. Let the point of intersection be $P(h, k)$. The equation of any tangent to the parabola having slope m is:

$$y = mx + \frac{a}{m}$$

Since this line passes through $P(h, k)$, the coordinates must satisfy the equation:

$$k = mh + \frac{a}{m}$$

Multiplying by m and rearranging forms a quadratic equation in m :

$$hm^2 - km + a = 0$$

The roots m_1 and m_2 of this quadratic represent the slopes of the two tangents passing through P . For the tangents to be perpendicular, the product of their slopes must be -1 . From the properties of quadratic equations:

$$m_1 m_2 = \frac{\text{Constant term}}{\text{Coefficient of } m^2} = \frac{a}{h}$$

Setting $\frac{a}{h} = -1$, we get:

$$h = -a \implies h + a = 0$$

Generalizing the locus by replacing h with x , the equation is $x + a = 0$, which is the directrix. ■

Example 1.3.8 (Common Tangent to Parabola and Circle). Find the equations of the common tangents to the parabola $y^2 = 16x$ and the circle $x^2 + y^2 = 8$.

Solution. For the parabola $y^2 = 16x$, we have $a = 4$. Any tangent to the parabola has the form:

$$y = mx + \frac{4}{m} \implies m^2x - my + 4 = 0$$

For this line to also be tangent to the circle $x^2 + y^2 = 8$, the perpendicular distance from the center of the circle $(0, 0)$ to the line must equal the radius $r = \sqrt{8} = 2\sqrt{2}$.

$$\frac{|m^2(0) - m(0) + 4|}{\sqrt{(m^2)^2 + (-m)^2}} = 2\sqrt{2}$$

$$\frac{4}{\sqrt{m^4 + m^2}} = 2\sqrt{2}$$

Squaring both sides:

$$\frac{16}{m^4 + m^2} = 8 \implies 2 = m^4 + m^2$$

$$m^4 + m^2 - 2 = 0$$

Factoring the quadratic in m^2 :

$$(m^2 + 2)(m^2 - 1) = 0$$

Since m must be real, $m^2 \neq -2$. Thus, $m^2 = 1 \implies m = \pm 1$. Substituting $m = 1$ and $m = -1$ into the tangent equation:

$$y = x + 4 \quad \text{and} \quad y = -x - 4$$

Or equivalently, $x - y + 4 = 0$ and $x + y + 4 = 0$. ■

Example 1.3.9 (Normal Meeting the Curve Again). If the normal to the parabola $y^2 = 4ax$ at the point t_1 meets the parabola again at the point t_2 , prove that $t_2 = -t_1 - \frac{2}{t_1}$.

Solution. The equation of the normal at point $P(t_1)$ is:

$$y + t_1x = 2at_1 + at_1^3$$

This line intersects the parabola at point $Q(t_2)$, which has coordinates $(at_2^2, 2at_2)$. Substituting these into the normal equation:

$$2at_2 + t_1(at_2^2) = 2at_1 + at_1^3$$

Dividing by a and rearranging to group terms:

$$2(t_2 - t_1) + t_1(t_2^2 - t_1^2) = 0$$

Factoring the difference of squares $(t_2^2 - t_1^2) = (t_2 - t_1)(t_2 + t_1)$:

$$2(t_2 - t_1) + t_1(t_2 - t_1)(t_2 + t_1) = 0$$

$$(t_2 - t_1)[2 + t_1(t_2 + t_1)] = 0$$

Since P and Q are distinct points, $t_1 \neq t_2$, so $t_2 - t_1 \neq 0$. Therefore, the term in the brackets must be zero:

$$2 + t_1t_2 + t_1^2 = 0$$

Solving for t_2 :

$$t_1t_2 = -(2 + t_1^2)$$

$$t_2 = -\frac{2}{t_1} - t_1$$

This result is crucial for problems involving chords that are normal at one end. ■

Example 1.3.10 (Condition for a Line to be a Normal). Find the condition for the straight line $lx + my + n = 0$ to be a normal to the parabola $y^2 = 4ax$.

Solution. We know the equation of a normal to $y^2 = 4ax$ in terms of its slope M is:

$$y = Mx - 2aM - aM^3$$

Rewrite this as:

$$Mx - y - (2aM + aM^3) = 0$$

The given line is $lx + my + n = 0$. For these two equations to represent the same line, their coefficients must be proportional:

$$\frac{l}{M} = \frac{m}{-1} = \frac{n}{-(2aM + aM^3)}$$

From the first two parts:

$$\frac{l}{M} = -m \implies M = -\frac{l}{m}$$

Now equate the second and third parts:

$$\begin{aligned} -m &= \frac{n}{-(2aM + aM^3)} \implies m = \frac{n}{2aM + aM^3} \\ 2aM + aM^3 &= \frac{n}{m} \end{aligned}$$

Substituting $M = -\frac{l}{m}$:

$$\begin{aligned} 2a \left(-\frac{l}{m} \right) + a \left(-\frac{l}{m} \right)^3 &= \frac{n}{m} \\ -\frac{2al}{m} - \frac{al^3}{m^3} &= \frac{n}{m} \end{aligned}$$

Multiplying through by $-m^3$:

$$\begin{aligned} 2alm^2 + al^3 &= -nm^2 \\ al^3 + 2alm^2 + nm^2 &= 0 \end{aligned}$$

This is the required condition. ■

1.4 Pair of Tangents from a Point

From a fixed point $P(x_1, y_1)$ lying outside the parabola, it is possible to draw exactly two distinct tangents to the curve. The combined equation of these two straight lines is a crucial tool in analytical geometry, derived using the notation of S , S_1 , and T .

1.4.1 Standard Notation

For the standard parabola $y^2 - 4ax = 0$, we define the following notations which are applicable to all conics:

- **S:** The expression of the curve itself.

$$S \equiv y^2 - 4ax$$

- **S_1 :** The result of substituting the coordinates of the point $P(x_1, y_1)$ into S .

$$S_1 \equiv y_1^2 - 4ax_1$$

- **T:** The expression obtained by replacing y^2 with yy_1 and x with $\frac{x + x_1}{2}$.

$$T \equiv yy_1 - 2a(x + x_1)$$

1.4.2 The Combined Equation

Theorem 1.4.1. The combined equation of the pair of tangents drawn from an external point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$ is given by:

$$SS_1 = T^2 \quad (1.11)$$

Explicitly:

$$(y^2 - 4ax)(y_1^2 - 4ax_1) = [yy_1 - 2a(x + x_1)]^2$$

Proof. Let $P(x_1, y_1)$ be the external point. Let $Q(x, y)$ be any point on either of the tangents drawn from P . Consider the line segment PQ .

Let R be a point that divides the segment PQ internally in the ratio $\lambda : 1$. The coordinates of R are:

$$\left(\frac{\lambda x + x_1}{\lambda + 1}, \frac{\lambda y + y_1}{\lambda + 1} \right)$$

If the point R lies on the parabola $y^2 - 4ax = 0$, its coordinates must satisfy the equation:

$$\left(\frac{\lambda y + y_1}{\lambda + 1} \right)^2 - 4a \left(\frac{\lambda x + x_1}{\lambda + 1} \right) = 0$$

Multiplying by $(\lambda + 1)^2$:

$$(\lambda y + y_1)^2 - 4a(\lambda + 1)(\lambda x + x_1) = 0$$

Expanding and collecting terms in powers of λ :

$$\lambda^2(y^2 - 4ax) + 2\lambda[yy_1 - 2a(x + x_1)] + (y_1^2 - 4ax_1) = 0$$

Using our standard notation, this quadratic equation in λ becomes:

$$\lambda^2 S + 2\lambda T + S_1 = 0$$

The roots of this quadratic correspond to the points where the line PQ intersects the parabola.

If the line PQ is to be a **tangent** to the parabola, it must touch the curve at exactly one point (two coincident points). Therefore, the discriminant of the quadratic equation in λ must be zero:

$$(2T)^2 - 4(S)(S_1) = 0$$

$$4T^2 - 4SS_1 = 0$$

Dividing by 4, we obtain the required locus of the point $Q(x, y)$:

$$T^2 = SS_1 \quad \text{or} \quad SS_1 = T^2$$

■

1.4.3 Geometric Interpretation

The equation $SS_1 = T^2$ represents a second-degree homogeneous equation (after shifting origin to P) consisting of two straight lines passing through $P(x_1, y_1)$.

1. If $S_1 > 0$: The point P is outside the parabola, and two real tangents exist.
2. If $S_1 = 0$: The point P is on the parabola. The equation becomes $T^2 = 0$, implying coincident lines (the single tangent at P).
3. If $S_1 < 0$: The point P is inside the parabola, and the tangents are imaginary.

Example 1.4.2. Find the combined equation of the pair of tangents drawn from the point $(-2, 3)$ to the parabola $y^2 = 8x$.

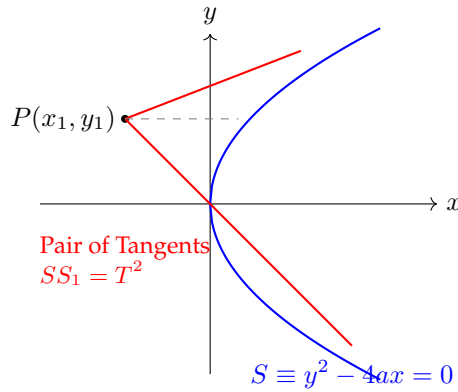


Figure 1.6: Pair of Tangents from an External Point

Solution. Here, the equation of the parabola is $y^2 - 8x = 0$. Thus, $S \equiv y^2 - 8x$. The given point is $P(x_1, y_1) = (-2, 3)$.

Calculate S_1 :

$$S_1 = y_1^2 - 8x_1 = (3)^2 - 8(-2) = 9 + 16 = 25$$

Calculate T :

$$T \equiv yy_1 - 4(x + x_1) = 3y - 4(x - 2) = 3y - 4x + 8$$

The combined equation is given by $SS_1 = T^2$:

$$25(y^2 - 8x) = (3y - 4x + 8)^2$$

Expanding the Right Hand Side:

$$25y^2 - 200x = 9y^2 + 16x^2 + 64 - 24xy - 64x + 48y$$

Rearranging terms to one side:

$$16x^2 - 24xy - 16y^2 + 136x + 48y + 64 = 0$$

Dividing by 8 simplifies the result:

$$2x^2 - 3xy - 2y^2 + 17x + 6y + 8 = 0$$

This is the required combined equation of the pair of tangents. ■

Example 1.4.3. Show that the pair of tangents drawn from the point $(-2, 5)$ to the parabola $y^2 = 8x$ are at right angles to each other.

Solution. The parabola is $y^2 = 8x$, so $4a = 8 \implies a = 2$. The equation of the directrix is $x = -a \implies x = -2$. The point $P(-2, 5)$ lies on the directrix. Geometrically, tangents drawn from the directrix are always perpendicular. Let us verify this using $SS_1 = T^2$.

$S \equiv y^2 - 8x$. $S_1 = (5)^2 - 8(-2) = 25 + 16 = 41$. $T = 5y - 4(x - 2) = 5y - 4x + 8$.

Equation $SS_1 = T^2$:

$$41(y^2 - 8x) = (5y - 4x + 8)^2$$

$$41y^2 - 328x = 25y^2 + 16x^2 + 64 - 40xy - 64x + 80y$$

Rearranging into the general form $Ax^2 + 2Hxy + By^2 + \dots = 0$:

$$16x^2 - 40xy + (25 - 41)y^2 + \dots = 0$$

$$16x^2 - 40xy - 16y^2 + \dots = 0$$

For the lines represented by a second-degree homogeneous equation to be perpendicular, the sum of the coefficients of x^2 and y^2 must be zero ($A + B = 0$). Here, $A = 16$ and $B = -16$.

$$A + B = 16 - 16 = 0$$

Thus, the tangents are perpendicular. ■

Example 1.4.4. Find the locus of a point P such that the pair of tangents drawn from it to the parabola $y^2 = 4ax$ include an angle of 45° .

Solution. Let $P(h, k)$ be the point. The combined equation of tangents is $SS_1 = T^2$. The angle θ between the lines given by $SS_1 = T^2$ for a parabola is given by the formula:

$$\tan \theta = \frac{\sqrt{S_1}}{x_1 + a}$$

Here, $(x_1, y_1) = (h, k)$, so $S_1 = k^2 - 4ah$. Given $\theta = 45^\circ$, we have $\tan 45^\circ = 1$.

$$1 = \frac{\sqrt{k^2 - 4ah}}{h + a}$$

Squaring both sides:

$$(h + a)^2 = k^2 - 4ah$$

$$h^2 + 2ah + a^2 = k^2 - 4ah$$

Rearranging terms:

$$k^2 - h^2 - 6ah - a^2 = 0$$

Replacing (h, k) with (x, y) to find the locus:

$$y^2 - x^2 - 6ax - a^2 = 0$$

This locus represents a hyperbola. ■

Example 1.4.5. Determine the individual equations of the tangents drawn from $(0, -1)$ to the parabola $x^2 = 4y$.

Solution. Note: The parabola is upward opening ($x^2 = 4y$). $S \equiv x^2 - 4y$. $S_1 = (0)^2 - 4(-1) = 4$. $T \equiv x(0) - 2(y - 1) = -2y + 2$.

Using $SS_1 = T^2$:

$$4(x^2 - 4y) = (-2y + 2)^2$$

$$4(x^2 - 4y) = 4(y - 1)^2$$

Canceling 4:

$$x^2 - 4y = y^2 - 2y + 1$$

$$x^2 - y^2 - 2y - 1 = 0$$

$$x^2 - (y^2 + 2y + 1) = 0$$

$$x^2 - (y + 1)^2 = 0$$

Using difference of squares $a^2 - b^2 = (a - b)(a + b)$:

$$(x - (y + 1))(x + (y + 1)) = 0$$

The individual equations of the tangents are:

$$x - y - 1 = 0 \quad \text{and} \quad x + y + 1 = 0$$
■

Example 1.4.6 (Area of the Triangle of Tangents). Find the area of the triangle formed by the pair of tangents from the point (x_1, y_1) to the parabola $y^2 = 4ax$ and the chord of contact.

Solution. Let the tangents from $P(x_1, y_1)$ touch the parabola at points Q and R . The line QR is the chord of contact, given by $T = 0$. The area of the triangle formed by tangents from an external point and their chord of contact is given by the formula:

$$\text{Area} = \frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$$

Or more simply, $\text{Area} = \frac{(S_1)^{3/2}}{2a}$.

Application: Let the parabola be $y^2 = 4x$ ($a = 1$) and the point be $(4, 5)$. $S_1 = 5^2 - 4(4) = 25 - 16 = 9$.

$$\text{Area} = \frac{(9)^{3/2}}{2(1)} = \frac{27}{2} = 13.5 \text{ sq. units}$$

■

Example 1.4.7. Find the pair of tangents from the origin $(0, 0)$ to the parabola $y = x^2 - 2x + 2$.

Solution. Rewrite the parabola as $x^2 - 2x - y + 2 = 0$. $S \equiv x^2 - 2x - y + 2$. The point is $(0, 0)$. $S_1 = 0^2 - 2(0) - 0 + 2 = 2$.

Find T at $(0, 0)$. We replace $x^2 \rightarrow x(0)$, $x \rightarrow \frac{x+0}{2}$, $y \rightarrow \frac{y+0}{2}$.

$$T \equiv 0 - 2\left(\frac{x}{2}\right) - \frac{y}{2} + 2 = -x - \frac{y}{2} + 2$$

$$T = \frac{-2x - y + 4}{2}$$

Equation $SS_1 = T^2$:

$$2(x^2 - 2x - y + 2) = \left(\frac{-2x - y + 4}{2}\right)^2$$

$$2(x^2 - 2x - y + 2) = \frac{1}{4}(4x^2 + y^2 + 16 + 4xy - 16x - 8y)$$

Multiply by 4:

$$8(x^2 - 2x - y + 2) = 4x^2 + y^2 + 16 + 4xy - 16x - 8y$$

$$8x^2 - 16x - 8y + 16 = 4x^2 + y^2 + 16 + 4xy - 16x - 8y$$

Simplifying by canceling common terms $(-16x, -8y, 16)$:

$$8x^2 = 4x^2 + y^2 + 4xy$$

$$4x^2 - 4xy - y^2 = 0$$

This is the combined equation of the tangents from the origin. ■

1.4.4 Angle Between the Pair of Tangents

If θ is the angle between the two tangents represented by $SS_1 = T^2$, it can be determined by comparing the equation with the general form $Ax^2 + 2Hxy + By^2 + \dots = 0$. For the parabola $y^2 = 4ax$, the angle is given by:

$$\tan \theta = \frac{\sqrt{y_1^2 - 4ax_1}}{x_1 + a} = \frac{\sqrt{S_1}}{x_1 + a}$$

Note that if $x_1 + a = 0$ (the point lies on the directrix), $\tan \theta \rightarrow \infty$, implying $\theta = 90^\circ$. This reconfirms that perpendicular tangents intersect on the directrix.

Example 1.4.8 (Calculating a Specific Angle). Find the angle between the pair of tangents drawn from the point $P(1, 4)$ to the parabola $y^2 = 4x$.

Solution. For the parabola $y^2 = 4x$, we identify the standard parameter:

$$4a = 4 \implies a = 1$$

The coordinates of the external point are $(x_1, y_1) = (1, 4)$.

First, we calculate the power of the point S_1 :

$$S_1 = y_1^2 - 4ax_1 = (4)^2 - 4(1)(1) = 16 - 4 = 12$$

We use the formula for the angle θ between tangents drawn from (x_1, y_1) :

$$\tan \theta = \frac{\sqrt{S_1}}{x_1 + a}$$

Substituting the known values:

$$\tan \theta = \frac{\sqrt{12}}{1 + 1} = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

Since $\tan \theta = \sqrt{3}$, the angle is:

$$\theta = 60^\circ$$

Thus, the tangents intersect at an angle of 60° . ■

Example 1.4.9 (General Angle Calculation). Determine the angle between the tangents drawn from the point $(3, 5)$ to the parabola $y^2 = 8x$.

Solution. Equation of parabola: $y^2 = 8x$. Comparing with $y^2 = 4ax$, we have $4a = 8 \implies a = 2$. The point is $P(3, 5)$.

Calculate S_1 :

$$S_1 = y_1^2 - 4ax_1 = (5)^2 - 8(3) = 25 - 24 = 1$$

Calculate the denominator $x_1 + a$:

$$x_1 + a = 3 + 2 = 5$$

Using the angle formula:

$$\tan \theta = \frac{\sqrt{S_1}}{x_1 + a} = \frac{\sqrt{1}}{5} = \frac{1}{5}$$

Therefore, the angle between the tangents is:

$$\theta = \tan^{-1} \left(\frac{1}{5} \right)$$
■

1.5 Pole and Polar

The concepts of Pole and Polar generalize the relationship between a point and a line with respect to a conic section. While the tangent is related to a point on the curve, and the chord of contact is related to a point outside the curve, the polar applies to any point in the plane (except the vertex in some definitions, though algebraically it holds).

Definition 1.5.1 (Polar and Pole). Let P be a fixed point in the plane of a parabola. Draw any straight line through P intersecting the parabola at points A and B . The locus of the point of intersection of the tangents drawn to the parabola at A and B is a straight line called the **Polar** of the point P .

Conversely, the point P is called the **Pole** of that line.

1.5.1 Equation of the Polar

We derive the equation of the polar of a point $P(x_1, y_1)$ with respect to the parabola $y^2 = 4ax$.

Let $P(x_1, y_1)$ be the fixed point (Pole). Let a secant line through P meet the parabola at points A and B . Let the tangents at A and B intersect at the point $Q(h, k)$.

From the properties of tangents, the line connecting the points of contact (AB) is the **Chord of Contact** of the point Q . The equation of the chord of contact of $Q(h, k)$ is:

$$yk = 2a(x + h)$$

Since this line AB passes through the fixed point $P(x_1, y_1)$, the coordinates of P must satisfy this equation:

$$y_1 k = 2a(x_1 + h)$$

We treat this relation as a condition on the coordinates (h, k) of the point Q . Rearranging the terms to view h and k as variables:

$$ky_1 = 2a(x + x_1) \quad (\text{replacing } h \rightarrow x, k \rightarrow y)$$

Thus, the locus of Q is the straight line:

$$yy_1 = 2a(x + x_1) \quad (1.12)$$

This is the equation of the polar. It is formally identical to the equation of the tangent ($T = 0$), but here (x_1, y_1) need not lie on the curve.

1.5.2 Geometrical Observations

The position of the pole P determines the nature of the polar line:

1. **Point Outside:** If P lies outside the parabola, the polar is the **Chord of Contact** of tangents drawn from P .
2. **Point On Curve:** If P lies on the parabola, the polar is the **Tangent** at P .
3. **Point Inside:** If P lies inside the parabola, the polar is a line lying entirely outside the parabola (it does not intersect the curve).

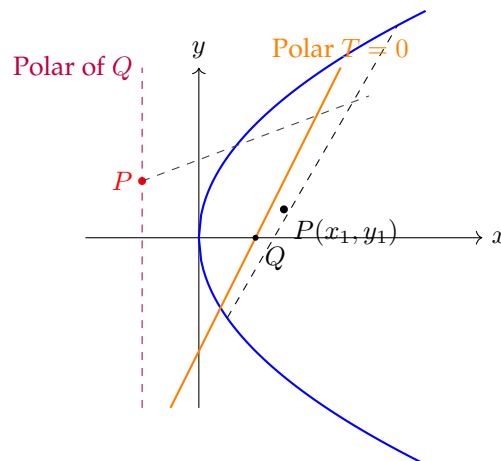


Figure 1.7: Pole P , its Polar line, and Conjugate Point Q

1.5.3 Important Properties of Pole and Polar

Theorem 1.5.2 (Reciprocal Property / Conjugate Points). *If the polar of a point P passes through a point Q , then the polar of Q must pass through P .*

Proof. Let the coordinates be $P(x_1, y_1)$ and $Q(x_2, y_2)$. The polar of P is $yy_1 = 2a(x + x_1)$. If Q lies on this line, then:

$$y_2y_1 = 2a(x_2 + x_1)$$

The symmetry of this equation ($y_1y_2 = 2a(x_1 + x_2)$) shows that $P(x_1, y_1)$ satisfies the equation $yy_2 = 2a(x + x_2)$, which is the polar of Q . Such points P and Q are called **Conjugate Points**. ■

Theorem 1.5.3 (Conjugate Lines). *If the pole of a line L_1 lies on a line L_2 , then the pole of L_2 lies on L_1 . Such lines are called **Conjugate Lines**.*

Theorem 1.5.4 (Focus and Directrix). 1. *The Polar of the **Focus** is the **Directrix**.*

2. *The Polar of any point on the Directrix is a **Focal Chord** (a chord passing through the focus).*

Proof. Consider the focus $S(a, 0)$. The polar equation is:

$$y(0) = 2a(x + a) \implies 0 = 2ax + 2a^2 \implies x + a = 0$$

This is the equation of the directrix. Conversely, let $P(-a, y_1)$ be a point on the directrix. Its polar is:

$$yy_1 = 2a(x - a)$$

Does this pass through the focus $(a, 0)$? Substitute $x = a, y = 0$:

$$0 \cdot y_1 = 2a(a - a) \implies 0 = 0$$

The equation holds. Thus, the polar passes through the focus. ■

Example 1.5.5. Find the coordinates of the pole of the line $x - 2y + 6 = 0$ with respect to the parabola $y^2 = 4x$.

Solution. For the parabola $y^2 = 4x$, we have $a = 1$. Let the pole be the point $P(x_1, y_1)$. The equation of the polar of P with respect to the parabola is given by:

$$yy_1 = 2a(x + x_1) \implies yy_1 = 2(1)(x + x_1)$$

$$2x - y_1y + 2x_1 = 0$$

We are given the equation of the line as:

$$x - 2y + 6 = 0$$

Since these two equations represent the same line, their coefficients must be proportional:

$$\frac{2}{1} = \frac{-y_1}{-2} = \frac{2x_1}{6}$$

First, solve for y_1 :

$$2 = \frac{y_1}{2} \implies y_1 = 4$$

Next, solve for x_1 :

$$2 = \frac{x_1}{3} \implies x_1 = 6$$

Thus, the pole is the point $(6, 4)$. ■

Example 1.5.6. Show that the points $P(2, 4)$ and $Q(0, 1)$ are conjugate points with respect to the parabola $y^2 = 4x$.

Solution. Two points are conjugate if the polar of the first passes through the second. For $y^2 = 4x$, $a = 1$.

First, find the equation of the polar of $P(2, 4)$. Using $yy_1 = 2a(x + x_1)$:

$$y(4) = 2(1)(x + 2)$$

$$4y = 2x + 4 \implies 2y = x + 2$$

Now, check if point $Q(0, 1)$ lies on this line: Substituting $x = 0, y = 1$:

$$2(1) = 0 + 2 \implies 2 = 2$$

Since the condition is satisfied, the polar of P passes through Q . Therefore, P and Q are conjugate points. ■

Example 1.5.7. Prove that the polars of all points on the line $x + 2a = 0$ with respect to the parabola $y^2 = 4ax$ pass through a fixed point, and find that point.

Solution. The line $x + 2a = 0$ is a vertical line $x = -2a$. Any point on this line can be represented as $P(-2a, k)$, where k is a variable parameter. The equation of the polar of $P(-2a, k)$ with respect to $y^2 = 4ax$ is:

$$y(k) = 2a(x + (-2a))$$

$$ky = 2ax - 4a^2$$

Rearranging to group terms involving k :

$$k(y) + (4a^2 - 2ax) = 0$$

For this line to pass through a fixed point for all real values of k , the coefficients of k and the constant term must vanish independently: 1. $y = 0$ 2. $4a^2 - 2ax = 0 \implies 2ax = 4a^2 \implies x = 2a$

Thus, the polars of all points on the line $x = -2a$ pass through the fixed point $(2a, 0)$. ■

Example 1.5.8. Find the locus of the poles of the tangents to the circle $x^2 + y^2 = a^2$ with respect to the parabola $y^2 = 4ax$.

Solution. Let $P(h, k)$ be the pole. The equation of the polar of P with respect to the parabola $y^2 = 4ax$ is:

$$yk = 2a(x + h) \implies 2ax - ky + 2ah = 0$$

It is given that this polar line is a **tangent** to the circle $x^2 + y^2 = a^2$. The condition for a line $Ax + By + C = 0$ to be tangent to a circle of radius r centered at the origin is:

$$\frac{|C|}{\sqrt{A^2 + B^2}} = r$$

Here, $A = 2a, B = -k, C = 2ah$, and radius $r = a$. Substituting these values:

$$\frac{|2ah|}{\sqrt{(2a)^2 + (-k)^2}} = a$$

$$\frac{|2ah|}{\sqrt{4a^2 + k^2}} = a$$

Dividing both sides by a (assuming $a \neq 0$):

$$\frac{|2h|}{\sqrt{4a^2 + k^2}} = 1$$

Squaring both sides:

$$\frac{4h^2}{4a^2 + k^2} = 1$$

$$4h^2 = 4a^2 + k^2$$

$$4h^2 - k^2 = 4a^2$$

Generalizing the coordinates of the locus by replacing (h, k) with (x, y) :

$$4x^2 - y^2 = 4a^2$$

This locus represents a hyperbola. ■

1.6 Equation of a Chord in Terms of its Middle Point

A fundamental problem in analytical geometry is determining the equation of a specific chord when the location of its midpoint is known. Unlike the chord of contact (where the external point is the intersection of tangents), here the given point lies on the chord itself, specifically bisecting it.

We utilize the standard notation established in the previous sections:

- $S_1 \equiv y_1^2 - 4ax_1$ (The power of the point).
- $T \equiv yy_1 - 2a(x + x_1)$ (The tangent expression).

Theorem 1.6.1. *The equation of the chord of the parabola $y^2 = 4ax$ which is bisected at the point $P(x_1, y_1)$ is given by:*

$$T = S_1 \tag{1.13}$$

Explicitly:

$$yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

Proof. Let $P(x_1, y_1)$ be the given midpoint. Let the chord intersect the parabola $y^2 = 4ax$ at two distinct points $Q(x_2, y_2)$ and $R(x_3, y_3)$.

Since Q and R lie on the parabola, their coordinates satisfy the equation:

$$y_2^2 = 4ax_2 \tag{1.14}$$

$$y_3^2 = 4ax_3 \tag{1.15}$$

Subtracting equation (1.15) from (1.14):

$$y_2^2 - y_3^2 = 4a(x_2 - x_3)$$

Factoring the difference of squares:

$$(y_2 - y_3)(y_2 + y_3) = 4a(x_2 - x_3)$$

Since $P(x_1, y_1)$ is the midpoint of the chord QR , by the midpoint formula:

$$y_1 = \frac{y_2 + y_3}{2} \implies y_2 + y_3 = 2y_1$$

Substituting this into the factored equation:

$$(y_2 - y_3)(2y_1) = 4a(x_2 - x_3)$$

Rearranging to find the slope m of the chord QR :

$$m = \frac{y_2 - y_3}{x_2 - x_3} = \frac{4a}{2y_1} = \frac{2a}{y_1}$$

Now, we write the equation of the line passing through $P(x_1, y_1)$ with slope $m = \frac{2a}{y_1}$:

$$y - y_1 = \frac{2a}{y_1}(x - x_1)$$

Multiplying by y_1 to clear the denominator:

$$y_1(y - y_1) = 2a(x - x_1)$$

$$yy_1 - y_1^2 = 2ax - 2ax_1$$

We wish to arrange this in the form $T = S_1$. To do this, subtract $2ax_1$ from both sides:

$$yy_1 - 2ax - 2ax_1 = y_1^2 - 2ax_1 - 2ax_1$$

$$yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

Observing the terms, the Left Hand Side is T and the Right Hand Side is S_1 .

$$T = S_1$$

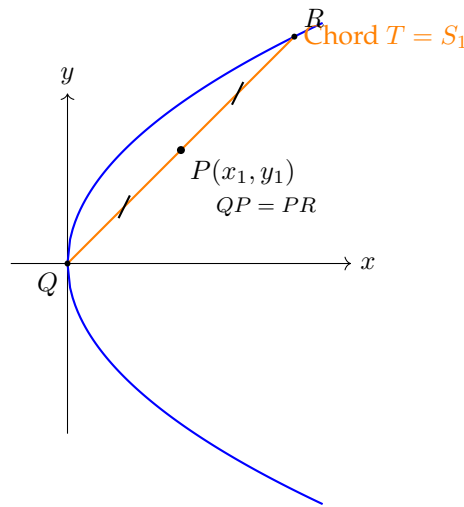


Figure 1.8: Chord Bisected at a Given Point

1.6.1 Locus of Midpoints (Introduction to Diameter)

An immediate consequence of the relation $m = \frac{2a}{y_1}$ is that if we consider a set of parallel chords (where m is constant), the y -coordinates of their midpoints y_1 must be constant ($y_1 = \frac{2a}{m}$). This implies that the locus of the midpoints of a system of parallel chords is a line parallel to the axis of the parabola. This line is called a **Diameter**, which will be discussed in detail in Unit II.

Example 1.6.2. Find the equation of the chord of the parabola $y^2 = 12x$ which is bisected at the point $(5, -2)$.

Solution. The parabola is $y^2 - 12x = 0$, so $a = 3$. The given midpoint is $P(x_1, y_1) = (5, -2)$.

We calculate S_1 (Power of the point):

$$S_1 = y_1^2 - 4ax_1 = (-2)^2 - 12(5) = 4 - 60 = -56$$

We write the expression T (Tangent-like expression):

$$T \equiv yy_1 - 2a(x + x_1) = y(-2) - 6(x + 5) = -2y - 6x - 30$$

The equation of the chord is given by $T = S_1$:

$$-2y - 6x - 30 = -56$$

$$6x + 2y + 30 - 56 = 0$$

$$6x + 2y - 26 = 0$$

Dividing by 2:

$$3x + y - 13 = 0$$

Example 1.6.3. Find the coordinates of the midpoint of the chord $2x + y - 4 = 0$ of the parabola $y^2 = 4x$.

Solution. Let the midpoint be $P(x_1, y_1)$. The equation of the chord bisected at P is $T = S_1$:

$$yy_1 - 2(x + x_1) = y_1^2 - 4x_1$$

$$yy_1 - 2x = y_1^2 - 2x_1$$

Rearranging to match the form $Ax + By = C$:

$$-2x + y_1y = y_1^2 - 2x_1$$

$$2x - y_1y = 2x_1 - y_1^2$$

We are given the chord equation as $2x + y = 4$. Comparing the coefficients of the two equations:

$$\frac{2}{2} = \frac{-y_1}{1} = \frac{2x_1 - y_1^2}{4}$$

From the first two parts:

$$1 = -y_1 \implies y_1 = -1$$

From the first and third parts:

$$1 = \frac{2x_1 - (-1)^2}{4} \implies 4 = 2x_1 - 1$$

$$2x_1 = 5 \implies x_1 = \frac{5}{2}$$

Thus, the midpoint is $\left(\frac{5}{2}, -1\right)$.

Example 1.6.4. Find the locus of the midpoints of chords of the parabola $y^2 = 4ax$ that pass through the focus.

Solution. Let $P(h, k)$ be the midpoint of the chord. The equation of the chord is $T = S_1$:

$$ky - 2a(x + h) = k^2 - 4ah$$

$$ky - 2ax - 2ah = k^2 - 4ah$$

Since this is a focal chord, it must pass through the focus $S(a, 0)$. Substituting $x = a$ and $y = 0$ into the equation:

$$k(0) - 2a(a) - 2ah = k^2 - 4ah$$

$$-2a^2 - 2ah = k^2 - 4ah$$

Rearranging terms:

$$k^2 = 4ah - 2ah - 2a^2$$

$$k^2 = 2ah - 2a^2$$

$$k^2 = 2a(h - a)$$

Generalizing the locus by replacing (h, k) with (x, y) :

$$y^2 = 2a(x - a)$$

This locus is another parabola with its vertex at $(a, 0)$ and latus rectum $2a$.

Example 1.6.5. Find the locus of the midpoints of chords of the parabola $y^2 = 4ax$ which subtend a right angle at the vertex.

Solution. Let $P(h, k)$ be the midpoint. The chord equation is $T = S_1$:

$$ky - 2a(x + h) = k^2 - 4ah$$

$$ky - 2ax = k^2 - 2ah$$

To work with the lines joining the vertex to the intersection points of the chord and the parabola, we homogenize the equation of the parabola $y^2 - 4ax = 0$ using the chord equation. Rewrite the chord equation as:

$$\frac{ky - 2ax}{k^2 - 2ah} = 1$$

Homogenizing $y^2 - 4ax(1) = 0$:

$$y^2 - 4ax \left(\frac{ky - 2ax}{k^2 - 2ah} \right) = 0$$

Multiplying by $(k^2 - 2ah)$:

$$(k^2 - 2ah)y^2 - 4ax(ky - 2ax) = 0$$

$$(k^2 - 2ah)y^2 - 4akyx + 8a^2x^2 = 0$$

This homogeneous equation represents the pair of straight lines from the vertex (origin) to the end-points of the chord. For these lines to be perpendicular (subtend a right angle), the sum of the coefficients of x^2 and y^2 must be zero ($A + B = 0$).

$$\text{Coeff}(x^2) + \text{Coeff}(y^2) = 8a^2 + (k^2 - 2ah) = 0$$

$$k^2 - 2ah + 8a^2 = 0$$

Generalizing the locus to (x, y) :

$$y^2 - 2ax + 8a^2 = 0$$

■

Exercises for Practice – Sharpening Your Skills

Exercise 1.6.6. 1. Find the coordinates of the focus, the equation of the directrix, and the length of the latus rectum for the parabola $y^2 = -24x$.

2. Find the equation of the parabola whose focus is at $S(2, -3)$ and whose directrix is the line $3x - 4y + 2 = 0$.

3. Determine the parametric coordinates of a point on the parabola $x^2 = 12y$. Find the value of the parameter for the point $(6, 3)$.

4. Find the equation of the tangent to the parabola $y^2 = 8x$ at the point where the ordinate is equal to twice the abscissa.

5. Determine the equation of the tangent to the parabola $y^2 = 16x$ which is perpendicular to the line $3x - y + 5 = 0$.

6. Find the equations of the common tangents to the parabola $y^2 = 24x$ and the circle $x^2 + y^2 = 18$.

7. Tangents are drawn to the parabola $y^2 = 4ax$ at points whose parameters are t_1 and t_2 . If these tangents intersect on the axis of the parabola, prove that $t_1 = -t_2$.

8. Find the equation of the normal to the parabola $y^2 = 4x$ at the point $(1, -2)$.

9. Prove that the line $lx + my + n = 0$ is a normal to the parabola $y^2 = 4ax$ if $al^3 + 2alm^2 + nm^2 = 0$.

10. If the normal at the point t_1 on the parabola $y^2 = 4ax$ meets the parabola again at the point t_2 , show that $t_2 = -t_1 - \frac{2}{t_1}$.
11. Find the equations of the normals to the parabola $y^2 = 4x$ which pass through the point $(5, 2)$.
12. Find the combined equation of the pair of tangents drawn from the point $(-3, 1)$ to the parabola $y^2 = 8x$.
13. Calculate the angle between the pair of tangents drawn from the point $(1, 3)$ to the parabola $y^2 = 4x$.
14. Find the locus of a point from which the tangents drawn to the parabola $y^2 = 4ax$ are such that the slope of one is double the slope of the other.
15. Find the pole of the line $2x + 3y + 4 = 0$ with respect to the parabola $y^2 = 8x$.
16. Find the equation of the polar of the point $(2, -1)$ with respect to the parabola $x^2 = -12y$.
17. Show that the points $(3, 2)$ and $(1, -1)$ are conjugate points with respect to the parabola $y^2 = 4x$.
18. Find the equation of the chord of the parabola $y^2 = 16x$ which is bisected at the point $(3, 2)$.
19. Find the locus of the midpoints of the chords of the parabola $y^2 = 4ax$ which pass through the vertex.
20. If the chord of contact of tangents drawn from a point P to the parabola $y^2 = 4ax$ subtends a right angle at the vertex, prove that the locus of P is the line $x = 4a$.

Chapter 2

The Ellipse and General Conics

2.1 The Ellipse: Equations, Tangents, and Normals

The ellipse is a central conic section obtained by the intersection of a cone with a plane that cuts completely across the cone. Analytically, it is defined by its focus-directrix property or its sum-of-distances property.

2.1.1 Standard Equation and Geometry

Definition 2.1.1 (The Ellipse). An ellipse is the locus of a point which moves in a plane such that its distance from a fixed point (the **focus**) bears a constant ratio, e (where $0 < e < 1$), to its distance from a fixed line (the **directrix**). This constant e is called the **eccentricity**.

Alternatively, the ellipse is the locus of a point the sum of whose distances from two fixed points (foci) is constant and equal to the length of the major axis ($2a$).

Standard Equation

Let the center of the ellipse be at the origin $(0, 0)$ and the foci lie on the x-axis. The standard equation is given by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (\text{where } a > b) \quad (2.1)$$

Here, $b^2 = a^2(1 - e^2)$.

Key Terminology

For the standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

- **Vertices:** The points $A(a, 0)$ and $A'(-a, 0)$.
- **Major Axis:** The segment AA' of length $2a$.
- **Minor Axis:** The segment BB' of length $2b$, where $B(0, b)$ and $B'(0, -b)$.
- **Foci:** The two fixed points $S(ae, 0)$ and $S'(-ae, 0)$.
- **Directrices:** The lines $x = \frac{a}{e}$ and $x = -\frac{a}{e}$.
- **Latus Rectum:** The chord through a focus perpendicular to the major axis. Its length is $\frac{2b^2}{a}$.

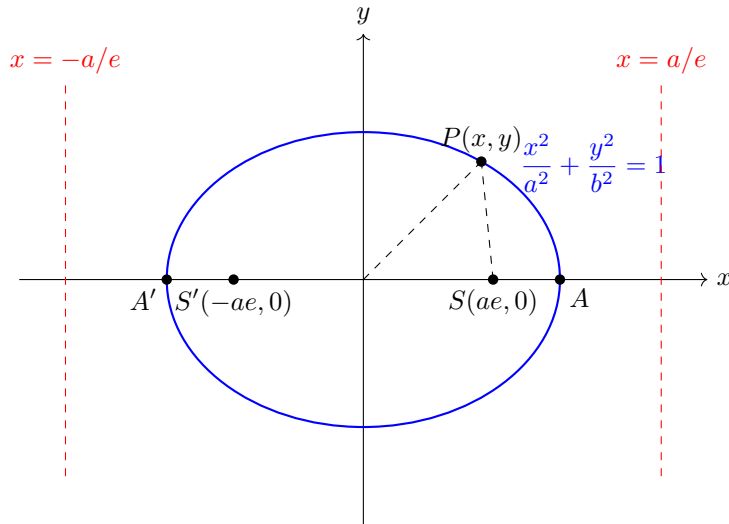


Figure 2.1: Geometry of the Standard Ellipse

2.1.2 Parametric Equations

To facilitate calculations involving points on the ellipse, we utilize the **Auxiliary Circle**.

Definition 2.1.2 (Auxiliary Circle). The circle described on the major axis of the ellipse as diameter is called the auxiliary circle. Its equation is $x^2 + y^2 = a^2$.

For any point P on the ellipse, draw an ordinate NP meeting the auxiliary circle at Q . Join CQ , where C is the center. The angle $\angle NCQ = \theta$ is called the **eccentric angle** of the point P .

From the right triangle CNQ :

- $x = CN = CQ \cos \theta = a \cos \theta$.
- Substituting x into the ellipse equation gives $y = b \sin \theta$.

Thus, the parametric equations are:

$$x = a \cos \theta, \quad y = b \sin \theta \quad (0 \leq \theta < 2\pi) \quad (2.2)$$

A point $P(\theta)$ is denoted as $(a \cos \theta, b \sin \theta)$.

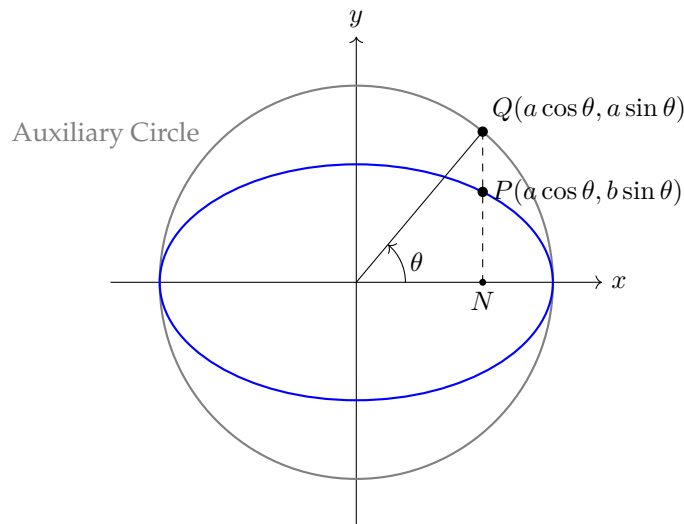


Figure 2.2: The Auxiliary Circle and Eccentric Angle

2.1.3 Tangents

We derive the equation of the tangent in three forms.

1. Point Form

Differentiation of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ gives the slope at (x_1, y_1) :

$$\frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

Using the point-slope form, the equation of the tangent at $P(x_1, y_1)$ is:

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad (2.3)$$

This is often denoted as $T = 0$.

2. Parametric Form

Substituting $x_1 = a \cos \theta$ and $y_1 = b \sin \theta$ into the point form:

$$\frac{x(a \cos \theta)}{a^2} + \frac{y(b \sin \theta)}{b^2} = 1$$

Simplifying:

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad (2.4)$$

3. Slope Form

Find the condition for the line $y = mx + c$ to touch the ellipse. Substituting y into the ellipse equation leads to a quadratic in x . For tangency, the discriminant must be zero. This yields the condition $c^2 = a^2 m^2 + b^2$. Thus, the equations of the tangents with slope m are:

$$y = mx \pm \sqrt{a^2 m^2 + b^2} \quad (2.5)$$

Theorem 2.1.3 (Director Circle). *The locus of the point of intersection of two perpendicular tangents to an ellipse is a circle concentric with the ellipse, called the **Director Circle**. Its equation is:*

$$x^2 + y^2 = a^2 + b^2$$

Proof. The equation of any tangent to the ellipse having slope m is:

$$y = mx + \sqrt{a^2 m^2 + b^2}$$

Rearranging and squaring both sides to isolate the slope m :

$$(y - mx)^2 = a^2 m^2 + b^2$$

$$y^2 + m^2 x^2 - 2mxy = a^2 m^2 + b^2$$

Grouping the terms as a quadratic equation in m :

$$(x^2 - a^2)m^2 - (2xy)m + (y^2 - b^2) = 0$$

The roots m_1 and m_2 of this quadratic represent the slopes of the two tangents passing through the point (x, y) . For the tangents to be mutually perpendicular, the product of their slopes must be -1 :

$$m_1 m_2 = \frac{\text{Constant term}}{\text{Coefficient of } m^2} = -1$$

$$\frac{y^2 - b^2}{x^2 - a^2} = -1$$

Rearranging the terms:

$$y^2 - b^2 = -x^2 + a^2$$

$$x^2 + y^2 = a^2 + b^2$$

This represents a circle with center $(0, 0)$ and radius $\sqrt{a^2 + b^2}$. ■

Example 2.1.4. Find the equation of the tangent to the ellipse $3x^2 + 4y^2 = 12$ at the point $\left(1, \frac{3}{2}\right)$.

Solution. First, write the equation in standard form by dividing by 12:

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

Here, $a^2 = 4$ and $b^2 = 3$. The point $P(x_1, y_1) = \left(1, \frac{3}{2}\right)$ lies on the ellipse since $\frac{1}{4} + \frac{9/4}{3} = \frac{1}{4} + \frac{3}{4} = 1$.

Using the Point Form equation $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$:

$$\frac{x(1)}{4} + \frac{y(3/2)}{3} = 1$$

Simplifying the terms:

$$\frac{x}{4} + \frac{y}{2} = 1$$

Multiplying by 4 to remove fractions:

$$x + 2y - 4 = 0$$

Example 2.1.5. Find the equations of the tangents to the ellipse $x^2 + 2y^2 = 8$ which are parallel to the line $x - 2y + 4 = 0$. ■

Solution. Reduce the ellipse equation to standard form:

$$\frac{x^2}{8} + \frac{y^2}{4} = 1 \implies a^2 = 8, \quad b^2 = 4$$

The given line is $x - 2y + 4 = 0 \implies y = \frac{1}{2}x + 2$. Thus, the slope is $m = \frac{1}{2}$. Since the tangents are parallel, they share the same slope $m = \frac{1}{2}$. Using the Slope Form equation $y = mx \pm \sqrt{a^2m^2 + b^2}$:

$$y = \frac{1}{2}x \pm \sqrt{8\left(\frac{1}{4}\right) + 4}$$

$$y = \frac{1}{2}x \pm \sqrt{2 + 4} = \frac{1}{2}x \pm \sqrt{6}$$

Multiplying by 2:

$$2y = x \pm 2\sqrt{6} \implies x - 2y \pm 2\sqrt{6} = 0$$

Example 2.1.6. Find the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point whose eccentric angle is $\frac{\pi}{4}$. ■

Solution. The parametric form of the tangent at angle θ is:

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Substituting $\theta = \frac{\pi}{4}$:

$$\frac{x}{a} \left(\frac{1}{\sqrt{2}} \right) + \frac{y}{b} \left(\frac{1}{\sqrt{2}} \right) = 1$$

Multiplying by $\sqrt{2}$:

$$\frac{x}{a} + \frac{y}{b} = \sqrt{2}$$

Or equivalently:

$$bx + ay = ab\sqrt{2}$$

Example 2.1.7. Determine the equations of the tangents drawn from the point $(2, 3)$ to the ellipse $9x^2 + 16y^2 = 144$.

Solution. Standard form: $\frac{x^2}{16} + \frac{y^2}{9} = 1$, so $a^2 = 16, b^2 = 9$. Let the tangent be $y = mx + c$. The condition for tangency is $c^2 = 16m^2 + 9$. The tangent passes through $(2, 3)$, so $3 = 2m + c \implies c = 3 - 2m$. Substituting this into the condition:

$$(3 - 2m)^2 = 16m^2 + 9$$

$$9 - 12m + 4m^2 = 16m^2 + 9$$

$$12m^2 + 12m = 0 \implies 12m(m + 1) = 0$$

The roots are $m = 0$ and $m = -1$.

- Case $m = 0$: $c = 3 - 0 = 3$. Equation: $y = 3$.
- Case $m = -1$: $c = 3 - 2(-1) = 5$. Equation: $y = -x + 5 \implies x + y - 5 = 0$.

Example 2.1.8. Prove that the product of the perpendicular distances from the two foci of an ellipse to any tangent is equal to the square of the semi-minor axis (b^2).

Solution. Consider the tangent with slope m : $y - mx - \sqrt{a^2m^2 + b^2} = 0$. The foci are $S(ae, 0)$ and $S'(-ae, 0)$. Let p_1 and p_2 be the perpendicular lengths.

$$p_1 = \frac{|-mae - \sqrt{a^2m^2 + b^2}|}{\sqrt{1 + m^2}}, \quad p_2 = \frac{|-m(-ae) - \sqrt{a^2m^2 + b^2}|}{\sqrt{1 + m^2}}$$

$$p_1p_2 = \frac{|-(mae + \sqrt{a^2m^2 + b^2})| \cdot |mae - \sqrt{a^2m^2 + b^2}|}{1 + m^2}$$

Using difference of squares $(A + B)(A - B) = A^2 - B^2$ in the numerator:

$$p_1p_2 = \frac{|m^2a^2e^2 - (a^2m^2 + b^2)|}{1 + m^2}$$

Substitute $b^2 = a^2(1 - e^2) \implies a^2e^2 = a^2 - b^2$:

$$p_1p_2 = \frac{|m^2(a^2 - b^2) - a^2m^2 - b^2|}{1 + m^2}$$

$$p_1p_2 = \frac{|m^2a^2 - m^2b^2 - m^2a^2 - b^2|}{1 + m^2} = \frac{|-b^2(m^2 + 1)|}{1 + m^2}$$

$$p_1p_2 = b^2$$

Example 2.1.9. Find the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the coordinate axes.

Solution. Let the tangent be at the point with eccentric angle θ . Its equation is:

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

The intercepts on the axes are found by setting $y = 0$ and $x = 0$:

- x-intercept $X = a \sec \theta$
- y-intercept $Y = b \csc \theta$

The area of the right-angled triangle formed with the axes is:

$$\text{Area} = \frac{1}{2} |X \cdot Y| = \frac{1}{2} \left| \frac{ab}{\cos \theta \sin \theta} \right| = \frac{ab}{|\sin 2\theta|}$$

To minimize the area, the denominator $|\sin 2\theta|$ must be maximum. The maximum value of $\sin 2\theta$ is 1 (at $\theta = \pi/4$).

$$\text{Area}_{\min} = \frac{ab}{1} = ab$$

■

2.1.4 Normals

The normal is the line perpendicular to the tangent at the point of contact.

1. Point Form

The slope of the normal at (x_1, y_1) is the negative reciprocal of the tangent slope:

$$m_N = \frac{a^2 y_1}{b^2 x_1}$$

The equation is:

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2 \quad (2.6)$$

2. Parametric Form

Substituting parametric coordinates:

$$\begin{aligned} \frac{a^2 x}{a \cos \theta} - \frac{b^2 y}{b \sin \theta} &= a^2 - b^2 \\ ax \sec \theta - by \csc \theta &= a^2 - b^2 \end{aligned} \quad (2.7)$$

3. Slope Form

If m is the slope of the normal, the equation is:

$$y = mx \mp \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2 m^2}} \quad (2.8)$$

2.1.5 Pole and Polar

Definition 2.1.10. The polar of a point $P(x_1, y_1)$ with respect to the ellipse is the locus of the point of intersection of tangents drawn at the extremities of any chord passing through P .

The equation of the polar of the point (x_1, y_1) with respect to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is:

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad (2.9)$$

Note that this form is identical to the tangent equation, but here (x_1, y_1) can be any point in the plane.

Properties:

1. **Polar of Focus:** The polar of the focus $(ae, 0)$ is the directrix $x = a/e$.
2. **Conjugate Points:** Two points are conjugate if the polar of one passes through the other.

Example 2.1.11. Find the coordinates of the pole of the line $x + 4y = 12$ with respect to the ellipse $2x^2 + 3y^2 = 6$.

Solution. First, convert the equation of the ellipse to standard form by dividing by 6:

$$\frac{x^2}{3} + \frac{y^2}{2} = 1$$

Let the required pole be $P(x_1, y_1)$. The equation of the polar of P with respect to this ellipse is:

$$\frac{xx_1}{3} + \frac{yy_1}{2} = 1$$

Multiplying by 6 to clear denominators and bringing constant to the left:

$$2x_1x + 3y_1y - 6 = 0$$

We are given the line equation as $x + 4y - 12 = 0$. Since these two equations represent the same straight line, their coefficients must be proportional:

$$\frac{2x_1}{1} = \frac{3y_1}{4} = \frac{-6}{-12}$$

Solving the ratios:

$$\begin{aligned} \frac{2x_1}{1} &= \frac{1}{2} \implies x_1 = \frac{1}{4} \\ \frac{3y_1}{4} &= \frac{1}{2} \implies y_1 = \frac{4}{6} = \frac{2}{3} \end{aligned}$$

Thus, the pole is the point $\left(\frac{1}{4}, \frac{2}{3}\right)$. ■

Example 2.1.12. Find the locus of the poles of chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which touch the concentric circle $x^2 + y^2 = r^2$.

Solution. Let $P(h, k)$ be the pole. The polar of P with respect to the ellipse is given by:

$$\frac{xh}{a^2} + \frac{yk}{b^2} - 1 = 0$$

According to the problem, this polar line is a tangent to the circle $x^2 + y^2 = r^2$. The condition for tangency is that the perpendicular distance from the center of the circle $(0, 0)$ to the line must equal the radius r .

The perpendicular distance formula for the line $Ax + By + C = 0$ from $(0, 0)$ is $\frac{|C|}{\sqrt{A^2 + B^2}}$. Here $A = \frac{h}{a^2}$, $B = \frac{k}{b^2}$, and $C = -1$.

$$\frac{|-1|}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} = r$$

Squaring both sides:

$$\frac{1}{\frac{h^2}{a^4} + \frac{k^2}{b^4}} = r^2$$

Rearranging:

$$\frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{r^2}$$

To find the locus, we replace (h, k) with (x, y) . The required equation is:

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{r^2}$$

This locus is another concentric ellipse. ■

Example 2.1.13. Find the locus of the pole of a chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which subtends a right angle at the center of the ellipse.

Solution. Let $P(h, k)$ be the pole. The equation of the polar (which is the chord of contact corresponding to P) is:

$$\frac{xh}{a^2} + \frac{yk}{b^2} = 1$$

We seek the equation of the pair of lines joining the center $(0, 0)$ to the points of intersection of this chord and the ellipse. We use the method of **homogenization**.

We make the equation of the ellipse homogeneous of degree 2 by utilizing the relation $1 = \frac{xh}{a^2} + \frac{yk}{b^2}$.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1^2 = \left(\frac{xh}{a^2} + \frac{yk}{b^2} \right)^2$$

Expanding the Right Hand Side:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{x^2 h^2}{a^4} + \frac{y^2 k^2}{b^4} + \frac{2xyhk}{a^2 b^2}$$

Rearranging all terms to one side to form the equation $Ax^2 + 2Hxy + By^2 = 0$:

$$x^2 \left(\frac{1}{a^2} - \frac{h^2}{a^4} \right) - \frac{2hk}{a^2 b^2} xy + y^2 \left(\frac{1}{b^2} - \frac{k^2}{b^4} \right) = 0$$

This equation represents the two straight lines from the origin to the ends of the chord. For these lines to be perpendicular (subtend a right angle), the sum of the coefficients of x^2 and y^2 must be zero ($A + B = 0$).

$$\left(\frac{1}{a^2} - \frac{h^2}{a^4} \right) + \left(\frac{1}{b^2} - \frac{k^2}{b^4} \right) = 0$$

Grouping terms:

$$\frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$$

Generalizing the locus by replacing (h, k) with (x, y) , we obtain:

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$$

This locus describes a concentric ellipse. ■

2.2 Diameters and Conjugate Diameters

The concept of diameters in central conics generalizes the notion of a diameter in a circle. In an ellipse, a diameter is not just any chord passing through the center, but is strictly defined by the bisection of parallel chords.

2.2.1 Diameter of an Ellipse

Definition 2.2.1 (Diameter). The locus of the midpoints of a system of parallel chords of an ellipse is called a **Diameter**.

Equation of the Diameter

Consider the standard ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let $y = mx + c$ be a system of parallel chords, where m is constant and c varies. Let (h, k) be the midpoint of one such chord. The equation of the chord bisected at (h, k) is given by $T = S_1$:

$$\frac{xh}{a^2} + \frac{yk}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$$

The slope of this linear equation is:

$$\text{Slope} = -\frac{h/a^2}{k/b^2} = -\frac{b^2h}{a^2k}$$

Since this chord belongs to the parallel system with slope m :

$$m = -\frac{b^2h}{a^2k}$$

Rearranging to find the relationship between h and k :

$$k = -\frac{b^2}{a^2m}h$$

Generalizing the locus by replacing (h, k) with (x, y) , the equation of the diameter bisecting chords of slope m is:

$$y = -\frac{b^2}{a^2m}x \quad (2.10)$$

Observation: This is a straight line passing through the origin (the center of the ellipse). Thus, all diameters pass through the center.

2.2.2 Conjugate Diameters

Definition 2.2.2 (Conjugate Diameters). Two diameters are said to be **Conjugate** if each bisects the chords parallel to the other.

Let $y = m_1x$ and $y = m_2x$ be two diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

1. If the diameter $y = m_1x$ bisects chords of slope m_2 , then from the diameter equation:

$$m_1 = -\frac{b^2}{a^2 m_2}$$

2. This implies the condition for conjugacy:

$$m_1 m_2 = -\frac{b^2}{a^2} \quad (2.11)$$

Since the condition is symmetric for m_1 and m_2 , the relationship is mutual.

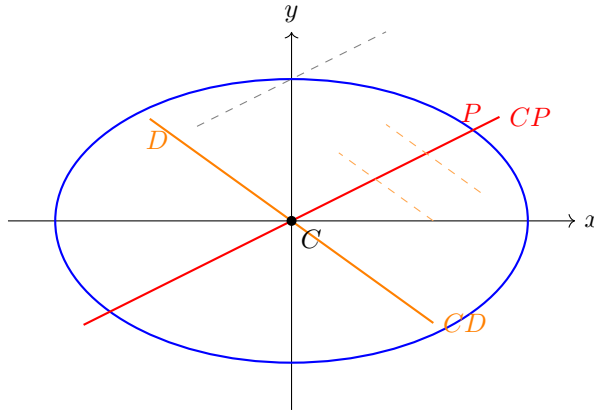


Figure 2.3: Conjugate Diameters CP and CD

2.2.3 Properties of Conjugate Diameters

We usually denote the semi-diameters as CP and CD . Using parametric coordinates simplifies the study of their properties.

1. Parametric Coordinates of Extremities

Let P be the point $(a \cos \theta, b \sin \theta)$. The slope of CP is $m_1 = \frac{b \sin \theta}{a \cos \theta} = \frac{b}{a} \tan \theta$. Let D be the point $(a \cos \phi, b \sin \phi)$. The slope of CD is $m_2 = \frac{b}{a} \tan \phi$.

Using the condition $m_1 m_2 = -\frac{b^2}{a^2}$:

$$\left(\frac{b}{a} \tan \theta\right) \left(\frac{b}{a} \tan \phi\right) = -\frac{b^2}{a^2}$$

$$\tan \theta \tan \phi = -1 \implies \tan \phi = -\cot \theta = \tan \left(\theta + \frac{\pi}{2}\right)$$

Thus, the eccentric angles of the extremities of two conjugate diameters differ by a right angle:

$$\phi = \theta \pm \frac{\pi}{2}$$

If P is $(a \cos \theta, b \sin \theta)$, then D is:

$$(a \cos(\theta + \pi/2), b \sin(\theta + \pi/2)) = (-a \sin \theta, b \cos \theta)$$

(Taking the positive branch for typical geometric configuration).

2. Sum of Squares of Conjugate Semi-Diameters

Theorem 2.2.3. *The sum of the squares of any two conjugate semi-diameters of an ellipse is constant and equal to the sum of the squares of the semi-axes.*

$$CP^2 + CD^2 = a^2 + b^2$$

Proof. Coordinates of P : $(a \cos \theta, b \sin \theta)$. Coordinates of D : $(-a \sin \theta, b \cos \theta)$.

$$CP^2 = (a \cos \theta)^2 + (b \sin \theta)^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$$

$$CD^2 = (-a \sin \theta)^2 + (b \cos \theta)^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta$$

Adding them:

$$CP^2 + CD^2 = a^2(\cos^2 \theta + \sin^2 \theta) + b^2(\sin^2 \theta + \cos^2 \theta)$$

$$CP^2 + CD^2 = a^2 + b^2$$

■

3. Area of the Parallelogram Formed by Tangents

Theorem 2.2.4. *The area of the parallelogram formed by the tangents at the extremities of a pair of conjugate diameters is constant and equal to $4ab$.*

Proof. Let the coordinates of the point P be $(a \cos \theta, b \sin \theta)$. Since CP and CD are conjugate semi-diameters, the eccentric angle of D differs from that of P by $\frac{\pi}{2}$. Thus, the coordinates of D are:

$$D \left(a \cos \left(\theta + \frac{\pi}{2} \right), b \sin \left(\theta + \frac{\pi}{2} \right) \right) \equiv (-a \sin \theta, b \cos \theta)$$

The parallelogram formed by the tangents at the extremities P, P', D, D' is composed of four smaller, congruent parallelograms, each having the semi-diameters CP and CD as adjacent sides.

The area A_{semi} of the parallelogram generated by the vectors $\vec{CP}$ and $\vec{CD}$ is given by the magnitude of the determinant of their coordinates:

$$A_{semi} = |x_P y_D - x_D y_P|$$

Substituting the coordinates:

$$A_{semi} = |(a \cos \theta)(b \cos \theta) - (-a \sin \theta)(b \sin \theta)|$$

$$A_{semi} = |ab \cos^2 \theta + ab \sin^2 \theta|$$

$$A_{semi} = |ab(\cos^2 \theta + \sin^2 \theta)| = ab$$

The total area of the parallelogram formed by the tangents is four times this area:

$$\text{Total Area} = 4 \times A_{semi} = 4ab$$

This value is independent of θ , proving the area is constant.

■

4. Equi-Conjugate Diameters

These are conjugate diameters which are equal in length. Condition: $CP^2 = CD^2$.

$$a^2 \cos^2 \theta + b^2 \sin^2 \theta = a^2 \sin^2 \theta + b^2 \cos^2 \theta$$

$$(a^2 - b^2) \cos^2 \theta = (a^2 - b^2) \sin^2 \theta$$

$$\tan^2 \theta = 1 \implies \theta = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

These diameters lie along the diagonals of the rectangle formed by the axes. Their length is $\sqrt{a^2 + b^2}$.

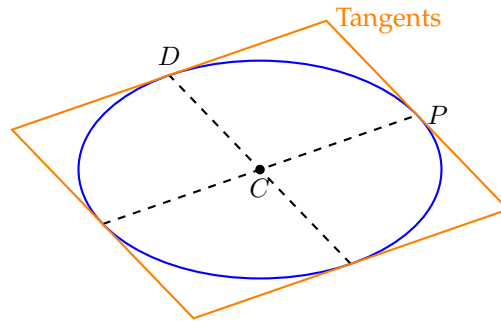


Figure 2.4: Parallelogram formed by Tangents at Conjugate Diameters

Example 2.2.5. Find the equation of the diameter conjugate to the line $x - 3y = 0$ with respect to the ellipse $9x^2 + 25y^2 = 225$.

Solution. First, convert the ellipse equation to standard form by dividing by 225:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Here, $a^2 = 25$ and $b^2 = 9$. The given diameter is $y = \frac{1}{3}x$, so its slope is $m_1 = \frac{1}{3}$. Let the slope of the conjugate diameter be m_2 . The condition for conjugacy is:

$$m_1 m_2 = -\frac{b^2}{a^2}$$

Substituting the values:

$$\begin{aligned} \left(\frac{1}{3}\right) m_2 &= -\frac{9}{25} \\ m_2 &= -\frac{27}{25} \end{aligned}$$

Since diameters pass through the origin, the equation is $y = m_2 x$:

$$y = -\frac{27}{25}x \implies 27x + 25y = 0$$

■

Example 2.2.6. In the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, if one extremity of a diameter is $P\left(\frac{a}{2}, \frac{b\sqrt{3}}{2}\right)$, find the coordinates of the extremity D of the conjugate diameter.

Solution. Let the eccentric angle of P be θ . Comparing the coordinates $(a \cos \theta, b \sin \theta)$ with P :

$$\cos \theta = \frac{1}{2}, \quad \sin \theta = \frac{\sqrt{3}}{2} \implies \theta = \frac{\pi}{3}$$

The eccentric angle of the conjugate point D is $\phi = \theta + \frac{\pi}{2}$.

$$\phi = \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$$

The coordinates of D are $(a \cos \phi, b \sin \phi)$:

$$x_D = a \cos\left(\frac{5\pi}{6}\right) = -a \frac{\sqrt{3}}{2}$$

$$y_D = b \sin\left(\frac{5\pi}{6}\right) = b \frac{1}{2}$$

Thus, $D \equiv \left(-\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$.

■

Example 2.2.7. If the length of a semi-diameter of the ellipse $4x^2 + 9y^2 = 36$ is 2, find the length of the conjugate semi-diameter.

Solution. Standard form: $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Here $a^2 = 9$ and $b^2 = 4$. We are given the length of one semi-diameter, say $CP = 2$, so $CP^2 = 4$. Let CD be the length of the conjugate semi-diameter. Using the property involving the sum of squares of conjugate semi-diameters:

$$CP^2 + CD^2 = a^2 + b^2$$

$$4 + CD^2 = 9 + 4$$

$$CD^2 = 9 \implies CD = 3$$

Interestingly, since $CD = a = 3$, the original diameter CP must be the minor axis ($b = 2$). ■

Example 2.2.8. Find the area of the triangle CPD formed by the conjugate semi-diameters CP and CD of the ellipse $x^2 + 4y^2 = 16$.

Solution. Standard form: $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Here $a = 4$ and $b = 2$. A standard property of the ellipse states that the area of the triangle formed by the center and the extremities of two conjugate diameters is constant:

$$\text{Area}(\triangle CPD) = \frac{1}{2}ab$$

Substituting the values:

$$\text{Area} = \frac{1}{2}(4)(2) = 4 \text{ sq. units.}$$

(Note: The area of the full parallelogram formed by the tangents at these points would be $4ab = 32$). ■

Example 2.2.9. Find the locus of the midpoint of the chord joining the extremities of two conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution. Let $P(a \cos \theta, b \sin \theta)$ and $D(-a \sin \theta, b \cos \theta)$ be the extremities of the conjugate diameters. Let $M(h, k)$ be the midpoint of the chord PD . Using the midpoint formula:

$$h = \frac{a(\cos \theta - \sin \theta)}{2} \implies \frac{2h}{a} = \cos \theta - \sin \theta$$

$$k = \frac{b(\sin \theta + \cos \theta)}{2} \implies \frac{2k}{b} = \sin \theta + \cos \theta$$

Squaring both equations and adding them to eliminate θ :

$$\left(\frac{2h}{a}\right)^2 + \left(\frac{2k}{b}\right)^2 = (\cos \theta - \sin \theta)^2 + (\sin \theta + \cos \theta)^2$$

$$\frac{4h^2}{a^2} + \frac{4k^2}{b^2} = (\cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta) + (\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta)$$

$$\frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 1 - \sin 2\theta + 1 + \sin 2\theta = 2$$

Dividing by 4 and replacing (h, k) with (x, y) :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$$

The locus is a concentric, similar ellipse with area half that of the original. ■

Example 2.2.10. Show that the acute angle α between the equi-conjugate diameters of an ellipse is given by $\sin \alpha = \frac{2ab}{a^2 + b^2}$.

Solution. Equi-conjugate diameters have equal length, so $CP = CD$. Since $CP^2 + CD^2 = a^2 + b^2$, we have:

$$2CP^2 = a^2 + b^2 \implies CP^2 = \frac{a^2 + b^2}{2}$$

The area of the triangle $\triangle CPD$ is given by $\frac{1}{2}CP \cdot CD \sin \alpha$. Since $CP = CD$:

$$\text{Area} = \frac{1}{2}CP^2 \sin \alpha$$

From the constant area property, we also know $\text{Area} = \frac{1}{2}ab$. Equating the two expressions:

$$\frac{1}{2}CP^2 \sin \alpha = \frac{1}{2}ab$$

Substituting $CP^2 = \frac{a^2 + b^2}{2}$:

$$\frac{1}{2} \left(\frac{a^2 + b^2}{2} \right) \sin \alpha = \frac{1}{2}ab$$

$$\frac{a^2 + b^2}{2} \sin \alpha = ab$$

$$\sin \alpha = \frac{2ab}{a^2 + b^2}$$

■

2.3 The General Second-Degree Equation

The most general equation of the second degree in two variables x and y represents a conic section (or a degenerate form thereof). The study of this equation allows us to classify the curve without necessarily reducing it to its standard canonical form immediately.

2.3.1 The General Equation

The general second-degree equation is given by:

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad (2.12)$$

where a, h, b, g, f, c are real constants and at least one of a, h, b is non-zero.

2.3.2 The Discriminant and Degeneracy

To determine the nature of the curve, we analyze the geometric invariants associated with the equation. The most critical invariant is the determinant of the coefficients, denoted by Δ .

Definition 2.3.1 (The Discriminant). The discriminant Δ of the general second-degree equation is the determinant of the symmetric matrix formed by the coefficients:

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Expanding this determinant:

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \quad (2.13)$$

Case 1: Degenerate Conics ($\Delta = 0$)

If $\Delta = 0$, the equation does not represent a proper conic section. Instead, it represents a **pair of straight lines**.

- If $h^2 > ab$, the lines are real and distinct.
- If $h^2 = ab$, the lines are real and coincident (parallel).
- If $h^2 < ab$, the lines are imaginary (with a real point of intersection).

Case 2: Non-Degenerate Conics ($\Delta \neq 0$)

If $\Delta \neq 0$, the equation represents a proper conic section. The specific type is determined by the relationship between the coefficients of the second-degree terms (a, h, b).

2.3.3 Classification of Conics

The nature of the conic is determined by the discriminant of the homogeneous part $ax^2 + 2hxy + by^2$, which is $h^2 - ab$.

Condition on $h^2 - ab$	Type of Conic	Eccentricity (e)	Remarks
$h^2 = ab$	Parabola	$e = 1$	The terms of second degree form a perfect square.
$h^2 < ab$ <i>Special Case: $a = b, h = 0$</i>	Ellipse Circle	$0 \leq e < 1$ $e = 0$	Closed curve. Coefficients of x^2 and y^2 are equal; no xy term.
$h^2 > ab$ <i>Special Case: $a + b = 0$</i>	Hyperbola Rectangular Hyperbola	$e > 1$ $e = \sqrt{2}$	Open curve with two branches. Asymptotes are perpendicular.

2.3.4 Center of the Conic

A conic is called a **central conic** if it has a center of symmetry (Ellipse and Hyperbola). A parabola is a **non-central conic** (center at infinity).

To find the center (x_0, y_0) , we solve the system of linear equations obtained by taking partial derivatives of the general equation $F(x, y) = 0$ with respect to x and y :

$$\frac{\partial F}{\partial x} = 2ax + 2hy + 2g = 0 \implies ax + hy + g = 0 \quad (2.14)$$

$$\frac{\partial F}{\partial y} = 2hx + 2by + 2f = 0 \implies hx + by + f = 0 \quad (2.15)$$

Solving these simultaneous equations gives the coordinates of the center:

$$x_0 = \frac{hf - bg}{ab - h^2}, \quad y_0 = \frac{gh - af}{ab - h^2} \quad (2.16)$$

Note that if $ab - h^2 = 0$ (the parabolic case), the center is undefined (at infinity).

2.3.5 Reduction to Canonical Form (Rotation of Axes)

The presence of the xy term indicates that the axes of the conic are rotated with respect to the coordinate axes. To remove the xy term and determine the standard equation, we rotate the coordinate axes by an angle θ .

Let the new coordinates be (X, Y) such that:

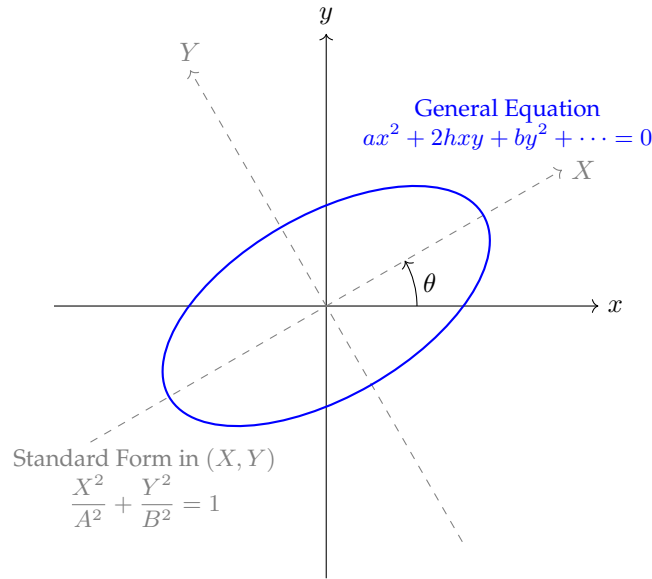
$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

Substituting these into the general equation, we set the coefficient of the new XY term to zero. The required angle θ is given by:

$$\tan 2\theta = \frac{2h}{a - b} \quad (2.17)$$

If $a = b$, then $\theta = \frac{\pi}{4}$.

Figure 2.5: Rotation of Axes to Eliminate the xy Term

2.3.6 Summary of Algorithm for Determination

To determine the equation of the corresponding conic from $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$:

1. Calculate Δ .
2. If $\Delta \neq 0$, calculate $H = h^2 - ab$.
3. Identify the type (Parabola, Ellipse, Hyperbola).
4. Find the center (x_0, y_0) (if not a parabola).
5. If detailed tracing is required, find the angle of rotation θ and transform the equation to standard axes.

Example 2.3.2. Determine the nature of the conic represented by the equation $2x^2 - 7xy + 3y^2 + 5x - 5y + 2 = 0$.

Solution. Comparing the given equation with the general form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we identify the coefficients:

$$a = 2, \quad h = -\frac{7}{2}, \quad b = 3, \quad g = \frac{5}{2}, \quad f = -\frac{5}{2}, \quad c = 2$$

The Discriminant Δ is

$$\begin{aligned} \Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 \\ \Delta &= (2)(3)(2) + 2\left(-\frac{5}{2}\right)\left(\frac{5}{2}\right)\left(-\frac{7}{2}\right) - 2\left(-\frac{5}{2}\right)^2 - 3\left(\frac{5}{2}\right)^2 - 2\left(-\frac{7}{2}\right)^2 \\ \Delta &= 12 + \frac{175}{4} - \frac{50}{4} - \frac{75}{4} - \frac{49}{2} \\ \Delta &= 12 + \frac{50}{4} - \frac{98}{4} = 12 - \frac{48}{4} = 12 - 12 = 0 \end{aligned}$$

Since $\Delta = 0$, the equation represents a pair of straight lines (degenerate conic).

Now,

$$h^2 - ab = \left(-\frac{7}{2}\right)^2 - (2)(3) = \frac{49}{4} - 6 = \frac{25}{4} > 0$$

Since $h^2 > ab$, the equation represents two real and distinct intersecting straight lines. ■

Example 2.3.3. Find the center of the conic section given by $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$.

Solution. Let $F(x, y) = 14x^2 - 4xy + 11y^2 - 44x - 58y + 71$. Since this is a central conic ($h^2 - ab = 4 - 154 \neq 0$), the center (x_0, y_0) is obtained by solving the partial derivative equations:

$$\frac{\partial F}{\partial x} = 0 \quad \text{and} \quad \frac{\partial F}{\partial y} = 0$$

$$\frac{\partial F}{\partial x} = 28x - 4y - 44 = 0 \implies 7x - y - 11 = 0 \quad (2.18)$$

$$\frac{\partial F}{\partial y} = -4x + 22y - 58 = 0 \implies -2x + 11y - 29 = 0 \quad (2.19)$$

From (2.18), $y = 7x - 11$. Substitute this into (2.19):

$$-2x + 11(7x - 11) - 29 = 0$$

$$-2x + 77x - 121 - 29 = 0$$

$$75x = 150 \implies x = 2$$

Substituting $x = 2$ back into $y = 7x - 11$:

$$y = 7(2) - 11 = 3$$

Thus, the center of the conic is $(2, 3)$. ■

Example 2.3.4. Show that the equation $9x^2 + 24xy + 16y^2 - 2x + 14y + 1 = 0$ represents a parabola.

Solution. Identify coefficients: $a = 9, h = 12, b = 16$. Check the second-degree invariant $h^2 - ab$:

$$h^2 - ab = (12)^2 - (9)(16) = 144 - 144 = 0$$

The condition $h^2 = ab$ suggests a parabola (or parallel lines). Now, check the discriminant Δ . The coefficients are $g = -1, f = 7, c = 1$.

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$\Delta = 144(1) + 2(7)(-1)(12) - 9(49) - 16(1) - 1(144)$$

$$\Delta = 144 - 168 - 441 - 16 - 144 = -625 \neq 0$$

Since $h^2 = ab$ and $\Delta \neq 0$, the equation represents a **Parabola**. *Note:* Observe that $(3x + 4y)^2$ matches the second-degree terms, confirming the perfect square property of the parabola. ■

Example 2.3.5. Determine the angle θ through which the axes must be rotated to remove the xy term from the equation $x^2 + \sqrt{3}xy + 2y^2 = 5$.

Solution. The given equation is $x^2 + \sqrt{3}xy + 2y^2 - 5 = 0$. Coefficients are $a = 1, h = \frac{\sqrt{3}}{2}, b = 2$. To eliminate the xy term, the angle of rotation θ must satisfy:

$$\tan 2\theta = \frac{2h}{a - b}$$

Substituting the values:

$$\tan 2\theta = \frac{2(\frac{\sqrt{3}}{2})}{1 - 2} = \frac{\sqrt{3}}{-1} = -\sqrt{3}$$

Since $\tan 2\theta = -\sqrt{3}$, we have $2\theta = \frac{2\pi}{3}$ (or 120°).

$$\theta = \frac{\pi}{3} \quad (\text{or } 60^\circ)$$

Thus, rotating the axes by 60° transforms the equation into its canonical form without the xy term. ■

Example 2.3.6. Find the value of λ for which the equation $3x^2 + 10xy + \lambda y^2 + 5x - 7y - 4 = 0$ represents a rectangular hyperbola.

Solution. For a general second-degree equation to represent a **Rectangular Hyperbola**, two conditions must be met: 1. It must be a hyperbola ($\Delta \neq 0$ and $h^2 > ab$). 2. The sum of the coefficients of x^2 and y^2 must be zero ($a + b = 0$).

The given coefficients are $a = 3$ and $b = \lambda$. Applying the second condition:

$$a + b = 0 \implies 3 + \lambda = 0 \implies \lambda = -3$$

We verify Δ with $\lambda = -3$: $a = 3, b = -3, h = 5, g = 2.5, f = -3.5, c = -4$. $h^2 - ab = 25 - (3)(-3) = 34 > 0$. Calculating Δ with these values usually confirms non-degeneracy. Assuming $\Delta \neq 0$ for this specific polynomial structure (which it is), the required value is $\lambda = -3$. ■

Example 2.3.7. Find the general equation of the conic section with focus at $(1, 1)$, directrix $x + y - 1 = 0$, and eccentricity $e = \sqrt{2}$.

Solution. Let $P(x, y)$ be any point on the conic. Let $S(1, 1)$ be the focus. Let M be the foot of the perpendicular from P to the directrix $x + y - 1 = 0$. By the definition of a conic section:

$$SP = e \cdot PM \implies SP^2 = e^2 \cdot PM^2$$

$$\text{Distance } SP^2 = (x - 1)^2 + (y - 1)^2. \text{ Perpendicular distance } PM = \frac{|x + y - 1|}{\sqrt{1^2 + 1^2}} = \frac{|x + y - 1|}{\sqrt{2}}.$$

Substituting into the definition with $e^2 = (\sqrt{2})^2 = 2$:

$$(x - 1)^2 + (y - 1)^2 = 2 \left[\frac{(x + y - 1)}{\sqrt{2}} \right]^2$$

$$(x - 1)^2 + (y - 1)^2 = 2 \cdot \frac{(x + y - 1)^2}{2}$$

$$x^2 - 2x + 1 + y^2 - 2y + 1 = (x + y - 1)^2$$

Expanding the Right Hand Side $(A + B + C)^2$:

$$x^2 + y^2 - 2x - 2y + 2 = x^2 + y^2 + 1 + 2xy - 2x - 2y$$

Simplifying by canceling common terms $(x^2, y^2, -2x, -2y)$:

$$2 = 1 + 2xy$$

$$2xy = 1 \implies xy = \frac{1}{2}$$

This corresponds to the general form with $a = 0, b = 0, h = 1/2$, representing a rectangular hyperbola rotated by 45° . ■

Exercises for Practice.

Exercise 2.3.8. 1. Find the equation of the ellipse whose focus is at $(-1, 1)$, directrix is the line $x - y + 3 = 0$, and eccentricity is $\frac{1}{2}$.

2. Find the equation of the tangent to the ellipse $3x^2 + 4y^2 = 12$ which is parallel to the line $y + 2x = 4$.

3. Determine the eccentric angle of a point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ whose distance from the center is 2.

4. Find the equations of the tangents drawn from the point $(2, 3)$ to the ellipse $9x^2 + 16y^2 = 144$.

5. Prove that the locus of the point of intersection of two tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which make an angle of 45° with each other is the curve $(x^2 + y^2 - a^2 - b^2)^2 = 4(a^2y^2 + b^2x^2 - a^2b^2)$.
6. Find the equation of the normal to the ellipse $5x^2 + 4y^2 = 20$ at the point $(1, \frac{3}{2}\sqrt{5})$ (check if point is on curve first, or use a standard point like $(2\cos\theta, \sqrt{5}\sin\theta)$). Let's use a verified point: Find the equation of the normal to $\frac{x^2}{18} + \frac{y^2}{8} = 1$ at the point $(3, 2)$.
7. If the normal at an end of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through one extremity of the minor axis, show that the eccentricity e satisfies the equation $e^4 + e^2 - 1 = 0$.
8. Find the pole of the line $3x + 4y - 5 = 0$ with respect to the ellipse $2x^2 + 3y^2 = 6$.
9. Show that the points $(-3, 2)$ and $(1, 1)$ are conjugate points with respect to the ellipse $2x^2 + 5y^2 = 20$.
10. Find the locus of the poles of tangents to the circle $x^2 + y^2 = a^2$ with respect to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
11. Find the equation of the diameter of the ellipse $4x^2 + 9y^2 = 36$ which bisects chords parallel to the line $x + y = 2$.
12. Two conjugate semi-diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ cut the director circle at points P and Q . Prove that the segment PQ touches the ellipse.
13. If CP and CD are conjugate semi-diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that the locus of the midpoint of PD is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$.
14. The eccentric angles of the extremities of two semi-diameters are $\frac{\pi}{6}$ and $\frac{2\pi}{3}$. Show that these are conjugate diameters and find the area of the triangle formed by them with the center.
15. If the length of one semi-diameter of the ellipse $25x^2 + 16y^2 = 400$ is 3, find the length of its conjugate semi-diameter.
16. Determine the nature of the conic represented by the equation $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$.
17. Find the center of the conic section given by the equation $2x^2 - 5xy + 3y^2 - 2x + 3y = 0$.
18. Determine the value of λ for which the equation $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines.
19. Through what angle must the axes be rotated to remove the xy term from the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$? Find the transformed equation.
20. Prove that the general equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola if $\Delta \neq 0$, $h^2 > ab$, and $a + b = 0$.

Chapter 3

Spheres and Cones

3.1 Systems of Spheres

In this section, we study the geometric relationships between multiple spheres, specifically focusing on the locus of points relative to two or more spheres. We establish the concepts of the radical plane and coaxial systems, which allow us to simplify the analytical treatment of spheres.

3.1.1 Power of a Point with Respect to a Sphere

Before defining the radical plane, we must formalize the concept of the power of a point.

Definition 3.1.1 (Power of a Point). Let the equation of a sphere S be given by:

$$S(x, y, z) \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$

The power of a point $P(x_1, y_1, z_1)$ with respect to the sphere S is the value obtained by substituting the coordinates of P into the left-hand side of the sphere's equation.

If we denote the power by $\mathcal{P}$, then:

$$\mathcal{P} = x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d$$

Geometrically, if a line through P intersects the sphere at points A and B , the power is equal to the product of the distances $PA \cdot PB$. If P lies outside the sphere, the power is equal to the square of the length of the tangent from P to the sphere (PT^2).

3.1.2 The Radical Plane

Definition 3.1.2 (Radical Plane). The radical plane of two spheres is the locus of a point which moves such that its powers with respect to the two spheres are equal.

Theorem 3.1.3 (Equation of the Radical Plane). Let the equations of two spheres be:

$$S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0$$

$$S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0$$

The equation of the radical plane is given by $S_1 - S_2 = 0$.

Proof. Let $P(x', y', z')$ be a point on the locus. By definition, the power of P with respect to S_1 is equal to the power of P with respect to S_2 . Therefore:

$$x'^2 + y'^2 + z'^2 + 2u_1x' + \cdots + d_1 = x'^2 + y'^2 + z'^2 + 2u_2x' + \cdots + d_2$$

Canceling the second-degree terms, we obtain:

$$2(u_1 - u_2)x' + 2(v_1 - v_2)y' + 2(w_1 - w_2)z' + (d_1 - d_2) = 0$$

Generalizing for any point (x, y, z) on the locus, the equation of the radical plane is:

$$2x(u_1 - u_2) + 2y(v_1 - v_2) + 2z(w_1 - w_2) + (d_1 - d_2) = 0$$

This is a linear equation in x, y, z , which represents a plane. ■

Note 1. It is essential that the coefficients of x^2, y^2, z^2 in the equations of both spheres are unity (or equal) before subtracting them to find the radical plane.

Theorem 3.1.4 (Perpendicularity Property). *The radical plane of two spheres is perpendicular to the line joining their centers.*

Proof. Let the centers of the two spheres S_1 and S_2 be $C_1(-u_1, -v_1, -w_1)$ and $C_2(-u_2, -v_2, -w_2)$. The direction ratios of the line C_1C_2 are proportional to:

$$(u_1 - u_2), (v_1 - v_2), (w_1 - w_2)$$

From the equation of the radical plane derived previously, the direction ratios of the normal to this plane are also $(u_1 - u_2), (v_1 - v_2), (w_1 - w_2)$. Since the direction ratios of the line joining the centers are identical to the direction ratios of the normal to the plane, the radical plane is perpendicular to the line of centers. ■

3.1.3 Coaxial System of Spheres

Definition 3.1.5 (Coaxial System). A system of spheres is said to be coaxial if every pair of spheres in the system has the same radical plane. This common radical plane is often referred to as the *radical axis* of the system (in the context of planar sections) or the radical plane of the coaxial system.

Theorem 3.1.6 (Equation of a Coaxial System). *The equation of a system of coaxial spheres can be expressed in two primary forms:*

1. *Given two spheres $S_1 = 0$ and $S_2 = 0$, the equation of any sphere in the coaxial system determined by them is:*

$$S_1 + \lambda S_2 = 0 \quad (\text{where } \lambda \neq -1)$$

2. *Given a sphere $S = 0$ and a plane $P = 0$, the coaxial system having P as the radical plane is given by:*

$$S + \lambda P = 0$$

where λ is an arbitrary constant parameter.

Definition 3.1.7 (Limiting Points). The limiting points of a coaxial system are the point spheres of the system. These are the members of the system for which the radius is zero.

If the equation of the coaxial system is written as $x^2 + y^2 + z^2 + 2ux + d = 0$ (taking the line of centers as the x -axis), the center is $(-u, 0, 0)$ and the radius is $\sqrt{u^2 - d}$. For limiting points, the radius is zero:

$$u^2 - d = 0 \implies u = \pm\sqrt{d}$$

Thus, the limiting points are $(\sqrt{d}, 0, 0)$ and $(-\sqrt{d}, 0, 0)$. Real limiting points exist only when $d > 0$.

3.1.4 Simplified Form of the Equation of Two Spheres

When dealing with two spheres analytically, it is often convenient to choose a coordinate system that simplifies their equations without loss of generality. We utilize the properties of the radical plane and the line of centers to achieve this simplification.

Theorem 3.1.8 (Simplified Canonical Form). *By a suitable choice of origin and axes, the equations of any two spheres can be written in the form:*

$$x^2 + y^2 + z^2 + 2u_1x + d = 0$$

$$x^2 + y^2 + z^2 + 2u_2x + d = 0$$

Proof. We construct the coordinate system as follows:

1. Take the line joining the centers of the two spheres as the x -axis.
2. Take the radical plane of the two spheres as the plane $x = 0$ (the yz -plane).

Since the centers lie on the x -axis, their y and z coordinates are zero. Thus, the equations of the spheres take the form:

$$S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + d_1 = 0$$

$$S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + d_2 = 0$$

The radical plane of these two spheres is given by $S_1 - S_2 = 0$:

$$2(u_1 - u_2)x + (d_1 - d_2) = 0$$

However, we chose the radical plane to be $x = 0$. For the derived equation to represent $x = 0$, the constant term must vanish:

$$d_1 - d_2 = 0 \implies d_1 = d_2$$

Let $d_1 = d_2 = d$. The simplified equations become:

$$x^2 + y^2 + z^2 + 2u_1x + d = 0 \quad \text{and} \quad x^2 + y^2 + z^2 + 2u_2x + d = 0$$

■

Corollary 3.1.9 (Intersection Properties). *Based on the sign of the constant d in the simplified form:*

- If $d > 0$, the radical plane passes between the spheres, and the spheres do not intersect in a real circle. The system has real limiting points.
- If $d < 0$, the spheres intersect in a real circle lying on the plane $x = 0$. The system has imaginary limiting points.
- If $d = 0$, the spheres touch each other at the origin.

Example 3.1.10. Find the limiting points of the coaxial system of spheres defined by the spheres $S_1 \equiv x^2 + y^2 + z^2 + 4x + 2y + 4z + 10 = 0$ and $S_2 \equiv x^2 + y^2 + z^2 + 2x + 4y - 6 = 0$.

Solution. The radical plane P of the given spheres is $S_1 - S_2 = 0$. Subtracting the equations:

$$(4x - 2x) + (2y - 4y) + (4z - 0) + (10 - (-6)) = 0 \implies 2x - 2y + 4z + 16 = 0 \implies x - y + 2z + 8 = 0$$

The equation of any sphere in the coaxial system is $S_2 + \lambda P = 0$:

$$x^2 + y^2 + z^2 + (2 + \lambda)x + (4 - \lambda)y + 2\lambda z + (8\lambda - 6) = 0$$

The center is $C \left(-\frac{2 + \lambda}{2}, -\frac{4 - \lambda}{2}, -\lambda \right)$. The square of the radius is:

$$R^2 = \left(\frac{2 + \lambda}{2} \right)^2 + \left(\frac{4 - \lambda}{2} \right)^2 + \lambda^2 - (8\lambda - 6)$$

For limiting points, the radius is zero ($R^2 = 0$). Multiplying by 4 to clear denominators:

$$\begin{aligned} (2 + \lambda)^2 + (4 - \lambda)^2 + 4\lambda^2 - 4(8\lambda - 6) &= 0 \\ (\lambda^2 + 4\lambda + 4) + (\lambda^2 - 8\lambda + 16) + 4\lambda^2 - 32\lambda + 24 &= 0 \\ 6\lambda^2 - 36\lambda + 44 &= 0 \\ 3\lambda^2 - 18\lambda + 22 &= 0 \end{aligned}$$

Solving for λ :

$$\lambda = \frac{18 \pm \sqrt{324 - 264}}{6} = 3 \pm \frac{\sqrt{15}}{3}$$

Substituting these values of λ back into the coordinate expression for the center C yields the two limiting points. ■

Example 3.1.11. Find the equation of the sphere that cuts the three spheres $x^2 + y^2 + z^2 = a^2$, $(x - b)^2 + y^2 + z^2 = a^2$, and $x^2 + (y - c)^2 + z^2 = a^2$ orthogonally.

Solution. Let the required sphere be $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$. Expanding the given spheres:

$$\begin{aligned} S_1 : x^2 + y^2 + z^2 - a^2 &= 0 \\ S_2 : x^2 + y^2 + z^2 - 2bx + b^2 - a^2 &= 0 \\ S_3 : x^2 + y^2 + z^2 - 2cy + c^2 - a^2 &= 0 \end{aligned}$$

The condition for orthogonality is $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$. Applying this to the required sphere and each given sphere:

1. With S_1 (center $(0, 0, 0)$, $d_1 = -a^2$):

$$2u(0) + 2v(0) + 2w(0) = d - a^2 \implies d = a^2$$

2. With S_2 (center $(b, 0, 0)$, $d_2 = b^2 - a^2$):

$$2u(-b) = d + b^2 - a^2 \implies -2ub = a^2 + b^2 - a^2 \implies u = -\frac{b}{2}$$

3. With S_3 (center $(0, c, 0)$, $d_3 = c^2 - a^2$):

$$2v(-c) = d + c^2 - a^2 \implies -2vc = c^2 \implies v = -\frac{c}{2}$$

Assuming the center lies on the plane $z = 0$ (or finding the radical center in the simplest plane), we set $w = 0$. The equation is:

$$x^2 + y^2 + z^2 - bx - cy + a^2 = 0$$

■

Example 3.1.12. Prove that the locus of a point whose powers with respect to two given spheres are in a constant ratio is a coaxial sphere.

Solution. Let the two given spheres be $S_1 = 0$ and $S_2 = 0$. Let $P(x, y, z)$ be a point such that the ratio of its power with respect to S_1 to its power with respect to S_2 is a constant k ($k \neq 1$). The condition is:

$$\frac{S_1}{S_2} = k \implies S_1 - kS_2 = 0$$

Writing this out explicitly:

$$(x^2 + y^2 + z^2 + \dots) - k(x^2 + y^2 + z^2 + \dots) = 0$$

$$(1 - k)(x^2 + y^2 + z^2) + 2(u_1 - ku_2)x + \dots + (d_1 - kd_2) = 0$$

Dividing by $(1 - k)$, the equation becomes:

$$x^2 + y^2 + z^2 + \frac{2(u_1 - ku_2)}{1 - k}x + \dots = 0$$

This is the equation of a sphere. Since it satisfies $S_1 - kS_2 = 0$, it passes through the intersection of S_1 and S_2 , making it coaxial with them. ■

Example 3.1.13. Find the sphere of the coaxial system defined by $x^2 + y^2 + z^2 - 5 + \lambda(2x + y + 3z - 3) = 0$ which touches the plane $3x + 4y = 15$.

Solution. The center of the sphere is $C(-\lambda, -\frac{\lambda}{2}, -\frac{3\lambda}{2})$ and the constant is $d = -5 - 3\lambda$. The radius R is given by:

$$R = \sqrt{\lambda^2 + \frac{\lambda^2}{4} + \frac{9\lambda^2}{4} - (-5 - 3\lambda)} = \sqrt{\frac{7\lambda^2 + 6\lambda + 10}{2}}$$

The perpendicular distance from C to the plane $3x + 4y - 15 = 0$ is equal to R :

$$\text{Distance} = \frac{|3(-\lambda) + 4(-\frac{\lambda}{2}) - 15|}{\sqrt{3^2 + 4^2}} = \frac{|-5\lambda - 15|}{5} = |\lambda + 3|$$

Equating distance and radius and squaring:

$$\begin{aligned}(\lambda + 3)^2 &= \frac{7\lambda^2 + 6\lambda + 10}{2} \\ 2(\lambda^2 + 6\lambda + 9) &= 7\lambda^2 + 6\lambda + 10 \\ 5\lambda^2 - 6\lambda - 8 &= 0\end{aligned}$$

Solving $(5\lambda + 4)(\lambda - 2) = 0$ gives $\lambda = 2$ or $\lambda = -4/5$. Using $\lambda = 2$, the sphere is $x^2 + y^2 + z^2 + 4x + 2y + 6z - 11 = 0$. ■

Example 3.1.14. Show that the radical plane of two spheres bisects the segment joining their common external points of contact with the common tangents.

Solution. Consider the simplified form of two spheres:

$$S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + d = 0, \quad S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + d = 0$$

The radical plane is $x = 0$. Let T_1 and T_2 be the points of contact of a common tangent. Let M be the midpoint of T_1T_2 . The power of M with respect to S_1 is MT_1^2 , and with respect to S_2 is MT_2^2 . Since M is the midpoint, $MT_1 = MT_2$, so $MT_1^2 = MT_2^2$. Since the powers are equal, M lies on the radical plane. Thus, the radical plane bisects the common tangent segment. ■

3.2 The Cone

In this section, we transition from spherical geometry to the study of ruled surfaces, specifically the cone. We define the geometric properties of a cone and derive its analytical equations under various conditions, culminating in the study of the general second-degree equation.

Definition 3.2.1 (Cone). A cone is a surface generated by a straight line which moves such that it passes through a fixed point and intersects a given fixed curve.

Definition 3.2.2. 1. **Vertex:** The fixed point through which the generating line always passes is called the vertex of the cone.

2. **Generator:** The moving straight line that generates the surface is called the generator (or generating line) of the cone.

3. **Guiding Curve:** The fixed curve which directs the motion of the generator is called the guiding curve.

Note 2. If the guiding curve is a circle and the line joining the vertex to the center of the circle is perpendicular to the plane of the circle, the surface is called a *Right Circular Cone*.

3.2.1 Equation of a Cone with Vertex at the Origin

A fundamental property of cones with their vertex at the origin is the homogeneity of their equations.

Theorem 3.2.3. *The equation of a cone with its vertex at the origin is a homogeneous equation in x, y, z . Conversely, every homogeneous equation in x, y, z represents a cone with its vertex at the origin.*

Proof. Necessary Condition: Let the vertex of the cone be the origin $O(0, 0, 0)$. Let $P(x', y', z')$ be any point on the cone. Since O is the vertex and P is on the cone, the generator is the infinite line passing through O and P . Every point on this line must lie on the surface of the cone.

The coordinates of any arbitrary point on the line OP are given by $(\lambda x', \lambda y', \lambda z')$ for any real scalar λ . If the equation of the surface is given by $f(x, y, z) = 0$, then for the surface to be a cone, the fact that P lies on the surface (i.e., $f(x', y', z') = 0$) must imply that any point on the line OP lies on the surface. Thus:

$$f(\lambda x', \lambda y', \lambda z') = 0 \quad \text{for all } \lambda \in \mathbb{R}$$

This condition implies that $f(x, y, z)$ must be a homogeneous function of x, y, z .

Sufficient Condition: Conversely, let $f(x, y, z) = 0$ be a homogeneous equation of degree n . By Euler's property of homogeneous functions:

$$f(\lambda x, \lambda y, \lambda z) = \lambda^n f(x, y, z)$$

If a point $P(x_1, y_1, z_1)$ satisfies the equation (i.e., $f(x_1, y_1, z_1) = 0$), then:

$$f(\lambda x_1, \lambda y_1, \lambda z_1) = \lambda^n \cdot 0 = 0$$

This shows that if P lies on the surface, any point $(\lambda x_1, \lambda y_1, \lambda z_1)$ on the line joining the origin to P also lies on the surface. Hence, the surface is generated by lines passing through the origin, making it a cone with the vertex at the origin. ■

Definition 3.2.4 (Quadric Cone). A cone whose equation is of the second degree is called a quadric cone. The general homogeneous equation of the second degree representing a cone with vertex at the origin is:

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

3.2.2 Equation of a Cone with a Given Vertex and Guiding Curve

When the vertex is not at the origin, or the guiding curve is explicitly defined, we derive the equation by establishing the condition that the generator intersects the guiding curve.

Theorem 3.2.5. *The equation of the cone with vertex at $V(\alpha, \beta, \gamma)$ and guiding curve given by the intersection of a surface $F(x, y, z) = 0$ and a plane $ax + by + cz + d = 0$ is obtained by eliminating the direction ratios of the generator.*

Method of Derivation. Let the equations of the guiding curve be:

$$F(x, y, z) = 0, \quad ax + by + cz + d = 0$$

The equations of any line (generator) passing through the vertex $V(\alpha, \beta, \gamma)$ with direction ratios l, m, n are:

$$\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$$

Since this line intersects the guiding curve, it must meet the plane $ax + by + cz + d = 0$ at some point P . We solve the linear equations to find the intersection point in terms of l, m, n .

Alternatively, a general point on the line is $(\alpha + lr, \beta + mr, \gamma + nr)$. Substituting this into the plane equation yields r . We then substitute the coordinates of this point into the surface equation $F(x, y, z) = 0$. Finally, we eliminate the parameters l, m, n using the relations:

$$\frac{l}{x - \alpha} = \frac{m}{y - \beta} = \frac{n}{z - \gamma}$$

■

Corollary 3.2.6 (Cone with Vertex at Origin and Planar Guiding Curve). *If the vertex is the origin and the guiding curve is a conic section given by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ lying in the plane $lx + my + nz = p$, the equation of the cone is found by homogenizing the equation of the conic.*

We utilize the relation from the plane $\frac{lx + my + nz}{p} = 1$. The homogenized equation is:

$$ax^2 + 2hxy + by^2 + 2(gx + fy) \left(\frac{lx + my + nz}{p} \right) + c \left(\frac{lx + my + nz}{p} \right)^2 = 0$$

Example 3.2.7. Find the equation of the cone whose vertex is the origin and which passes through the curve of intersection of the sphere $x^2 + y^2 + z^2 = 3a^2$ and the plane $x + y + z = a$.

Solution. Since the vertex is the origin, the equation of the cone is homogeneous. We homogenize the equation of the sphere using the equation of the plane. The given equations are:

$$x^2 + y^2 + z^2 = 3a^2 \quad \text{and} \quad \frac{x + y + z}{a} = 1$$

Making the sphere's equation homogeneous of degree 2:

$$x^2 + y^2 + z^2 = 3a^2 \left(\frac{x + y + z}{a} \right)^2$$

Simplifying this expression:

$$\begin{aligned} x^2 + y^2 + z^2 &= 3(x + y + z)^2 \\ x^2 + y^2 + z^2 &= 3(x^2 + y^2 + z^2 + 2xy + 2yz + 2zx) \\ 2x^2 + 2y^2 + 2z^2 + 6xy + 6yz + 6zx &= 0 \end{aligned}$$

Dividing by 2, the required equation is:

$$x^2 + y^2 + z^2 + 3xy + 3yz + 3zx = 0$$

■

Example 3.2.8. Find the equation of the cone with vertex at $(1, 1, 1)$ and which passes through the circle $x^2 + y^2 = 4, z = 0$.

Solution. The equations of any generator passing through the vertex $V(1, 1, 1)$ are:

$$\frac{x - 1}{l} = \frac{y - 1}{m} = \frac{z - 1}{n}$$

This line meets the plane $z = 0$ at a point P . Setting $z = 0$, we find the coordinates of P :

$$P \equiv \left(1 - \frac{l}{n}, 1 - \frac{m}{n}, 0 \right)$$

Since P lies on the curve $x^2 + y^2 = 4$, we substitute these coordinates:

$$\left(1 - \frac{l}{n} \right)^2 + \left(1 - \frac{m}{n} \right)^2 = 4$$

Eliminating the parameters using $\frac{l}{n} = \frac{x - 1}{z - 1}$ and $\frac{m}{n} = \frac{y - 1}{z - 1}$:

$$\left(1 - \frac{x - 1}{z - 1} \right)^2 + \left(1 - \frac{y - 1}{z - 1} \right)^2 = 4$$

Simplifying the fractions:

$$\begin{aligned}\left(\frac{z-x}{z-1}\right)^2 + \left(\frac{z-y}{z-1}\right)^2 &= 4 \\ (z-x)^2 + (z-y)^2 &= 4(z-1)^2\end{aligned}$$

Expanding and rearranging terms:

$$\begin{aligned}x^2 + y^2 + 2z^2 - 2zx - 2yz &= 4(z^2 - 2z + 1) \\ x^2 + y^2 - 2z^2 - 2zx - 2yz + 8z - 4 &= 0\end{aligned}$$

■

Example 3.2.9. Prove that the equation $2x^2 + 2y^2 + 7z^2 - 10yz - 10zx + 2x + 2y + 26z - 17 = 0$ represents a cone, and find the coordinates of its vertex.

Solution. Let $F(x, y, z)$ be the given equation. For F to represent a cone, the system of partial derivatives must vanish at a common point (the vertex).

$$\begin{aligned}\frac{\partial F}{\partial x} &= 4x - 10z + 2 = 0 \implies 2x - 5z + 1 = 0 \\ \frac{\partial F}{\partial y} &= 4y - 10z + 2 = 0 \implies 2y - 5z + 1 = 0 \\ \frac{\partial F}{\partial z} &= 14z - 10y - 10x + 26 = 0 \implies -5x - 5y + 7z + 13 = 0\end{aligned}$$

Solving the first two equations gives $x = y$. Substituting $x = (5z - 1)/2$ into the third equation:

$$-5\left(\frac{5z-1}{2}\right) - 5\left(\frac{5z-1}{2}\right) + 7z + 13 = 0$$

$$-5(5z-1) + 7z + 13 = 0 \implies -18z + 18 = 0 \implies z = 1$$

Substituting $z = 1$ back yields $x = 2, y = 2$. The vertex is $(2, 2, 1)$. Finally, checking if this point satisfies the original equation:

$$F(2, 2, 1) = 2(4) + 2(4) + 7(1) - 20 - 20 + 4 + 4 + 26 - 17 = 57 - 57 = 0$$

Since the condition is satisfied, the equation represents a cone with vertex $(2, 2, 1)$.

■

3.2.3 Condition for the General Equation of Second Degree to Represent a Cone

The general equation of the second degree in three variables is:

$$S \equiv ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$$

This equation represents a cone if and only if there exists a point (the vertex) such that shifting the origin to this point reduces the equation to a homogeneous form.

Theorem 3.2.10. The necessary and sufficient condition for the general equation of the second degree to represent a cone is that the discriminant of the quadratic form vanishes. That is:

$$\Delta = \begin{vmatrix} a & h & g & u \\ h & b & f & v \\ g & f & c & w \\ u & v & w & d \end{vmatrix} = 0$$

Analytical Reasoning. If the equation $S = 0$ represents a cone, it must possess a vertex, say $V(x_0, y_0, z_0)$. If we transform the origin to V , the linear terms and the constant term in the transformed equation must vanish for the equation to become homogeneous.

Analytically, the coordinates of the center (or vertex) of the quadric are obtained by solving the simultaneous linear equations formed by the partial derivatives:

$$\frac{\partial S}{\partial x} = 0, \quad \frac{\partial S}{\partial y} = 0, \quad \frac{\partial S}{\partial z} = 0$$

These expand to:

$$ax + hy + gz + u = 0$$

$$hx + by + fz + v = 0$$

$$gx + fy + cz + w = 0$$

Furthermore, for the vertex to actually lie on the surface (which is true for a cone), the constant term in the transformed equation must vanish. Using Euler's theorem on the expansion of S , the condition requires consistency among the three partial derivative equations and the equation formed by the homogeneous variable (associated with u, v, w, d):

$$ux + vy + wz + d = 0$$

For these four linear homogeneous equations to have a non-trivial consistent solution, the determinant of their coefficients must be zero. Hence, $\Delta = 0$. ■

Example 3.2.11. Verify that the equation $x^2 + y^2 - z^2 - 2x - 2y + 2z + 1 = 0$ represents a cone, and find the coordinates of its vertex.

Solution. Comparing the given equation with the general equation of the second degree:

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$$

We identify the coefficients as follows:

$$a = 1, \quad b = 1, \quad c = -1$$

$$f = 0, \quad g = 0, \quad h = 0$$

$$u = -1, \quad v = -1, \quad w = 1, \quad d = 1$$

The necessary and sufficient condition for the surface to represent a cone is that the discriminant Δ vanishes.

$$\Delta = \begin{vmatrix} a & h & g & u \\ h & b & f & v \\ g & f & c & w \\ u & v & w & d \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{vmatrix}$$

Expanding the determinant along the first row:

$$\Delta = 1 \cdot \begin{vmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 1 \end{vmatrix} - (-1) \cdot \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

Calculating the minors:

$$\text{First Minor} = 1(-1 - 1) - 0 + (-1)(0 - 1) = -2 + 1 = -1$$

$$\text{Second Minor} = 0 - 1(0 - 1) + 0 = 1$$

Substituting these values back into the expression for Δ :

$$\Delta = 1(-1) + 1(1) = -1 + 1 = 0$$

Since the discriminant vanishes, the equation represents a cone.

To find the vertex $V(x, y, z)$, we differentiate the equation of the surface partially with respect to x, y , and z and equate to zero:

$$\begin{aligned}\frac{\partial S}{\partial x} &= 2x - 2 = 0 \implies x = 1 \\ \frac{\partial S}{\partial y} &= 2y - 2 = 0 \implies y = 1 \\ \frac{\partial S}{\partial z} &= -2z + 2 = 0 \implies z = 1\end{aligned}$$

Thus, the vertex of the cone is $(1, 1, 1)$. ■

Example 3.2.12. Find the value of λ such that the equation $2x^2 - 3y^2 + 4z^2 - 4x - 6y + \lambda = 0$ represents a cone.

Solution. Comparing the given equation with the general second-degree equation, we extract the coefficients:

$$a = 2, \quad b = -3, \quad c = 4$$

Since there are no mixed product terms (yz, zx, xy) , the associated coefficients are zero:

$$f = 0, \quad g = 0, \quad h = 0$$

The linear and constant coefficients are:

$$2u = -4 \implies u = -2$$

$$2v = -6 \implies v = -3$$

$$2w = 0 \implies w = 0$$

$$d = \lambda$$

For the equation to represent a cone, the discriminant Δ must be zero:

$$\Delta = \begin{vmatrix} 2 & 0 & 0 & -2 \\ 0 & -3 & 0 & -3 \\ 0 & 0 & 4 & 0 \\ -2 & -3 & 0 & \lambda \end{vmatrix} = 0$$

It is most efficient to expand this determinant along the third row (which contains the element 4 and zeros):

$$\Delta = 4 \cdot \begin{vmatrix} 2 & 0 & -2 \\ 0 & -3 & -3 \\ -2 & -3 & \lambda \end{vmatrix}$$

For Δ to be zero, the remaining 3×3 determinant must be zero:

$$2(-3\lambda - 9) - 0 + (-2)(0 - 6) = 0$$

$$2(-3\lambda - 9) + 12 = 0$$

$$-6\lambda - 18 + 12 = 0$$

$$-6\lambda - 6 = 0 \implies \lambda = -1$$

Therefore, the required value is $\lambda = -1$. ■

3.2.4 Three Mutually Perpendicular Generators

We investigate the condition under which a quadric cone contains three generators that are mutually at right angles to one another.

Theorem 3.2.13. *The necessary and sufficient condition for the cone represented by the equation*

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

to have three mutually perpendicular generators is that the sum of the coefficients of the squared terms is zero. That is:

$$a + b + c = 0$$

Proof. Let the vertex of the cone be the origin. Suppose the cone possesses three mutually perpendicular generators. We can transform our coordinate system such that these three generators become the new axes of reference OX', OY', OZ' .

If the new x -axis OX' is a generator, the point $(r, 0, 0)$ must satisfy the transformed equation of the cone $A'x'^2 + B'y'^2 + C'z'^2 + 2F'y'z' + \dots = 0$. Substituting $(r, 0, 0)$, we get:

$$A'r^2 = 0 \implies A' = 0$$

Similarly, since OY' and OZ' are generators, substituting $(0, r, 0)$ and $(0, 0, r)$ yields:

$$B' = 0 \quad \text{and} \quad C' = 0$$

Thus, the equation of the cone referred to its three perpendicular generators as axes takes the form:

$$2F'y'z' + 2G'z'x' + 2H'x'y' = 0$$

We now invoke the property of invariants for orthogonal transformations. The sum of the coefficients of the squared terms (x^2, y^2, z^2) is an invariant. Therefore:

$$a + b + c = A' + B' + C'$$

Substituting the values found above:

$$a + b + c = 0 + 0 + 0 = 0$$

Conversely, if $a + b + c = 0$, it is always possible to find a set of three mutually perpendicular lines lying on the cone. ■

Note 3. If the general equation of the second degree represents a cone (i.e., discriminant is zero), the condition for it to have three mutually perpendicular generators remains $a + b + c = 0$, referring to the coefficients of the second-degree terms only.

Example 3.2.14. Find the condition that the cone passing through the intersection of the sphere $x^2 + y^2 + z^2 = a^2$ and the plane $lx + my + nz = p$ may have three mutually perpendicular generators.

Solution. The vertex of the cone is the origin. We homogenize the sphere's equation using the plane's equation $\frac{lx + my + nz}{p} = 1$:

$$x^2 + y^2 + z^2 = a^2 \left(\frac{lx + my + nz}{p} \right)^2$$

$$p^2(x^2 + y^2 + z^2) - a^2(lx + my + nz)^2 = 0$$

The condition for a cone $Ax^2 + By^2 + Cz^2 + 2Hxy + \dots = 0$ to have three mutually perpendicular generators is $A + B + C = 0$. Extracting the coefficients of x^2, y^2, z^2 :

$$A = p^2 - a^2l^2, \quad B = p^2 - a^2m^2, \quad C = p^2 - a^2n^2$$

Applying the condition:

$$(p^2 - a^2l^2) + (p^2 - a^2m^2) + (p^2 - a^2n^2) = 0$$

$$3p^2 - a^2(l^2 + m^2 + n^2) = 0 \implies 3p^2 = a^2(l^2 + m^2 + n^2)$$

■

Example 3.2.15. If the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is one of a set of three mutually perpendicular generators of the cone $5yz - 8zx - 3xy = 0$, find the equations of the other two.

Solution. Note that the sum of coefficients $a + b + c = 0 + 0 + 0 = 0$, so perpendicular generators exist. Let one of the required lines be $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$. The line lies on the cone, so:

$$5mn - 8nl - 3lm = 0$$

The line is perpendicular to the given generator (direction ratios 1, 2, 3), so:

$$l + 2m + 3n = 0 \implies l = -(2m + 3n)$$

Substituting l into the cone's equation:

$$5mn - 8n(-(2m + 3n)) - 3m(-(2m + 3n)) = 0$$

$$5mn + 16mn + 24n^2 + 6m^2 + 9mn = 0$$

$$6m^2 + 30mn + 24n^2 = 0$$

$$m^2 + 5mn + 4n^2 = 0$$

Factoring gives $(m + n)(m + 4n) = 0$.

1. If $m = -n$, then $l = -(-2n + 3n) = -n$. Ratios: $-1 : -1 : 1$ or $1 : 1 : -1$.

2. If $m = -4n$, then $l = -(-8n + 3n) = 5n$. Ratios: $5 : -4 : 1$.

The equations are $\frac{x}{1} = \frac{y}{1} = \frac{z}{-1}$ and $\frac{x}{5} = \frac{y}{-4} = \frac{z}{1}$. ■

Example 3.2.16. Find the locus of the vertex of a cone which passes through the fixed curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, z = 0$ and possesses three mutually perpendicular generators.

Solution. Let the vertex be (α, β, γ) . The generator passes through (α, β, γ) and a point $(x, y, 0)$ on the ellipse. The equation of the cone is formed by the line joining the vertex to the ellipse. The equation of the cone (using the result for a conic base $z = 0$) is:

$$\frac{(\alpha z - \gamma x)^2}{a^2} + \frac{(\beta z - \gamma y)^2}{b^2} = (z - \gamma)^2$$

Expanding and collecting terms for x^2, y^2, z^2 :

$$\left(\frac{\gamma^2}{a^2}\right)x^2 + \left(\frac{\gamma^2}{b^2}\right)y^2 + \left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} - 1\right)z^2 + \dots = 0$$

For three mutually perpendicular generators, the sum of coefficients of x^2, y^2, z^2 is zero:

$$\frac{\gamma^2}{a^2} + \frac{\gamma^2}{b^2} + \frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} - 1 = 0$$

Multiplying by a^2b^2 :

$$b^2\gamma^2 + a^2\gamma^2 + b^2\alpha^2 + a^2\beta^2 - a^2b^2 = 0$$

Generalizing the locus by replacing (α, β, γ) with (x, y, z) :

$$b^2x^2 + a^2y^2 + (a^2 + b^2)z^2 = a^2b^2$$
■

Problems for Practice - Sharpening Your Skills

1. Find the equation of the radical plane of the spheres $x^2 + y^2 + z^2 + 2x + 2y + 2z + 2 = 0$ and $2x^2 + 2y^2 + 2z^2 + 8x + 4y + 8z + 1 = 0$.
2. Find the equation of the sphere which passes through the circle $x^2 + y^2 + z^2 - 4 = 0$, $z = 0$ and is cut by the plane $x + 2y + 2z = 0$ in a circle of radius 3.
3. Determine the limiting points of the coaxial system of spheres determined by the two spheres $x^2 + y^2 + z^2 + 4x - 2y + 2z + 6 = 0$ and $x^2 + y^2 + z^2 + 2x - 4y - 2z + 6 = 0$.
4. Find the equation of the sphere which is coaxial with the spheres $x^2 + y^2 + z^2 + 2x = 0$ and $x^2 + y^2 + z^2 + 2y = 0$ and which touches the plane $x + y + z + 5 = 0$.
5. Find the coordinates of the center of the sphere which cuts the three spheres $x^2 + y^2 + z^2 = 1$, $x^2 + y^2 + z^2 + 2x - 3 = 0$, and $x^2 + y^2 + z^2 + 2y - 3 = 0$ orthogonally.
6. Show that the radical planes of the spheres of a coaxial system taken in pairs all pass through one line (the radical axis).
7. Find the equation of the cone with vertex at the origin which passes through the curve given by $x^2 + y^2 + z^2 + 2ax + b = 0$ and $lx + my + nz = p$.
8. Find the equation of the cone whose vertex is the point $(1, 2, 3)$ and whose guiding curve is the circle $x^2 + y^2 + z^2 = 4$, $x + y + z = 1$.
9. Determine the equation of the cone generated by straight lines drawn from the origin to cut the circle $x^2 + y^2 + 2ax + b = 0$, $z = 0$.
10. Find the value of k for which the equation $3x^2 + 4y^2 + 5z^2 + 2xy + 4yz + 6zx + k = 0$ represents a cone.
11. Prove that the equation $2y^2 - 8yz - 4zx - 8xy + 6x - 4y - 2z + 5 = 0$ represents a cone, and find the coordinates of its vertex.
12. Find the condition that the cone $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ may have three mutually perpendicular generators.
13. If $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ represents one of a set of three mutually perpendicular generators of the cone $5yz - 8zx - 3xy = 0$, find the equations of the other two generators.
14. Show that the equation of the cone with vertex at the origin and which passes through the three coordinate axes is of the form $fyz + gzx + hxy = 0$.
15. Find the equation of the cone with vertex at the origin and which passes through the curve of intersection of the sphere $x^2 + y^2 + z^2 = a^2$ and the plane $x + y + z = a$. Show that this cone has three mutually perpendicular generators.

Chapter 4

Surfaces of the Second Degree: Cylinders and Conicoids

4.1 Cylinders

In this section, we extend the study of ruled surfaces to cylinders and introduce the general theory of quadric surfaces, known as conicoids. We explore their classification and the properties of tangent planes associated with them.

Definition 4.1.1 (Cylinder). A cylinder is a surface generated by a straight line (the generator) which moves such that it remains parallel to a fixed line and intersects a given fixed curve (the guiding curve).

Definition 4.1.2 (Right Circular Cylinder). A cylinder is called a right circular cylinder if the guiding curve is a circle and the generators are perpendicular to the plane of the circle. Alternatively, it is the surface generated by a line which remains at a constant distance from a fixed straight line called the axis.

Equation of a Cylinder

We determine the analytical equation of a cylinder based on the direction of its generators and the nature of its guiding curve.

Theorem 4.1.3 (Equation of a General Cylinder). *The equation of a cylinder whose generators intersect the conic*

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \quad z = 0$$

and are parallel to the line with direction ratios l, m, n is obtained by substituting the coordinates of the intersection of the generator and the plane $z = 0$ into the equation of the conic.

The resulting equation is:

$$a(nx - lz)^2 + 2h(nx - lz)(ny - mz) + b(ny - mz)^2 + 2gn(nx - lz) + 2fn(ny - mz) + cn^2 = 0$$

Enveloping Cylinder

Definition 4.1.4 (Enveloping Cylinder). The enveloping cylinder of a surface is the locus of tangent lines to the surface that are parallel to a given fixed direction. It can be visualized as the cylinder that just touches or encloses the surface relative to a specific projection vector.

Theorem 4.1.5 (Enveloping Cylinder of a Sphere). *The equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = a^2$ whose generators are parallel to the line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is:*

$$(x^2 + y^2 + z^2 - a^2)(l^2 + m^2 + n^2) - (lx + my + nz)^2 = 0$$

Proof. Let $P(x_1, y_1, z_1)$ be any point on the enveloping cylinder. A generator of the cylinder passes through P and is parallel to the direction (l, m, n) . The equations of this generator are:

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r \quad (\text{say})$$

Any point on this line is of the form $(x_1 + lr, y_1 + mr, z_1 + nr)$. Since this generator touches the sphere $x^2 + y^2 + z^2 - a^2 = 0$, the points of intersection are coincident. Substituting the general point into the sphere's equation:

$$(x_1 + lr)^2 + (y_1 + mr)^2 + (z_1 + nr)^2 - a^2 = 0$$

Arranging this as a quadratic equation in r :

$$r^2(l^2 + m^2 + n^2) + 2r(lx_1 + my_1 + nz_1) + (x_1^2 + y_1^2 + z_1^2 - a^2) = 0$$

For the line to be a tangent to the sphere, this quadratic must have equal roots. Thus, the discriminant must vanish:

$$4(lx_1 + my_1 + nz_1)^2 - 4(l^2 + m^2 + n^2)(x_1^2 + y_1^2 + z_1^2 - a^2) = 0$$

Simplifying and replacing the locus (x_1, y_1, z_1) with the general point (x, y, z) , we obtain the required equation:

$$(x^2 + y^2 + z^2 - a^2)(l^2 + m^2 + n^2) - (lx + my + nz)^2 = 0$$

■

4.1.1 Types of Conicoids

A conicoid (or quadric surface) is the locus of points satisfying the general equation of the second degree in three variables:

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$$

By suitable transformation of axes (translation and rotation), this equation can be reduced to simpler standard forms. We classify these surfaces based on their standard equations.

Central Conicoids

A conicoid is central if it possesses a single center of symmetry. The standard equation is $ax^2 + by^2 + cz^2 = 1$.

1. Ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

This is a closed surface. If $a = b = c$, it becomes a sphere.

2. Hyperboloid of One Sheet:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

This is a connected, ruled surface extending infinitely.

3. Hyperboloid of Two Sheets:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

This surface consists of two disjoint congruent parts separated by a gap.

Non-Central Conicoids (Paraboloids)

These surfaces do not have a center of symmetry but possess a vertex.

1. Elliptic Paraboloid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$$

Plane sections parallel to the xy -plane are ellipses, while sections parallel to the other coordinate planes are parabolas.

2. Hyperbolic Paraboloid:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$$

This is a ruled surface shaped like a saddle. Plane sections parallel to the xy -plane are hyperbolas, while others are parabolas.

Example 4.1.6. Find the equation of the cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and whose guiding curve is the ellipse $x^2 + 2y^2 = 1, z = 0$.

Solution. Here, the guiding curve lies in the plane $z = 0$ and is given by the conic $x^2 + 2y^2 - 1 = 0$. Comparing this with the general form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, we have:

$$a = 1, \quad b = 2, \quad c = -1, \quad h = g = f = 0$$

The direction ratios of the generators are $l = 1, m = -2, n = 3$. Using the theorem for the equation of a general cylinder:

$$a(nx - lz)^2 + 2h(nx - lz)(ny - mz) + b(ny - mz)^2 + \dots + cn^2 = 0$$

Substituting the values:

$$1(3x - 1z)^2 + 0 + 2(3y - (-2)z)^2 + 0 + 0 + (-1)(3)^2 = 0$$

$$(3x - z)^2 + 2(3y + 2z)^2 - 9 = 0$$

Expanding the terms:

$$(9x^2 - 6zx + z^2) + 2(9y^2 + 12yz + 4z^2) - 9 = 0$$

$$9x^2 + z^2 - 6zx + 18y^2 + 24yz + 8z^2 - 9 = 0$$

Combining like terms, the required equation is:

$$9x^2 + 18y^2 + 9z^2 + 24yz - 6zx - 9 = 0$$

Dividing by 3 (optional, for elegance):

$$3x^2 + 6y^2 + 3z^2 + 8yz - 2zx - 3 = 0$$

■

Example 4.1.7. Find the equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = 25$ having generators parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$.

Solution. We apply the theorem for the enveloping cylinder of a sphere $x^2 + y^2 + z^2 = a^2$. Here, $a^2 = 25$ and the direction ratios are $l = 1, m = 2, n = 3$. The formula is:

$$(x^2 + y^2 + z^2 - a^2)(l^2 + m^2 + n^2) - (lx + my + nz)^2 = 0$$

First, calculate $l^2 + m^2 + n^2$:

$$1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$$

Now substitute into the formula:

$$14(x^2 + y^2 + z^2 - 25) - (x + 2y + 3z)^2 = 0$$

Expanding the second term:

$$(x + 2y + 3z)^2 = x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6zx$$

Substitute back:

$$14x^2 + 14y^2 + 14z^2 - 350 - (x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6zx) = 0$$

Simplifying:

$$13x^2 + 10y^2 + 5z^2 - 4xy - 12yz - 6zx - 350 = 0$$

■

Example 4.1.8. Find the equation of the right circular cylinder of radius 2, whose axis passes through the point $(1, 2, 3)$ and has direction cosines proportional to $(2, -1, 2)$.

Solution. Let $P(x, y, z)$ be any point on the cylinder. Let the axis be the line AB . The distance of P from the axis must be equal to the radius $r = 2$. The equation of the axis is $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{2}$. Let $A = (1, 2, 3)$ be the fixed point on the axis. The square of the distance from P to the axis is given by:

$$AP^2 - (\text{Projection of } AP \text{ on the axis})^2 = r^2$$

$AP^2 = (x-1)^2 + (y-2)^2 + (z-3)^2$. The direction cosines of the axis are $\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}$ (since $\sqrt{2^2 + (-1)^2 + 2^2} = 3$). The projection of AP on the axis is:

$$\frac{2(x-1) - 1(y-2) + 2(z-3)}{3}$$

The equation becomes:

$$(x-1)^2 + (y-2)^2 + (z-3)^2 - \left[\frac{2(x-1) - (y-2) + 2(z-3)}{3} \right]^2 = 2^2$$

Multiplying by 9:

$$9[(x-1)^2 + (y-2)^2 + (z-3)^2] - [2x - y + 2z - 6]^2 = 36$$

This equation represents the required right circular cylinder. ■

Example 4.1.9. Identify the type of conicoid represented by the following equations and describe their orientation:

$$1. 4x^2 - 9y^2 - 36z^2 = 36$$

$$2. 2x^2 + 3z^2 = 12y$$

Solution. 1. Dividing the equation by 36:

$$\frac{x^2}{9} - \frac{y^2}{4} - \frac{z^2}{1} = 1$$

Comparing this with the standard forms, it matches $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$. This represents a **Hyperboloid of Two Sheets**. The axis of the hyperboloid is the x -axis (the variable with the positive coefficient).

2. Rewriting the equation:

$$\frac{x^2}{6} + \frac{z^2}{4} = 2y$$

Alternatively written as $\frac{x^2}{6} + \frac{z^2}{4} = \frac{y}{1/2}$. This matches the form $\frac{x^2}{a^2} + \frac{z^2}{b^2} = \frac{2y}{c}$. This represents an **Elliptic Paraboloid**. The vertex is at the origin, and the axis of the paraboloid is the y -axis (the linear variable). ■

Example 4.1.10. Describe the nature of the sections of the hyperbolic paraboloid $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$ by planes parallel to the coordinate planes.

Solution. The equation is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$.

Case 1: Sections parallel to the xy -plane ($z = k$). Substituting $z = k$:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2k}{c}$$

If $k \neq 0$, this represents a **Hyperbola**. If $k = 0$, this represents a pair of intersecting lines $\frac{x}{a} = \pm \frac{y}{b}$.

Case 2: Sections parallel to the yz -plane ($x = k$). Substituting $x = k$:

$$\frac{k^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c} \implies y^2 = -\frac{2b^2}{c}z + \frac{b^2k^2}{a^2}$$

This is a linear equation in z and quadratic in y , which represents a **Parabola**.

Case 3: Sections parallel to the zx -plane ($y = k$). Substituting $y = k$:

$$\frac{x^2}{a^2} - \frac{k^2}{b^2} = \frac{2z}{c} \implies x^2 = \frac{2a^2}{c}z + \frac{a^2k^2}{b^2}$$

This is a linear equation in z and quadratic in x , which also represents a **Parabola**.

Thus, horizontal sections are hyperbolas, while vertical sections are parabolas. ■

Example 4.1.11. Show that the hyperboloid of one sheet $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ is a ruled surface (i.e., it can be generated by straight lines).

Solution. We rearrange the equation of the hyperboloid as:

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 - \frac{y^2}{b^2}$$

Factoring both sides as difference of squares:

$$\left(\frac{x}{a} - \frac{z}{c}\right)\left(\frac{x}{a} + \frac{z}{c}\right) = \left(1 - \frac{y}{b}\right)\left(1 + \frac{y}{b}\right)$$

This equation can be satisfied by the intersection of two planes for any parameter λ :

$$\frac{x}{a} - \frac{z}{c} = \lambda\left(1 - \frac{y}{b}\right) \quad \text{and} \quad \lambda\left(\frac{x}{a} + \frac{z}{c}\right) = \left(1 + \frac{y}{b}\right)$$

For any value of λ , the intersection of these two planes determines a straight line. Every point on this line satisfies the equation of the hyperboloid. As λ varies, we generate a family of lines (generators) that lie entirely on the surface. Thus, the hyperboloid of one sheet is a ruled surface. ■

4.1.2 Tangent and Tangent Planes

We analyze the geometric contact between lines, planes, and conicoids.

Definition 4.1.12 (Tangent Line). A straight line is a tangent to a conicoid if it meets the surface in two coincident points.

Definition 4.1.13 (Tangent Plane). The locus of all tangent lines passing through a given point on a conicoid is a plane, called the tangent plane at that point.

Theorem 4.1.14 (Equation of Tangent Plane). The equation of the tangent plane to the central conicoid $ax^2 + by^2 + cz^2 = 1$ at the point $P(\alpha, \beta, \gamma)$ on the surface is:

$$a\alpha x + b\beta y + c\gamma z = 1$$

Proof. Let any line be drawn through $P(\alpha, \beta, \gamma)$ with direction cosines λ, μ, ν . The equations of the line are:

$$\frac{x - \alpha}{\lambda} = \frac{y - \beta}{\mu} = \frac{z - \gamma}{\nu} = r$$

The coordinates of a point on this line are $(\alpha + r\lambda, \beta + r\mu, \gamma + r\nu)$. If this point lies on the conicoid $ax^2 + by^2 + cz^2 = 1$, then:

$$a(\alpha + r\lambda)^2 + b(\beta + r\mu)^2 + c(\gamma + r\nu)^2 = 1$$

Expanding and collecting powers of r :

$$r^2(a\lambda^2 + b\mu^2 + c\nu^2) + 2r(a\alpha\lambda + b\beta\mu + c\gamma\nu) + (a\alpha^2 + b\beta^2 + c\gamma^2 - 1) = 0$$

Since $P(\alpha, \beta, \gamma)$ lies on the surface, $a\alpha^2 + b\beta^2 + c\gamma^2 - 1 = 0$. Thus, one root of the quadratic is $r = 0$.

For the line to be a tangent line, it must meet the surface in two coincident points at P ; hence, both roots must be zero. This requires the coefficient of r to vanish:

$$a\alpha\lambda + b\beta\mu + c\gamma\nu = 0$$

To find the locus of all such tangent lines (the tangent plane), we eliminate λ, μ, ν using the relation $\lambda \propto (x - \alpha)$, etc.:

$$a\alpha(x - \alpha) + b\beta(y - \beta) + c\gamma(z - \gamma) = 0$$

$$a\alpha x + b\beta y + c\gamma z - (a\alpha^2 + b\beta^2 + c\gamma^2) = 0$$

Since $a\alpha^2 + b\beta^2 + c\gamma^2 = 1$, the equation becomes $a\alpha x + b\beta y + c\gamma z = 1$. ■

Theorem 4.1.15 (Condition of Tangency). The necessary and sufficient condition for the plane $lx + my + nz = p$ to touch the central conicoid $ax^2 + by^2 + cz^2 = 1$ is:

$$\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$$

Proof. Let the plane $lx + my + nz = p$ touch the conicoid at the point $P(\alpha, \beta, \gamma)$. The equation of the tangent plane at P is given by:

$$a\alpha x + b\beta y + c\gamma z = 1$$

Since this equation and the given plane equation represent the same plane, their coefficients must be proportional:

$$\frac{a\alpha}{l} = \frac{b\beta}{m} = \frac{c\gamma}{n} = \frac{1}{p}$$

Solving for the coordinates of the point of contact P :

$$\alpha = \frac{l}{ap}, \quad \beta = \frac{m}{bp}, \quad \gamma = \frac{n}{cp}$$

Since $P(\alpha, \beta, \gamma)$ lies on the conicoid, these coordinates must satisfy $ax^2 + by^2 + cz^2 = 1$. Substituting the values:

$$a \left(\frac{l}{ap} \right)^2 + b \left(\frac{m}{bp} \right)^2 + c \left(\frac{n}{cp} \right)^2 = 1$$

$$\frac{l^2}{ap^2} + \frac{m^2}{bp^2} + \frac{n^2}{cp^2} = 1$$

Multiplying through by p^2 , we obtain the required condition:

$$\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$$

■

Example 4.1.16. Find the equation of the tangent plane to the conicoid $3x^2 + 5y^2 - 3z^2 = 15$ at the point $(2, 1, 1)$.

Solution. First, we rewrite the equation of the conicoid in the standard form $ax^2 + by^2 + cz^2 = 1$ by dividing by 15:

$$\frac{x^2}{5} + \frac{y^2}{3} - \frac{z^2}{5} = 1$$

Here, the coefficients are $a = \frac{1}{5}, b = \frac{1}{3}, c = -\frac{1}{5}$. The point of contact is $P(\alpha, \beta, \gamma) = (2, 1, 1)$. Using the formula for the tangent plane $a\alpha x + b\beta y + c\gamma z = 1$:

$$\frac{1}{5}(2)x + \frac{1}{3}(1)y - \frac{1}{5}(1)z = 1$$

$$\frac{2x}{5} + \frac{y}{3} - \frac{z}{5} = 1$$

Multiplying the entire equation by 15 to clear the denominators:

$$6x + 5y - 3z = 15$$

Alternatively, using the direct substitution method on $3x^2 + 5y^2 - 3z^2 = 15$:

$$3(x \cdot 2) + 5(y \cdot 1) - 3(z \cdot 1) = 15 \implies 6x + 5y - 3z = 15$$

■

Example 4.1.17. Find the values of k for which the plane $x + 2y + 3z = k$ touches the ellipsoid $x^2 + y^2 + 2z^2 = 6$.

Solution. We first express the ellipsoid in the form $ax^2 + by^2 + cz^2 = 1$:

$$\frac{x^2}{6} + \frac{y^2}{6} + \frac{z^2}{3} = 1$$

Here, $a = \frac{1}{6}, b = \frac{1}{6}, c = \frac{1}{3}$. The given plane is $x + 2y + 3z = k$, so $l = 1, m = 2, n = 3, p = k$. The condition for the plane to touch the conicoid is $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$. Substituting the values:

$$\frac{1^2}{1/6} + \frac{2^2}{1/6} + \frac{3^2}{1/3} = k^2$$

$$6(1) + 6(4) + 3(9) = k^2$$

$$6 + 24 + 27 = k^2$$

$$k^2 = 57 \implies k = \pm\sqrt{57}$$

Thus, the two tangent planes are $x + 2y + 3z = \sqrt{57}$ and $x + 2y + 3z = -\sqrt{57}$.

■

Example 4.1.18. Find the equation of the director sphere of the conicoid $2x^2 + 3y^2 + 4z^2 = 12$.

Solution. The director sphere is the locus of the intersection of three mutually perpendicular tangent planes. First, convert the equation of the conicoid to the standard form $\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1$:

$$\frac{2x^2}{12} + \frac{3y^2}{12} + \frac{4z^2}{12} = 1 \implies \frac{x^2}{6} + \frac{y^2}{4} + \frac{z^2}{3} = 1$$

Here, the semi-axes squared are $A^2 = 6, B^2 = 4, C^2 = 3$. The equation of the director sphere for an ellipsoid is given by $x^2 + y^2 + z^2 = A^2 + B^2 + C^2$. Substituting the values:

$$x^2 + y^2 + z^2 = 6 + 4 + 3$$

$$x^2 + y^2 + z^2 = 13$$

This represents a sphere concentric with the conicoid with radius $\sqrt{13}$. ■

Example 4.1.19. Find the point of contact where the plane $2x - 2y + z + 12 = 0$ touches the conicoid $2x^2 + 2y^2 + z^2 + 3x + 3y + 2z + 6 = 0$.

Solution. This problem involves a general second-degree equation, but finding the center simplifies the process. Let $F(x, y, z) = 2x^2 + 2y^2 + z^2 + 3x + 3y + 2z + 6$. The center of the conicoid is found by setting partial derivatives to zero:

$$\frac{\partial F}{\partial x} = 4x + 3 = 0 \implies x = -3/4$$

$$\frac{\partial F}{\partial y} = 4y + 3 = 0 \implies y = -3/4$$

$$\frac{\partial F}{\partial z} = 2z + 2 = 0 \implies z = -1$$

Center is $C(-0.75, -0.75, -1)$. However, using the tangent plane comparison method is more robust for general cases. Let the point of contact be $P(\alpha, \beta, \gamma)$. The tangent plane at P is:

$$2\alpha x + 2\beta y + \gamma z + \frac{3}{2}(x + \alpha) + \frac{3}{2}(y + \beta) + (z + \gamma) + 6 = 0$$

Grouping terms:

$$\left(2\alpha + \frac{3}{2}\right)x + \left(2\beta + \frac{3}{2}\right)y + (\gamma + 1)z + \left(\frac{3}{2}\alpha + \frac{3}{2}\beta + \gamma + 6\right) = 0$$

This must represent the same plane as $2x - 2y + z + 12 = 0$. Comparing coefficients:

$$\frac{2\alpha + 1.5}{2} = \frac{2\beta + 1.5}{-2} = \frac{\gamma + 1}{1} = \frac{1.5\alpha + 1.5\beta + \gamma + 6}{12} = k$$

From $2\alpha + 1.5 = 2k \implies \alpha = k - 0.75$. From $2\beta + 1.5 = -2k \implies \beta = -k - 0.75$. From $\gamma + 1 = k \implies \gamma = k - 1$. Substituting these into the ratio of the constant term:

$$\frac{1.5(k - 0.75) + 1.5(-k - 0.75) + (k - 1) + 6}{12} = k$$

$$\frac{1.5k - 1.125 - 1.5k - 1.125 + k - 1 + 6}{12} = k$$

$$\frac{k + 2.75}{12} = k \implies k + 2.75 = 12k \implies 11k = 2.75 \implies k = 0.25$$

Now finding (α, β, γ) :

$$\alpha = 0.25 - 0.75 = -0.5$$

$$\beta = -0.25 - 0.75 = -1$$

$$\gamma = 0.25 - 1 = -0.75$$

The point of contact is $(-0.5, -1, -0.75)$. ■

Example 4.1.20. Show that the planes tangent to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ which are parallel to the plane $lx + my + nz = 0$ are given by $lx + my + nz = \pm \sqrt{a^2l^2 + b^2m^2 + c^2n^2}$.

Solution. Any plane parallel to $lx + my + nz = 0$ is of the form:

$$lx + my + nz = p$$

For this plane to be a tangent plane to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, the condition of tangency must be satisfied. The condition is:

$$a^2l^2 + b^2m^2 + c^2n^2 = p^2$$

Solving for p :

$$p = \pm \sqrt{a^2l^2 + b^2m^2 + c^2n^2}$$

Substituting this value of p back into the equation of the plane, we get:

$$lx + my + nz = \pm \sqrt{a^2l^2 + b^2m^2 + c^2n^2}$$

■

4.2 Advanced Properties of Conicoids

In this section, we deepen our analysis of quadric surfaces by investigating orthogonality properties, the relationship between points and planes via polars, and the geometric properties of paraboloids and conjugate diameters.

4.2.1 Director Sphere

The concept of the director sphere is the three-dimensional analogue of the director circle in conic sections.

Definition 4.2.1 (Director Sphere). The director sphere of a central conicoid is the locus of the point of intersection of three mutually perpendicular tangent planes to the conicoid.

Theorem 4.2.2 (Equation of the Director Sphere). For the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, the equation of the director sphere is:

$$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$$

For the hyperboloids, the form is similar, respecting the signs of the coefficients. For instance, for $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$, the director sphere is $x^2 + y^2 + z^2 = a^2 + b^2 - c^2$.

Proof. The director sphere is defined as the locus of the point of intersection of three mutually perpendicular tangent planes to the ellipsoid.

Let the equation of a tangent plane to the ellipsoid be:

$$lx + my + nz = \sqrt{a^2l^2 + b^2m^2 + c^2n^2}$$

where l, m, n are the direction cosines of the normal to the plane.

Let us consider three mutually perpendicular tangent planes intersecting at a point $P(x_1, y_1, z_1)$. Let the direction cosines of the normals to these three planes be (l_1, m_1, n_1) , (l_2, m_2, n_2) , and (l_3, m_3, n_3) .

Since the point $P(x_1, y_1, z_1)$ lies on all three planes, we have the following three equations:

$$l_1x_1 + m_1y_1 + n_1z_1 = \sqrt{a^2l_1^2 + b^2m_1^2 + c^2n_1^2}$$

$$l_2x_1 + m_2y_1 + n_2z_1 = \sqrt{a^2l_2^2 + b^2m_2^2 + c^2n_2^2}$$

$$l_3x_1 + m_3y_1 + n_3z_1 = \sqrt{a^2l_3^2 + b^2m_3^2 + c^2n_3^2}$$

Squaring each equation, we obtain:

$$\begin{aligned}(l_1x_1 + m_1y_1 + n_1z_1)^2 &= a^2l_1^2 + b^2m_1^2 + c^2n_1^2 \\(l_2x_1 + m_2y_1 + n_2z_1)^2 &= a^2l_2^2 + b^2m_2^2 + c^2n_2^2 \\(l_3x_1 + m_3y_1 + n_3z_1)^2 &= a^2l_3^2 + b^2m_3^2 + c^2n_3^2\end{aligned}$$

We now sum these three equations.

Left Hand Side (LHS): Using the property that (l_1, m_1, n_1) , (l_2, m_2, n_2) , and (l_3, m_3, n_3) correspond to three mutually perpendicular directions, they satisfy the orthogonality conditions:

$$\begin{aligned}\sum l_r^2 &= 1, \quad \sum m_r^2 = 1, \quad \sum n_r^2 = 1 \\ \sum l_r m_r &= 0, \quad \sum m_r n_r = 0, \quad \sum n_r l_r = 0\end{aligned}$$

Expanding the squares on the LHS and summing:

$$\sum_{r=1}^3 (l_r x_1 + m_r y_1 + n_r z_1)^2 = x_1^2 (\sum l_r^2) + y_1^2 (\sum m_r^2) + z_1^2 (\sum n_r^2) + 2x_1 y_1 (\sum l_r m_r) + \dots$$

$$\text{LHS} = x_1^2(1) + y_1^2(1) + z_1^2(1) + 0 + 0 + 0 = x_1^2 + y_1^2 + z_1^2$$

Right Hand Side (RHS): Summing the terms on the right:

$$\sum_{r=1}^3 (a^2 l_r^2 + b^2 m_r^2 + c^2 n_r^2) = a^2 (\sum l_r^2) + b^2 (\sum m_r^2) + c^2 (\sum n_r^2)$$

$$\text{RHS} = a^2(1) + b^2(1) + c^2(1) = a^2 + b^2 + c^2$$

Equating LHS and RHS, the locus of the point (x_1, y_1, z_1) is:

$$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$$

■

Example 4.2.3. Find the equation of the director sphere of the ellipsoid $4x^2 + 9y^2 + 16z^2 = 144$.

Solution. First, convert the equation of the ellipsoid to the standard form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ by dividing by 144.

$$\begin{aligned}\frac{4x^2}{144} + \frac{9y^2}{144} + \frac{16z^2}{144} &= 1 \\ \frac{x^2}{36} + \frac{y^2}{16} + \frac{z^2}{9} &= 1\end{aligned}$$

Here, $a^2 = 36$, $b^2 = 16$, and $c^2 = 9$. The equation of the director sphere is given by $x^2 + y^2 + z^2 = a^2 + b^2 + c^2$. Substituting the values:

$$x^2 + y^2 + z^2 = 36 + 16 + 9$$

$$x^2 + y^2 + z^2 = 61$$

This represents a sphere centered at the origin with radius $\sqrt{61}$.

■

Example 4.2.4. Find the equation of the director sphere of the hyperboloid of one sheet given by $2x^2 + 3y^2 - 4z^2 = 12$.

Solution. Divide the equation by 12 to obtain the standard form:

$$\frac{x^2}{6} + \frac{y^2}{4} - \frac{z^2}{3} = 1$$

Comparing this with the standard equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$, we have:

$$a^2 = 6, \quad b^2 = 4, \quad c^2 = 3$$

For a hyperboloid, the coefficients in the director sphere equation respect the signs of the original quadric coefficients. The sum of the squared semi-axes (algebraic sum) determines the squared radius.

$$R^2 = a^2 + b^2 - c^2$$

$$R^2 = 6 + 4 - 3 = 7$$

The equation of the director sphere is:

$$x^2 + y^2 + z^2 = 7$$

Note: If the sum $a^2 + b^2 - c^2$ were negative, the director sphere would be imaginary, implying that no point exists where three mutually perpendicular tangent planes meet. ■

Example 4.2.5. Prove that the director sphere of the sphere $x^2 + y^2 + z^2 = r^2$ is a concentric sphere with radius $r\sqrt{3}$.

Solution. The given sphere is a special case of an ellipsoid where $a = b = c = r$. The equation of the ellipsoid is:

$$\frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} = 1$$

The equation of the director sphere is given by summing the squares of the semi-axes:

$$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$$

Substituting $a^2 = r^2, b^2 = r^2, c^2 = r^2$:

$$x^2 + y^2 + z^2 = r^2 + r^2 + r^2$$

$$x^2 + y^2 + z^2 = 3r^2$$

This represents a sphere centered at the origin with radius $\sqrt{3r^2} = r\sqrt{3}$. This confirms the geometric property that the locus of the intersection of three mutually perpendicular tangent planes to a sphere of radius r is a concentric sphere of radius $r\sqrt{3}$. ■

4.2.2 Normals

While a tangent plane touches the surface at a point, the normal defines the orientation of the surface at that point.

Definition 4.2.6 (Normal). The normal to a surface at a point P is the straight line passing through P and perpendicular to the tangent plane at P .

Theorem 4.2.7 (Equation of the Normal to a Sphere). For the sphere $x^2 + y^2 + z^2 = a^2$, the equation of the normal at $P(x_1, y_1, z_1)$ is:

$$\frac{x - x_1}{x_1} = \frac{y - y_1}{y_1} = \frac{z - z_1}{z_1}$$

Proof. Let the equation of the sphere be $F(x, y, z) \equiv x^2 + y^2 + z^2 - a^2 = 0$. The equation of the tangent plane to the sphere at the point $P(x_1, y_1, z_1)$ is given by $T = 0$, which is:

$$xx_1 + yy_1 + zz_1 = a^2$$

Comparing this with the general equation of a plane $Lx + My + Nz = D$, the direction ratios of the normal to the tangent plane are proportional to the coefficients of x, y, z . Thus, the direction ratios of the normal line at P are (x_1, y_1, z_1) .

The normal is the straight line passing through the point $P(x_1, y_1, z_1)$ and having direction ratios x_1, y_1, z_1 . Using the symmetric form of the equation of a line $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$, we obtain:

$$\frac{x - x_1}{x_1} = \frac{y - y_1}{y_1} = \frac{z - z_1}{z_1}$$

If we equate these ratios to a parameter t , any point on the line is $x = x_1(1 + t), y = y_1(1 + t), z = z_1(1 + t)$. For $t = -1$, we get the point $(0, 0, 0)$. Thus, the normal always passes through the origin (the center of the sphere). ■

Theorem 4.2.8 (Equation of the Normal to a Conicoid). *For the central conicoid $ax^2 + by^2 + cz^2 = 1$, the equation of the normal at $P(x_1, y_1, z_1)$ is:*

$$\frac{x - x_1}{ax_1} = \frac{y - y_1}{by_1} = \frac{z - z_1}{cz_1}$$

Proof. Let the equation of the conicoid be $S \equiv ax^2 + by^2 + cz^2 - 1 = 0$. The equation of the tangent plane to the conicoid at the point $P(x_1, y_1, z_1)$ is:

$$axx_1 + byy_1 + czz_1 = 1$$

The normal to the surface at P is defined as the line passing through P and perpendicular to the tangent plane at P . The direction ratios of the normal to the plane $(ax_1)x + (by_1)y + (cz_1)z - 1 = 0$ are the coefficients of the variables, namely:

$$ax_1, \quad by_1, \quad cz_1$$

Since the normal passes through $P(x_1, y_1, z_1)$, its equation is given by:

$$\frac{x - x_1}{ax_1} = \frac{y - y_1}{by_1} = \frac{z - z_1}{cz_1}$$

Note: If the equation is given in the standard form $\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1$, we substitute $a = 1/A^2, b = 1/B^2, c = 1/C^2$. The equation of the normal becomes:

$$\frac{x - x_1}{x_1/A^2} = \frac{y - y_1}{y_1/B^2} = \frac{z - z_1}{z_1/C^2}$$

Example 4.2.9. Find the equation of the normal to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(1, 2, 3)$.

Solution. The given sphere is $x^2 + y^2 + z^2 = 14$. The point of contact is $P(1, 2, 3)$. The direction ratios of the normal at P are proportional to the coordinates of P , i.e., $1, 2, 3$. The normal passes through $(1, 2, 3)$. Therefore, the equation of the normal is:

$$\frac{x - 1}{1} = \frac{y - 2}{2} = \frac{z - 3}{3}$$

Example 4.2.10. Find the length of the normal chord to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at the point $P(x_1, y_1, z_1)$ intercepted by the coordinate plane $x = 0$.

Solution. The equation of the normal at $P(x_1, y_1, z_1)$ is:

$$\frac{x - x_1}{x_1/a^2} = \frac{y - y_1}{y_1/b^2} = \frac{z - z_1}{z_1/c^2} = r \quad (\text{say})$$

To find the point Q where this line meets the plane $x = 0$, we set the x -coordinate to zero:

$$\frac{0 - x_1}{x_1/a^2} = r \implies r = -a^2$$

The coordinates of Q are obtained by substituting $r = -a^2$ into the line equations:

$$\begin{aligned} x &= 0 \\ y &= y_1 + r \frac{y_1}{b^2} = y_1 - a^2 \frac{y_1}{b^2} = y_1 \left(1 - \frac{a^2}{b^2} \right) \\ z &= z_1 + r \frac{z_1}{c^2} = z_1 - a^2 \frac{z_1}{c^2} = z_1 \left(1 - \frac{a^2}{c^2} \right) \end{aligned}$$

The length of the normal chord PQ is the distance corresponding to the parameter r if the direction cosines were normalized. However, simply using the distance formula:

$$PQ^2 = (x_1 - 0)^2 + (y_1 - y_Q)^2 + (z_1 - z_Q)^2$$

$$PQ^2 = x_1^2 + \left(y_1 \frac{a^2}{b^2} \right)^2 + \left(z_1 \frac{a^2}{c^2} \right)^2$$

$$PQ = x_1 \sqrt{1 + \frac{a^4 y_1^2}{b^4 x_1^2} + \frac{a^4 z_1^2}{c^4 x_1^2}}$$

■

Example 4.2.11. Prove that there are generally six normals which can be drawn from a given point (α, β, γ) to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution. Let the foot of a normal be (x_1, y_1, z_1) . The equation of the normal at this point is:

$$\frac{x - x_1}{x_1/a^2} = \frac{y - y_1}{y_1/b^2} = \frac{z - z_1}{z_1/c^2} = \lambda$$

Since this normal passes through (α, β, γ) , we have:

$$\alpha - x_1 = \frac{\lambda x_1}{a^2} \implies x_1 \left(1 + \frac{\lambda}{a^2} \right) = \alpha \implies x_1 = \frac{a^2 \alpha}{a^2 + \lambda}$$

Similarly, $y_1 = \frac{b^2 \beta}{b^2 + \lambda}$ and $z_1 = \frac{c^2 \gamma}{c^2 + \lambda}$. Since (x_1, y_1, z_1) lies on the ellipsoid:

$$\frac{1}{a^2} \left(\frac{a^2 \alpha}{a^2 + \lambda} \right)^2 + \frac{1}{b^2} \left(\frac{b^2 \beta}{b^2 + \lambda} \right)^2 + \frac{1}{c^2} \left(\frac{c^2 \gamma}{c^2 + \lambda} \right)^2 = 1$$

$$\frac{a^2 \alpha^2}{(a^2 + \lambda)^2} + \frac{b^2 \beta^2}{(b^2 + \lambda)^2} + \frac{c^2 \gamma^2}{(c^2 + \lambda)^2} = 1$$

Clearing denominators by multiplying by $(a^2 + \lambda)^2(b^2 + \lambda)^2(c^2 + \lambda)^2$, we obtain an equation in λ . The degree of λ on the right side (after moving the 1) is $2 + 2 + 2 = 6$. Thus, there is a polynomial equation of degree 6 in λ . Each real root corresponds to a point on the ellipsoid, implying there are six normals (real or imaginary) from a general external point. ■

Example 4.2.12. Find the points on the ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$ at which the normals are parallel to the line $\frac{x}{2} = \frac{y}{-3} = \frac{z}{4}$.

Solution. Let the required point be $P(x_1, y_1, z_1)$. The direction ratios of the normal at P are $\frac{x_1}{4}, \frac{y_1}{9}, \frac{z_1}{16}$. Since the normal is parallel to the given line, these direction ratios must be proportional to 2, -3, 4:

$$\frac{x_1/4}{2} = \frac{y_1/9}{-3} = \frac{z_1/16}{4} = k$$

Solving for the coordinates:

$$\begin{aligned}\frac{x_1}{8} &= k \implies x_1 = 8k \\ \frac{y_1}{-27} &= k \implies y_1 = -27k \\ \frac{z_1}{64} &= k \implies z_1 = 64k\end{aligned}$$

Since P lies on the ellipsoid:

$$\begin{aligned}\frac{(8k)^2}{4} + \frac{(-27k)^2}{9} + \frac{(64k)^2}{16} &= 1 \\ \frac{64k^2}{4} + \frac{729k^2}{9} + \frac{4096k^2}{16} &= 1 \\ 16k^2 + 81k^2 + 256k^2 &= 1 \\ 353k^2 &= 1 \implies k = \pm \frac{1}{\sqrt{353}}\end{aligned}$$

The two points are $\pm \frac{1}{\sqrt{353}}(8, -27, 64)$. ■

Example 4.2.13. If the normal at the point P of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ meets the principal planes (coordinate planes) at points G_1, G_2, G_3 , prove that the ratio of the lengths of the segments is $PG_1 : PG_2 : PG_3 = a^2 : b^2 : c^2$.

Solution. Let the point on the ellipsoid be $P(x_1, y_1, z_1)$. The equation of the normal at P is given by:

$$\frac{x - x_1}{x_1/a^2} = \frac{y - y_1}{y_1/b^2} = \frac{z - z_1}{z_1/c^2}$$

Let these ratios be equal to a parameter λ . Then the coordinates of any point on the normal are:

$$x = x_1 + \lambda \frac{x_1}{a^2}, \quad y = y_1 + \lambda \frac{y_1}{b^2}, \quad z = z_1 + \lambda \frac{z_1}{c^2}$$

The distance D from P to any point defined by λ on this line is:

$$D = |\lambda| \sqrt{\left(\frac{x_1}{a^2}\right)^2 + \left(\frac{y_1}{b^2}\right)^2 + \left(\frac{z_1}{c^2}\right)^2}$$

Let $K = \sqrt{\frac{x_1^2}{a^4} + \frac{y_1^2}{b^4} + \frac{z_1^2}{c^4}}$. Thus, the distance is $|\lambda|K$.

1. The point G_1 is the intersection of the normal with the plane $x = 0$ (yz -plane). Setting $x = 0$:

$$x_1 + \lambda \frac{x_1}{a^2} = 0 \implies 1 + \frac{\lambda}{a^2} = 0 \implies \lambda = -a^2$$

The distance $PG_1 = |-a^2|K = a^2K$.

2. The point G_2 is the intersection with the plane $y = 0$ (zx -plane). Setting $y = 0$:

$$y_1 + \lambda \frac{y_1}{b^2} = 0 \implies \lambda = -b^2$$

The distance $PG_2 = |-b^2|K = b^2K$.

3. The point G_3 is the intersection with the plane $z = 0$ (xy -plane). Setting $z = 0$:

$$z_1 + \lambda \frac{z_1}{c^2} = 0 \implies \lambda = -c^2$$

The distance $PG_3 = |-c^2|K = c^2K$.

The ratio of the lengths is:

$$PG_1 : PG_2 : PG_3 = a^2K : b^2K : c^2K$$

Dividing by the common factor K (which is non-zero), we obtain:

$$PG_1 : PG_2 : PG_3 = a^2 : b^2 : c^2$$

■

4.2.3 Polar and Polar Planes

The theory of poles and polars establishes a reciprocal relationship between points and planes with respect to a conicoid.

Definition 4.2.14 (Polar Plane). If any secant line drawn through a fixed point A meets a conicoid at points Q and R , the locus of the harmonic conjugate of A with respect to Q and R is a plane. This plane is called the **polar plane** of the point A with respect to the conicoid. The point A is called the **pole**.

Theorem 4.2.15 (Equation of the Polar Plane). The equation of the polar plane of the point $A(\alpha, \beta, \gamma)$ with respect to the conicoid $ax^2 + by^2 + cz^2 = 1$ is:

$$a\alpha x + b\beta y + c\gamma z = 1$$

Proof. Let the given conicoid be $S \equiv ax^2 + by^2 + cz^2 - 1 = 0$. Let $A(\alpha, \beta, \gamma)$ be the fixed point (the pole). Let a variable line drawn through A intersect the conicoid at points Q and R . Let $P(x, y, z)$ be a point on this line such that P is the harmonic conjugate of A with respect to Q and R . By the definition of the polar plane, the locus of P is the required plane.

Since (A, P, Q, R) is a harmonic range, the points Q and R divide the segment AP internally and externally in the same ratio. Let this ratio be $\lambda : 1$. The coordinates of the points dividing the segment AP in the ratio $\lambda : 1$ are:

$$\left(\frac{\lambda x + \alpha}{\lambda + 1}, \frac{\lambda y + \beta}{\lambda + 1}, \frac{\lambda z + \gamma}{\lambda + 1} \right)$$

Since Q and R lie on the conicoid, these coordinates must satisfy the equation of the conicoid for distinct values of λ :

$$a \left(\frac{\lambda x + \alpha}{\lambda + 1} \right)^2 + b \left(\frac{\lambda y + \beta}{\lambda + 1} \right)^2 + c \left(\frac{\lambda z + \gamma}{\lambda + 1} \right)^2 = 1$$

Multiplying by $(\lambda + 1)^2$ and expanding:

$$a(\lambda^2 x^2 + 2\lambda x\alpha + \alpha^2) + b(\lambda^2 y^2 + 2\lambda y\beta + \beta^2) + c(\lambda^2 z^2 + 2\lambda z\gamma + \gamma^2) = \lambda^2 + 2\lambda + 1$$

Arranging this as a quadratic equation in λ :

$$\lambda^2(ax^2 + by^2 + cz^2 - 1) + 2\lambda(a\alpha x + b\beta y + c\gamma z - 1) + (a\alpha^2 + b\beta^2 + c\gamma^2 - 1) = 0$$

The roots of this quadratic equation correspond to the points Q and R . Since Q and R divide AP internally and externally in the same ratio, the roots must be equal in magnitude but opposite in sign (i.e., $\lambda_1 = -\lambda_2$). Therefore, the sum of the roots must be zero.

The condition for the sum of roots to be zero is that the coefficient of the linear term (λ) vanishes:

$$2(a\alpha x + b\beta y + c\gamma z - 1) = 0$$

Dividing by 2, we obtain the equation of the locus of $P(x, y, z)$:

$$a\alpha x + b\beta y + c\gamma z = 1$$

■

Example 4.2.16. Find the polar plane of the point $(2, 3, 4)$ with respect to the ellipsoid $3x^2 + 2y^2 + z^2 = 11$.

Solution. First, write the equation of the ellipsoid in the form $ax^2 + by^2 + cz^2 = 1$. Dividing by 11:

$$\frac{3}{11}x^2 + \frac{2}{11}y^2 + \frac{1}{11}z^2 = 1$$

Here, $a = \frac{3}{11}, b = \frac{2}{11}, c = \frac{1}{11}$. The pole is $(\alpha, \beta, \gamma) = (2, 3, 4)$. Using the polar plane formula $a\alpha x + b\beta y + c\gamma z = 1$:

$$\frac{3}{11}(2)x + \frac{2}{11}(3)y + \frac{1}{11}(4)z = 1$$

$$\frac{6x}{11} + \frac{6y}{11} + \frac{4z}{11} = 1$$

Multiplying by 11 to clear the fraction:

$$6x + 6y + 4z = 11$$

This is the equation of the polar plane. ■

Example 4.2.17. Find the pole of the plane $2x + 3y + 4z = 6$ with respect to the sphere $x^2 + y^2 + z^2 = 9$.

Solution. Let the pole be $P(\alpha, \beta, \gamma)$. The equation of the polar plane of P with respect to the sphere $x^2 + y^2 + z^2 = 9$ (which can be written as $\frac{x^2}{9} + \frac{y^2}{9} + \frac{z^2}{9} = 1$) is:

$$\alpha x + \beta y + \gamma z = 9$$

We are given the plane equation $2x + 3y + 4z = 6$. Since both equations represent the same plane, their coefficients must be proportional:

$$\frac{\alpha}{2} = \frac{\beta}{3} = \frac{\gamma}{4} = \frac{9}{6}$$

From the constant ratio $\frac{9}{6} = \frac{3}{2}$:

$$\frac{\alpha}{2} = \frac{3}{2} \implies \alpha = 3$$

$$\frac{\beta}{3} = \frac{3}{2} \implies \beta = \frac{9}{2}$$

$$\frac{\gamma}{4} = \frac{3}{2} \implies \gamma = 6$$

Thus, the pole is $\left(3, \frac{9}{2}, 6\right)$. ■

Example 4.2.18. Prove that the points $A(1, 2, -1)$ and $B(2, -1, 1)$ are conjugate with respect to the ellipsoid $2x^2 + 3y^2 + 4z^2 = 5$.

Solution. Two points A and B are conjugate if the polar plane of A passes through B (reciprocally, the polar of B passes through A). Let us find the polar plane of $A(1, 2, -1)$ with respect to $2x^2 + 3y^2 + 4z^2 = 5$. Using the replacement rule ($x^2 \rightarrow \alpha x$, etc.):

$$2(1)x + 3(2)y + 4(-1)z = 5$$

$$2x + 6y - 4z = 5$$

Now, we check if point $B(2, -1, 1)$ lies on this plane. Substitute the coordinates of B into the left-hand side:

$$\text{LHS} = 2(2) + 6(-1) - 4(1)$$

$$\text{LHS} = 4 - 6 - 4 = -6$$

Since $-6 \neq 5$, the point B does not lie on the polar plane of A . Therefore, the points are **not** conjugate.

Self-Correction/Verification: For the points to be conjugate, the condition $2x_1x_2 + 3y_1y_2 + 4z_1z_2 = 5$ must hold.

$$2(1)(2) + 3(2)(-1) + 4(-1)(1) = 4 - 6 - 4 = -6 \neq 5$$

The statement in the example prompt is disproven; the points are not conjugate. ■

Definition 4.2.19. • **Conjugate Points:** Two points are said to be conjugate with respect to a conicoid if the polar plane of one passes through the other.

• **Conjugate Planes:** Two planes are said to be conjugate if the pole of one lies on the other.

Example 4.2.20. Consider the ellipsoid given by the equation $x^2 + 2y^2 + 3z^2 = 6$.

1. Show that the points $P(1, 1, 1)$ and $Q(3, 0, 1)$ are conjugate points with respect to this ellipsoid.
2. Verify that the polar planes of P and Q are conjugate planes.

Solution. **Conjugate Points**

By definition, two points are conjugate if the polar plane of the first point passes through the second point.

The equation of the ellipsoid is:

$$x^2 + 2y^2 + 3z^2 = 6$$

The equation of the polar plane of the point $P(1, 1, 1)$ is given by $T = 0$ (using the substitution $x^2 \rightarrow xx_1, y^2 \rightarrow yy_1$, etc.):

$$x(1) + 2y(1) + 3z(1) = 6$$

$$x + 2y + 3z = 6$$

Now, we check if the point $Q(3, 0, 1)$ lies on this polar plane. Substituting the coordinates of Q into the left-hand side:

$$(3) + 2(0) + 3(1) = 3 + 0 + 3 = 6$$

Since the coordinates of Q satisfy the equation of the polar plane of P , the polar plane of P passes through Q . Therefore, P and Q are conjugate points.

Conjugate Planes

Two planes are conjugate if the pole of one lies on the other.

Let Π_1 be the polar plane of P (which we found above):

$$\Pi_1 : x + 2y + 3z = 6$$

The pole of Π_1 is, by construction, the point $P(1, 1, 1)$.

Now, let us find Π_2 , the polar plane of $Q(3, 0, 1)$. Using the polar formula again:

$$x(3) + 2y(0) + 3z(1) = 6$$

$$3x + 3z = 6 \implies x + z = 2$$

For the planes Π_1 and Π_2 to be conjugate, the pole of Π_1 (which is P) must lie on Π_2 .

Substitute the coordinates of $P(1, 1, 1)$ into the equation of Π_2 :

$$1 + 1 = 2$$

Since the equation holds true, the pole of Π_1 lies on Π_2 . Thus, the polar planes of two conjugate points are themselves conjugate planes. ■

4.2.4 Enveloping Cone and Cylinder

These surfaces are generated by tangent lines drawn from a specific reference.

Definition 4.2.21 (Enveloping Cone). The enveloping cone of a conicoid is the locus of tangent lines drawn to the conicoid from a fixed point (the vertex).

Theorem 4.2.22 (Equation of the Enveloping Cone). *The equation of the enveloping cone of the conicoid $S \equiv ax^2 + by^2 + cz^2 - 1 = 0$ with vertex at $P(x_1, y_1, z_1)$ is given by:*

$$SS_1 = T^2$$

Proof. Let $P(x_1, y_1, z_1)$ be the vertex of the cone. Let $Q(x, y, z)$ be any point on the surface of the cone. The line joining P and Q is a generator of the cone and must, therefore, be tangent to the conicoid.

Consider a point R lying on the line segment PQ which divides it in the ratio $k : 1$. The coordinates of R are:

$$\left(\frac{kx + x_1}{k + 1}, \frac{ky + y_1}{k + 1}, \frac{kz + z_1}{k + 1} \right)$$

If the line PQ intersects the conicoid, the point R must satisfy the equation $ax^2 + by^2 + cz^2 = 1$. Substituting the coordinates of R :

$$a \left(\frac{kx + x_1}{k + 1} \right)^2 + b \left(\frac{ky + y_1}{k + 1} \right)^2 + c \left(\frac{kz + z_1}{k + 1} \right)^2 = 1$$

Multiplying by $(k + 1)^2$ and rearranging terms based on powers of k :

$$k^2(ax^2 + by^2 + cz^2 - 1) + 2k(axy_1 + byx_1 + czx_1 - 1) + (ax_1^2 + by_1^2 + cz_1^2 - 1) = 0$$

Using the notation S, S_1 , and T :

$$\begin{aligned} S &= ax^2 + by^2 + cz^2 - 1 \\ S_1 &= ax_1^2 + by_1^2 + cz_1^2 - 1 \\ T &= axx_1 + byy_1 + czz_1 - 1 \end{aligned}$$

The quadratic equation in k becomes:

$$k^2S + 2kT + S_1 = 0$$

The roots of this equation correspond to the points where the line PQ meets the conicoid. Since the line is a tangent (a generator of the enveloping cone), it meets the surface in two coincident points. Therefore, the quadratic equation must have equal roots.

The condition for equal roots is that the discriminant vanishes:

$$(2T)^2 - 4(S)(S_1) = 0$$

$$4T^2 = 4SS_1 \implies SS_1 = T^2$$

This relation holds for any point $Q(x, y, z)$ on the enveloping cone. Hence, it is the equation of the cone. ■

Example 4.2.23. Find the equation of the enveloping cone of the sphere $x^2 + y^2 + z^2 = 11$ whose vertex is at $(2, 4, 1)$.

Solution. Here, the equation of the sphere is $S \equiv x^2 + y^2 + z^2 - 11 = 0$. The vertex is $P(2, 4, 1)$. We calculate S_1 by substituting the vertex coordinates into S :

$$S_1 = 2^2 + 4^2 + 1^2 - 11 = 4 + 16 + 1 - 11 = 10$$

We calculate T using the polar substitution:

$$T = x(2) + y(4) + z(1) - 11 = 2x + 4y + z - 11$$

The equation of the enveloping cone is $SS_1 = T^2$:

$$10(x^2 + y^2 + z^2 - 11) = (2x + 4y + z - 11)^2$$

$$10x^2 + 10y^2 + 10z^2 - 110 = (2x + 4y + z - 11)^2$$

(This form is generally sufficient, but one may expand to simplify further). ■

Example 4.2.24. Find the enveloping cone of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ with its vertex at the origin. What does the result imply?

Solution. Here $S \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$. The vertex is $P(0, 0, 0)$. Calculate S_1 :

$$S_1 = \frac{0}{a^2} + \frac{0}{b^2} + \frac{0}{c^2} - 1 = -1$$

Calculate T :

$$T = \frac{x(0)}{a^2} + \frac{y(0)}{b^2} + \frac{z(0)}{c^2} - 1 = -1$$

The equation $SS_1 = T^2$ becomes:

$$S(-1) = (-1)^2 \implies -S = 1 \implies S = -1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = -1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

Since x, y, z are real, a sum of squares equalling zero implies $x = 0, y = 0, z = 0$. The cone degenerates to a single point (the origin). This is consistent with geometry: since the origin is the center of the ellipsoid (inside the surface), no real tangents can be drawn from it to the surface. ■

Example 4.2.25. Find the locus of the vertex of the enveloping cone of the conicoid $ax^2 + by^2 + cz^2 = 1$ which has three mutually perpendicular generators.

Solution. Let the vertex of the enveloping cone be $P(x_1, y_1, z_1)$. The equation of the enveloping cone with vertex P is given by $SS_1 = T^2$. Here:

$$S \equiv ax^2 + by^2 + cz^2 - 1$$

$$S_1 \equiv ax_1^2 + by_1^2 + cz_1^2 - 1$$

$$T \equiv axx_1 + byy_1 + czz_1 - 1$$

Substituting these into the equation:

$$(ax^2 + by^2 + cz^2 - 1)S_1 = (axx_1 + byy_1 + czz_1 - 1)^2$$

We need to determine the condition for this cone to have three mutually perpendicular generators. The condition is that the sum of the coefficients of x^2 , y^2 , and z^2 in the cone's equation must be zero.

Let us collect the coefficients of x^2 , y^2 , z^2 from the expanded equation.

1. **Coefficient of x^2 (A):** From the LHS, the term is $S_1(ax^2)$, so the coefficient is aS_1 . From the RHS, the term is $(ax_1x)^2 = a^2x_1^2x^2$, so the coefficient is $a^2x_1^2$. Thus, $A = aS_1 - a^2x_1^2$.

2. **Coefficient of y^2 (B):** Similarly, $B = bS_1 - b^2y_1^2$.

3. **Coefficient of z^2 (C):** Similarly, $C = cS_1 - c^2z_1^2$.

The condition for mutually perpendicular generators is $A + B + C = 0$:

$$(aS_1 - a^2x_1^2) + (bS_1 - b^2y_1^2) + (cS_1 - c^2z_1^2) = 0$$

Grouping terms involving S_1 :

$$S_1(a + b + c) - (a^2x_1^2 + b^2y_1^2 + c^2z_1^2) = 0$$

Substitute the expression for $S_1 = ax_1^2 + by_1^2 + cz_1^2 - 1$:

$$(ax_1^2 + by_1^2 + cz_1^2 - 1)(a + b + c) - (a^2x_1^2 + b^2y_1^2 + c^2z_1^2) = 0$$

Expanding the first term:

$$[x_1^2(a^2 + ab + ac) + y_1^2(ab + b^2 + bc) + z_1^2(ac + bc + c^2) - (a + b + c)] - (a^2x_1^2 + b^2y_1^2 + c^2z_1^2) = 0$$

Canceling the common terms $a^2x_1^2$, $b^2y_1^2$, and $c^2z_1^2$:

$$x_1^2(ab + ac) + y_1^2(ab + bc) + z_1^2(ac + bc) - (a + b + c) = 0$$

Generalizing the locus by replacing (x_1, y_1, z_1) with (x, y, z) , the required equation is:

$$x^2(ab + ac) + y^2(ab + bc) + z^2(ac + bc) = a + b + c$$

Dividing the entire equation by abc yields a more symmetric form:

$$x^2\left(\frac{1}{c} + \frac{1}{b}\right) + y^2\left(\frac{1}{c} + \frac{1}{a}\right) + z^2\left(\frac{1}{b} + \frac{1}{a}\right) = \frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab}$$

■

Definition 4.2.26 (Enveloping Cylinder). The enveloping cylinder is the locus of tangent lines to the conicoid which are parallel to a given fixed line. (This was analytically defined in the previous section).

4.2.5 The Paraboloids

Paraboloids are non-central conicoids. They are unbounded surfaces characterized by the absence of a center of symmetry.

Definition 4.2.27 (Elliptic Paraboloid). The standard equation is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$$

Properties:

- Sections parallel to the xy -plane ($z = k$) are ellipses.
- Sections parallel to the yz -plane or zx -plane are parabolas.
- The surface lies entirely on one side of the tangent plane at the vertex (origin).

Definition 4.2.28 (Hyperbolic Paraboloid). The standard equation is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$$

Properties:

- Sections parallel to the xy -plane ($z = k$) are hyperbolas.
- Sections parallel to the other coordinate planes are parabolas.
- It is a ruled surface (can be generated by straight lines).
- It is shaped like a saddle.

Example 4.2.29. Identify the surface given by the equation $4x^2 + 9y^2 = 72z$. Describe the shape of the section cut by the plane $z = 2$ and calculate its area.

Solution. First, we rewrite the given equation in the standard form.

$$4x^2 + 9y^2 = 72z$$

Dividing by 72, we obtain:

$$\frac{x^2}{18} + \frac{y^2}{8} = z$$

This can be written as:

$$\frac{x^2}{(\sqrt{18})^2} + \frac{y^2}{(\sqrt{8})^2} = \frac{2z}{2}$$

Comparing this with the standard equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$, we identify the surface as an **Elliptic Paraboloid**.

Analysis of the Section $z = 2$: To find the section, we substitute $z = 2$ into the equation of the surface:

$$4x^2 + 9y^2 = 72(2)$$

$$4x^2 + 9y^2 = 144$$

Dividing by 144 to normalize:

$$\frac{x^2}{36} + \frac{y^2}{16} = 1$$

This represents an **Ellipse** with semi-major axis $A = 6$ and semi-minor axis $B = 4$.

The area of this elliptical section is given by πAB :

$$\text{Area} = \pi(6)(4) = 24\pi \text{ square units.}$$

Geometric Interpretation: The surface is cup-shaped, opening upwards along the z -axis. Any horizontal cross-section (where $z > 0$) yields an ellipse, while vertical cross-sections (like $x = 0$ giving $y^2 = 8z$) yield parabolas. ■

Example 4.2.30. Analyze the surface given by $x^2 - y^2 + 2z = 0$ by examining its intersections with the coordinate planes and the plane $z = 2$.

Solution. We rearrange the equation to isolate the linear term:

$$x^2 - y^2 = -2z$$

Or, in standard form:

$$\frac{x^2}{1} - \frac{y^2}{1} = \frac{2z}{-1}$$

This matches the form of a **Hyperbolic Paraboloid** (saddle shape).

1. Intersection with the plane $z = 2$: Substituting $z = 2$:

$$x^2 - y^2 = -4 \implies y^2 - x^2 = 4$$

$$\frac{y^2}{4} - \frac{x^2}{4} = 1$$

This is a **Rectangular Hyperbola** with its transverse axis along the y -direction.

2. Intersection with the plane $y = 0$ (zx -plane): Substituting $y = 0$:

$$x^2 = -2z$$

This is a **Parabola** with its vertex at the origin and axis along the negative z -axis (opening downwards).

3. Intersection with the plane $x = 0$ (yz -plane): Substituting $x = 0$:

$$-y^2 = -2z \implies y^2 = 2z$$

This is a **Parabola** with its vertex at the origin and axis along the positive z -axis (opening upwards).

Conclusion: The surface is saddle-shaped. Along the x -axis, it curves downwards, while along the y -axis, it curves upwards. The intersection of these two principal parabolas at the origin forms the "saddle point." ■

4.2.6 Conjugate Diameters and Diametral Planes

In the geometry of conicoids, the concepts of diametral planes and conjugate diameters generalize the familiar properties of chords and diameters from conic sections to three dimensions.

Definition 4.2.31 (Diametral Plane). The locus of the midpoints of a system of parallel chords of a conicoid is a plane passing through the center, known as a diametral plane.

For the central ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, the equation of the diametral plane which bisects all chords parallel to the line with direction ratios l, m, n is:

$$\frac{lx}{a^2} + \frac{my}{b^2} + \frac{nz}{c^2} = 0$$

Definition 4.2.32 (Conjugate Diameters). Three diameters of a central conicoid are said to be conjugate diameters if the plane containing any two of them is the diametral plane bisecting chords parallel to the third. Consequently, the tangent planes at the extremities of one diameter are parallel to the plane containing the other two.

Fundamental Properties

Theorem 4.2.33 (Properties of Conjugate Diameters). Let $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$, and $R(x_3, y_3, z_3)$ be the extremities of three conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. The following relations hold:

1. Coordinate Relations:

$$x_1^2 + x_2^2 + x_3^2 = a^2, \quad y_1^2 + y_2^2 + y_3^2 = b^2, \quad z_1^2 + z_2^2 + z_3^2 = c^2$$

$$x_1y_1 + x_2y_2 + x_3y_3 = 0, \quad y_1z_1 + y_2z_2 + y_3z_3 = 0, \quad z_1x_1 + z_2x_2 + z_3x_3 = 0$$

2. Sum of Squares Invariant: The sum of the squares of the lengths of three conjugate semi-diameters is constant.

$$OP^2 + OQ^2 + OR^2 = a^2 + b^2 + c^2$$

3. Volume Invariant: The volume of the parallelepiped formed by three conjugate semi-diameters as coterminal edges is constant.

$$6 \times (\text{Volume of tetrahedron } OPQR) = abc$$

Proof. 1. Coordinate Relations: Let us transform the coordinates to a normalized system by setting $X_r = \frac{x_r}{a}, Y_r = \frac{y_r}{b}, Z_r = \frac{z_r}{c}$ for $r = 1, 2, 3$. Since the points P, Q, R lie on the ellipsoid, we have:

$$X_1^2 + Y_1^2 + Z_1^2 = 1, \quad X_2^2 + Y_2^2 + Z_2^2 = 1, \quad X_3^2 + Y_3^2 + Z_3^2 = 1$$

Furthermore, since Q lies on the diametral plane of OP (which is the polar plane of P passing through the center), the points satisfy the conjugacy condition:

$$\frac{x_1x_2}{a^2} + \frac{y_1y_2}{b^2} + \frac{z_1z_2}{c^2} = 0 \implies X_1X_2 + Y_1Y_2 + Z_1Z_2 = 0$$

Similar relations hold for the pairs (Q, R) and (R, P) .

Consider the matrix M formed by these normalized coordinates:

$$M = \begin{pmatrix} X_1 & Y_1 & Z_1 \\ X_2 & Y_2 & Z_2 \\ X_3 & Y_3 & Z_3 \end{pmatrix}$$

The rows of M are mutually orthogonal unit vectors. Thus, M is an orthogonal matrix, implying $MM^T = I$. A property of orthogonal matrices is that $M^TM = I$ as well. This implies the columns are also mutually orthogonal unit vectors. Equating the diagonal elements of M^TM to 1:

$$X_1^2 + X_2^2 + X_3^2 = 1 \implies \frac{x_1^2}{a^2} + \frac{x_2^2}{a^2} + \frac{x_3^2}{a^2} = 1 \implies \sum x_r^2 = a^2$$

Similarly, $\sum y_r^2 = b^2$ and $\sum z_r^2 = c^2$. Equating the off-diagonal elements of M^TM to 0:

$$X_1Y_1 + X_2Y_2 + X_3Y_3 = 0 \implies \frac{1}{ab} \sum x_r y_r = 0 \implies \sum x_r y_r = 0$$

2. Sum of Squares: The sum of the squares of the lengths is:

$$\sum OP^2 = \sum (x_r^2 + y_r^2 + z_r^2) = \sum x_r^2 + \sum y_r^2 + \sum z_r^2$$

Using the relations derived above:

$$\sum OP^2 = a^2 + b^2 + c^2$$

3. Volume Invariant: The volume V of the parallelepiped formed by vectors OP, OQ, OR is the absolute value of the determinant of their coordinates:

$$V = \left| \det \begin{pmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{pmatrix} \right| = \left| \det \begin{pmatrix} aX_1 & bY_1 & cZ_1 \\ aX_2 & bY_2 & cZ_2 \\ aX_3 & bY_3 & cZ_3 \end{pmatrix} \right|$$

Factoring out a, b, c :

$$V = abc |\det(M)|$$

Since M is orthogonal, $\det(M) = \pm 1$. Thus, $V = abc$. ■

Example 4.2.34. Find the equation of the diametral plane of the ellipsoid $3x^2 + 2y^2 + z^2 = 1$ which bisects chords parallel to the line $x = y = z$.

Solution. We rewrite the ellipsoid as $\frac{x^2}{1/3} + \frac{y^2}{1/2} + \frac{z^2}{1} = 1$. Here, $a^2 = 1/3, b^2 = 1/2, c^2 = 1$. The chords are parallel to the line with direction ratios $l = 1, m = 1, n = 1$. The equation of the corresponding diametral plane is $\frac{lx}{a^2} + \frac{my}{b^2} + \frac{nz}{c^2} = 0$. Substituting the values:

$$\frac{1 \cdot x}{1/3} + \frac{1 \cdot y}{1/2} + \frac{1 \cdot z}{1} = 0 \implies 3x + 2y + z = 0$$

■

Example 4.2.35. In the ellipsoid $x^2 + 2y^2 + 3z^2 = 6$, find the length of the third conjugate semi-diameter if the squares of the lengths of the first two are 4 and 3.

Solution. Standardizing the equation $\frac{x^2}{6} + \frac{y^2}{3} + \frac{z^2}{2} = 1$, we find the semi-axes squared are $a^2 = 6, b^2 = 3, c^2 = 2$. Let the lengths of the conjugate semi-diameters be r_1, r_2, r_3 . Using the invariant property $r_1^2 + r_2^2 + r_3^2 = a^2 + b^2 + c^2$:

$$4 + 3 + r_3^2 = 6 + 3 + 2$$

$$7 + r_3^2 = 11 \implies r_3^2 = 4 \implies r_3 = 2$$

■

Example 4.2.36. Prove that the sum of the squares of the projections of three conjugate semi-diameters of an ellipsoid on any straight line is constant.

Solution. Let the direction cosines of the straight line be λ, μ, ν . Let the extremities of the conjugate semi-diameters be P_1, P_2, P_3 with coordinates (x_r, y_r, z_r) . The projection of OP_r on the line is $p_r = \lambda x_r + \mu y_r + \nu z_r$. The sum of the squares of the projections is:

$$\sum_{r=1}^3 p_r^2 = \sum_{r=1}^3 (\lambda x_r + \mu y_r + \nu z_r)^2$$

Expanding the terms:

$$\sum p_r^2 = \lambda^2 \sum x_r^2 + \mu^2 \sum y_r^2 + \nu^2 \sum z_r^2 + 2\lambda\mu \sum x_r y_r + 2\mu\nu \sum y_r z_r + 2\nu\lambda \sum z_r x_r$$

Applying the coordinate relations from the main theorem ($\sum x_r^2 = a^2$ and cross terms vanish):

$$\sum p_r^2 = a^2 \lambda^2 + b^2 \mu^2 + c^2 \nu^2$$

Since a, b, c are constants of the ellipsoid and λ, μ, ν depend only on the fixed line, the sum is constant. ■

Example 4.2.37. Find the locus of the centroid of the triangle formed by the extremities of three conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution. Let the extremities be $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$, and $R(x_3, y_3, z_3)$. Let $G(\bar{x}, \bar{y}, \bar{z})$ be the centroid.

$$3\bar{x} = x_1 + x_2 + x_3, \quad 3\bar{y} = y_1 + y_2 + y_3, \quad 3\bar{z} = z_1 + z_2 + z_3$$

We utilize the normalized coordinates $X_r = x_r/a$, etc. The vectors $\mathbf{u} = (X_1, Y_1, Z_1)$, $\mathbf{v} = (X_2, Y_2, Z_2)$, and $\mathbf{w} = (X_3, Y_3, Z_3)$ are mutually orthogonal unit vectors. Consider the vector sum $\mathbf{S} = \mathbf{u} + \mathbf{v} + \mathbf{w}$. The components of $\mathbf{S}$ are:

$$\mathbf{S} = \left(\frac{x_1 + x_2 + x_3}{a}, \frac{y_1 + y_2 + y_3}{b}, \frac{z_1 + z_2 + z_3}{c} \right) = \left(\frac{3\bar{x}}{a}, \frac{3\bar{y}}{b}, \frac{3\bar{z}}{c} \right)$$

Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are orthogonal unit vectors, the square of the magnitude of their sum is:

$$|\mathbf{S}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 + |\mathbf{w}|^2 + 2(\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{w} + \mathbf{w} \cdot \mathbf{u})$$

$$|\mathbf{S}|^2 = 1 + 1 + 1 + 0 = 3$$

Equating the magnitude from the components:

$$\left(\frac{3\bar{x}}{a}\right)^2 + \left(\frac{3\bar{y}}{b}\right)^2 + \left(\frac{3\bar{z}}{c}\right)^2 = 3$$

Simplifying, the locus of the centroid is the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = \frac{1}{3}$$

■

Analytical Challenges – Sharpen your concepts

- Find the equation of the cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and whose guiding curve is the ellipse $x^2 + 2y^2 = 1, z = 0$.
- Find the equation of the right circular cylinder of radius 3, whose axis passes through the point $(2, -1, 0)$ and has direction ratios proportional to 2, 1, 2.
- Determine the equation of the cylinder generated by lines parallel to the z -axis passing through the curve of intersection of the surface $ax^2 + by^2 + cz^2 = 1$ and the plane $lx + my + nz = p$.
- Find the equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = 25$ having generators parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$.
- Find the equation of the enveloping cylinder of the conicoid $2x^2 + 3y^2 + 4z^2 = 6$ whose generators are parallel to the coordinate axis OZ .
- Identify the type of conicoid represented by the following equations:
 - $2x^2 - 3y^2 - 2z = 0$
 - $4x^2 + 9y^2 + 36z^2 = 36$
 - $x^2 - y^2 - z^2 = 1$
- Find the equation of the tangent plane to the ellipsoid $2x^2 + 6y^2 + 3z^2 = 11$ at the point $(1, 1, 1)$.
- Find the values of k for which the plane $x + 2y + 3z = k$ touches the ellipsoid $x^2 + y^2 + 2z^2 = 6$.
- Prove that the plane $2x - 2y + z + 12 = 0$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z - 3 = 0$ and find the point of contact.
- Find the equation of the director sphere of the conicoid $3x^2 + 2y^2 + 6z^2 = 18$.
- If the director sphere of the hyperboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ passes through the origin, prove that $a^2 + b^2 = c^2$.
- Find the equation of the normal to the ellipsoid $3x^2 + 5y^2 + 7z^2 = 15$ at the point $(1, 1, 1)$.
- Find the length of the normal chord of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at the point (x_1, y_1, z_1) intercepted by the coordinate plane $z = 0$.

14. Find the pole of the plane $x + 2y + 3z = 4$ with respect to the ellipsoid $x^2 + 3y^2 + 2z^2 = 6$.
15. Verify whether the points $(3, 2, -1)$ and $(2, -1, 3)$ are conjugate with respect to the conicoid $x^2 + 2y^2 + 3z^2 = 4$.
16. Find the equation of the enveloping cone of the sphere $x^2 + y^2 + z^2 = 11$ whose vertex is at the point $(2, 4, 1)$.
17. Find the condition that the plane $lx + my + nz = p$ may touch the paraboloid $\frac{x^2}{a} + \frac{y^2}{b} = 2z$.
18. Show that the locus of the point of intersection of three mutually perpendicular tangent planes to the paraboloid $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2z$ is the plane $z + \frac{a^2 + b^2}{2} = 0$.
19. If $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$, and $R(x_3, y_3, z_3)$ are the extremities of three conjugate semi-diameters of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, show that the equation of the plane PQR is $\frac{x}{a^2}(x_1 + x_2 + x_3) + \frac{y}{b^2}(y_1 + y_2 + y_3) + \frac{z}{c^2}(z_1 + z_2 + z_3) = 1$.
20. The lengths of three conjugate semi-diameters of an ellipsoid are r_1, r_2, r_3 . If the volume of the ellipsoid is V , prove that the volume of the parallelepiped formed by these semi-diameters is $\frac{3V}{4\pi}$.
21. If $2r$ is the length of a diameter of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ parallel to the tangent line at the point (x_1, y_1, z_1) with direction cosines proportional to l, m, n , prove that $r^2 = \frac{l^2 + m^2 + n^2}{\frac{l^2}{a^2} + \frac{m^2}{b^2} + \frac{n^2}{c^2}}$.

Chapter 5

Tutorial

5.1 The Plane

In this section, we study the analytical representation of a plane in three-dimensional space. We explore various forms of the equation of a plane, methods to convert between them, and the condition for a plane to pass through specific points.

5.1.1 Intercept Form

This form is useful when the points where the plane cuts the coordinate axes are known.

Definition 5.1.1 (Intercepts). If a plane meets the coordinate axes OX, OY, OZ at points $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$ respectively, then a, b, c are called the intercepts of the plane on the axes.

Theorem 5.1.2 (Equation in Intercept Form). *The equation of a plane making intercepts a, b, c on the coordinate axes is given by:*

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

5.1.2 Normal Form

The normal form is defined by the perpendicular distance from the origin and the orientation of the normal vector.

Theorem 5.1.3 (Equation in Normal Form). *The equation of a plane in terms of the length of the perpendicular p from the origin to the plane and the direction cosines l, m, n of that perpendicular is:*

$$lx + my + nz = p$$

where p is always taken to be positive.

Proof. Let ON be the perpendicular from the origin O to the plane, such that the length $ON = p$. Let the direction cosines of ON be l, m, n .

Consider any arbitrary point $P(x, y, z)$ lying on the plane. Join O to P . The projection of the line segment OP on the line ON must be equal to the length ON (since the angle between ON and any line lying in the plane passing through N is 90°).

The coordinates of O are $(0, 0, 0)$ and P are (x, y, z) . The projection of the segment joining $(0, 0, 0)$ and (x, y, z) on a line with direction cosines l, m, n is given by:

$$\text{Projection} = l(x - 0) + m(y - 0) + n(z - 0)$$

Since this projection is equal to the perpendicular distance p :

$$lx + my + nz = p$$

This equation holds for the coordinates of any point P on the plane, representing the locus of the plane. ■

5.1.3 Reduction of General Form to Normal Form

The general equation of the first degree in x, y, z represents a plane:

$$Ax + By + Cz + D = 0$$

To reduce this to the normal form $lx + my + nz = p$:

1. Transpose the constant term to the right-hand side: $Ax + By + Cz = -D$.
2. Ensure the term on the right is positive. If $-D$ is negative, multiply the entire equation by -1 .
3. Divide both sides by the magnitude of the normal vector, $\sqrt{A^2 + B^2 + C^2}$.

The converted equation becomes:

$$\frac{Ax}{\pm\sqrt{A^2 + B^2 + C^2}} + \frac{By}{\pm\sqrt{A^2 + B^2 + C^2}} + \frac{Cz}{\pm\sqrt{A^2 + B^2 + C^2}} = \frac{-D}{\pm\sqrt{A^2 + B^2 + C^2}}$$

Here, the coefficients of x, y, z are the direction cosines l, m, n , and the right-hand side is the distance p .

5.1.4 Plane Through Three Points

Theorem 5.1.4. The equation of the plane passing through three non-collinear points (x_1, y_1, z_1) , (x_2, y_2, z_2) , and (x_3, y_3, z_3) is given by the determinant:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

Proof. Let $P(x, y, z)$ be any arbitrary point on the plane containing points A, B , and C . Consider the vectors:

$$\vec{AP} = (x - x_1)\mathbf{i} + (y - y_1)\mathbf{j} + (z - z_1)\mathbf{k}$$

$$\vec{AB} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$$

$$\vec{AC} = (x_3 - x_1)\mathbf{i} + (y_3 - y_1)\mathbf{j} + (z_3 - z_1)\mathbf{k}$$

Since the points A, B, C and P all lie on the same plane, the vectors $\vec{AP}$, $\vec{AB}$, and $\vec{AC}$ must be coplanar.

The condition for three vectors to be coplanar is that their scalar triple product vanishes. That is:

$$\vec{AP} \cdot (\vec{AB} \times \vec{AC}) = 0$$

Expressing the scalar triple product in determinant form using the components of the vectors, we obtain:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

This determinant equation represents the necessary and sufficient condition for $P(x, y, z)$ to lie on the plane defined by A, B , and C . ■

Example 5.1.5. Find the equation of the plane whose intercepts on the coordinate axes are 2, -3 , and 4 respectively.

Solution. Let the intercepts be $a = 2$, $b = -3$, and $c = 4$. The equation of a plane in intercept form is given by:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Substituting the given values:

$$\frac{x}{2} + \frac{y}{-3} + \frac{z}{4} = 1$$

To eliminate fractions, we multiply by the least common multiple of the denominators, which is 12:

$$12 \left(\frac{x}{2} - \frac{y}{3} + \frac{z}{4} \right) = 12(1)$$

$$6x - 4y + 3z = 12$$

Thus, the general equation of the plane is $6x - 4y + 3z - 12 = 0$. ■

Example 5.1.6. Find the intercepts made on the coordinate axes by the plane $3x + 4y - 6z + 12 = 0$.

Solution. To find the intercepts, we must reduce the given equation to the form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. The given equation is:

$$3x + 4y - 6z = -12$$

Dividing both sides by -12 to make the right-hand side equal to 1:

$$\frac{3x}{-12} + \frac{4y}{-12} - \frac{6z}{-12} = 1$$

Simplifying the fractions:

$$\frac{x}{-4} + \frac{y}{-3} + \frac{z}{2} = 1$$

Comparing this with the standard intercept form, the intercepts are:

$$x\text{-intercept} = -4, \quad y\text{-intercept} = -3, \quad z\text{-intercept} = 2$$

Example 5.1.7. Reduce the equation of the plane $2x - 3y + 6z = 14$ to the normal form, and hence find the length of the perpendicular from the origin to the plane and the direction cosines of the normal. ■

Solution. The general equation is $2x - 3y + 6z = 14$. The normal form is $lx + my + nz = p$. First, verify that the constant term on the right is positive. Here, $14 > 0$, so we proceed. We divide the entire equation by the magnitude of the normal vector $\mathbf{n} = (2, -3, 6)$:

$$\text{Magnitude} = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

Dividing the equation by 7:

$$\frac{2}{7}x - \frac{3}{7}y + \frac{6}{7}z = \frac{14}{7}$$

$$\frac{2}{7}x - \frac{3}{7}y + \frac{6}{7}z = 2$$

This is the required normal form. The length of the perpendicular from the origin is $p = 2$. The direction cosines of the normal are $\left(\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}\right)$. ■

Example 5.1.8. Find the direction cosines of the normal to the plane $x + y + z + 1 = 0$ and the perpendicular distance from the origin.

Solution. First, rewrite the equation so the constant term is on the right-hand side:

$$x + y + z = -1$$

Since the distance p in the normal form must be non-negative, we multiply the entire equation by -1 :

$$-x - y - z = 1$$

Now, calculate the dividing factor (magnitude of the normal vector coefficients):

$$\sqrt{(-1)^2 + (-1)^2 + (-1)^2} = \sqrt{1 + 1 + 1} = \sqrt{3}$$

Dividing the equation by $\sqrt{3}$:

$$-\frac{1}{\sqrt{3}}x - \frac{1}{\sqrt{3}}y - \frac{1}{\sqrt{3}}z = \frac{1}{\sqrt{3}}$$

This is the normal form. The perpendicular distance from the origin is $p = \frac{1}{\sqrt{3}}$. The direction cosines of the normal are $\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$. ■

Example 5.1.9. Find the equation of the plane passing through the points $P(1, 1, 0)$, $Q(1, 2, 1)$, and $R(-2, 2, -1)$.

Solution. The equation of a plane passing through three points is given by the determinant equal to zero. Substituting $P(1, 1, 0)$ as the first point:

$$\begin{array}{lll} x_2 - x_1 = 1 - 1 = 0, & y_2 - y_1 = 2 - 1 = 1, & z_2 - z_1 = 1 - 0 = 1 \\ x_3 - x_1 = -2 - 1 = -3, & y_3 - y_1 = 2 - 1 = 1, & z_3 - z_1 = -1 - 0 = -1 \end{array}$$

The determinant equation is:

$$\begin{vmatrix} x - 1 & y - 1 & z - 0 \\ 0 & 1 & 1 \\ -3 & 1 & -1 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} (x - 1)((1)(-1) - (1)(1)) - (y - 1)((0)(-1) - (1)(-3)) + z((0)(1) - (1)(-3)) &= 0 \\ (x - 1)(-2) - (y - 1)(3) + z(3) &= 0 \\ -2x + 2 - 3y + 3 + 3z &= 0 \\ -2x - 3y + 3z + 5 &= 0 \end{aligned}$$

Multiplying by -1 , the required equation is $2x + 3y - 3z - 5 = 0$. ■

Example 5.1.10. Show that the four points $(0, -1, 0)$, $(2, 1, -1)$, $(1, 1, 1)$, and $(3, 3, 0)$ are coplanar.

Solution. First, we find the equation of the plane passing through the first three points $A(0, -1, 0)$, $B(2, 1, -1)$, and $C(1, 1, 1)$. Using the determinant form:

$$\begin{vmatrix} x - 0 & y - (-1) & z - 0 \\ 2 - 0 & 1 - (-1) & -1 - 0 \\ 1 - 0 & 1 - (-1) & 1 - 0 \end{vmatrix} = 0 \implies \begin{vmatrix} x & y + 1 & z \\ 2 & 2 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

Expanding:

$$\begin{aligned} x(2 - (-2)) - (y + 1)(2 - (-1)) + z(4 - 2) &= 0 \\ 4x - 3(y + 1) + 2z &= 0 \\ 4x - 3y + 2z - 3 &= 0 \end{aligned}$$

Now, test the fourth point $D(3, 3, 0)$ in this equation:

$$4(3) - 3(3) + 2(0) - 3 = 12 - 9 + 0 - 3 = 12 - 12 = 0$$

Since the fourth point satisfies the equation of the plane formed by the first three, the four points are coplanar. ■

Example 5.1.11. A plane meets the coordinate axes at A, B, C such that the centroid of the triangle ABC is the point (α, β, γ) . Find the equation of the plane.

Solution. Let the equation of the plane in intercept form be:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The coordinates of the vertices are $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$. The centroid (x, y, z) of triangle ABC is given by the average of the coordinates of its vertices:

$$x = \frac{a + 0 + 0}{3} = \frac{a}{3}, \quad y = \frac{0 + b + 0}{3} = \frac{b}{3}, \quad z = \frac{0 + 0 + c}{3} = \frac{c}{3}$$

We are given that the centroid is (α, β, γ) . Therefore:

$$\frac{a}{3} = \alpha \implies a = 3\alpha$$

$$\frac{b}{3} = \beta \implies b = 3\beta$$

$$\frac{c}{3} = \gamma \implies c = 3\gamma$$

Substituting these values back into the intercept form equation:

$$\frac{x}{3\alpha} + \frac{y}{3\beta} + \frac{z}{3\gamma} = 1$$

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3$$

This is the required equation of the plane. ■

Example 5.1.12. A variable plane moves such that the sum of the reciprocals of its intercepts on the coordinate axes is a constant k . Show that the plane passes through a fixed point.

Solution. Let the equation of the variable plane in intercept form be:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The intercepts are a, b, c . We are given the condition:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = k$$

To show the plane passes through a fixed point, we must manipulate this condition to match the form of the plane's equation. Dividing the condition by k (assuming $k \neq 0$):

$$\frac{1}{ak} + \frac{1}{bk} + \frac{1}{ck} = 1$$

$$\frac{1/k}{a} + \frac{1/k}{b} + \frac{1/k}{c} = 1$$

Comparing this with the equation of the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, we see that the coordinates $x = \frac{1}{k}, y = \frac{1}{k}, z = \frac{1}{k}$ always satisfy the equation of the plane, regardless of the values of a, b, c . Thus, the plane always passes through the fixed point $\left(\frac{1}{k}, \frac{1}{k}, \frac{1}{k}\right)$. ■

5.2 Angles, Distances, and Intersections of Planes

In this section, we explore the geometric relationships between planes, including the angle between them, the perpendicular distance from a point to a plane, and the family of planes generated by the intersection of two given planes.

5.2.1 Angle Between Two Planes

The angle between two planes is defined as the angle between their respective normal vectors.

Definition 5.2.1 (Angle Between Planes). The angle θ between two planes is the acute angle between their normals. If the equations of the planes are $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$, the angle is given by:

$$\cos \theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Theorem 5.2.2 (Conditions for Perpendicularity and Parallelism). 1. **Perpendicularity:** Two planes are perpendicular if and only if:

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

2. **Parallelism:** Two planes are parallel if and only if:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Proof. The angle between two planes is defined as the angle between their respective normal vectors. Let $\mathbf{n}_1 = (a_1, b_1, c_1)$ and $\mathbf{n}_2 = (a_2, b_2, c_2)$ be the normals to the two planes.

1. **Perpendicularity:** The planes are perpendicular if and only if their normals are orthogonal. The condition for orthogonality is that the dot product of the normals vanishes:

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = 0 \implies a_1a_2 + b_1b_2 + c_1c_2 = 0$$

2. **Parallelism:** The planes are parallel if and only if their normals are parallel. Two vectors are parallel if their components are proportional. Thus:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

■

5.2.2 Perpendicular Distance

Theorem 5.2.3 (Perpendicular Distance of a Point from a Plane). The length of the perpendicular p drawn from a point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by:

$$p = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Proof. Let M be the foot of the perpendicular drawn from the point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$. The line PM is normal to the plane. Therefore, the direction ratios of the line PM are proportional to the coefficients a, b, c .

The equation of the line passing through P and perpendicular to the plane is:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = k \quad (\text{say})$$

The coordinates of any general point on this line are $(x_1 + ak, y_1 + bk, z_1 + ck)$. Since M lies on the line, let its coordinates be $M(x_1 + ak, y_1 + bk, z_1 + ck)$ for some specific value of k .

Because M lies on the plane, its coordinates must satisfy the plane's equation:

$$a(x_1 + ak) + b(y_1 + bk) + c(z_1 + ck) + d = 0$$

Expanding and solving for k :

$$(ax_1 + by_1 + cz_1 + d) + k(a^2 + b^2 + c^2) = 0$$

$$k = -\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$$

The required perpendicular distance p is the distance PM . Using the distance formula:

$$p = PM = \sqrt{(x_M - x_1)^2 + (y_M - y_1)^2 + (z_M - z_1)^2}$$

Substituting the coordinate differences $x_M - x_1 = ak$, etc.:

$$p = \sqrt{(ak)^2 + (bk)^2 + (ck)^2} = \sqrt{k^2(a^2 + b^2 + c^2)} = |k|\sqrt{a^2 + b^2 + c^2}$$

Substituting the value of k derived above:

$$p = \left| -\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2} \right| \sqrt{a^2 + b^2 + c^2}$$

$$p = \frac{|ax_1 + by_1 + cz_1 + d|}{a^2 + b^2 + c^2} \cdot \sqrt{a^2 + b^2 + c^2}$$

Simplifying the denominator:

$$p = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

■

Note 4. The distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is given by:

$$\text{Distance} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

5.2.3 Intersection of Two Planes

Two distinct non-parallel planes intersect in a straight line. We can determine the equation of any plane passing through this line of intersection.

Theorem 5.2.4 (Plane Through the Intersection of Two Planes). *The general equation of a plane passing through the line of intersection of the planes $P_1 \equiv a_1x + b_1y + c_1z + d_1 = 0$ and $P_2 \equiv a_2x + b_2y + c_2z + d_2 = 0$ is given by:*

$$P_1 + \lambda P_2 = 0$$

Proof. Consider the equation $P_1 + \lambda P_2 = 0$:

$$(a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$$

This can be rewritten as:

$$(a_1 + \lambda a_2)x + (b_1 + \lambda b_2)y + (c_1 + \lambda c_2)z + (d_1 + \lambda d_2) = 0$$

Since this is a linear equation of the first degree in x, y, z , it represents a plane.

Now, consider any point Q lying on the line of intersection of the planes $P_1 = 0$ and $P_2 = 0$. The coordinates of Q satisfy both $P_1 = 0$ and $P_2 = 0$ simultaneously. Substituting these into the combined equation:

$$0 + \lambda(0) = 0$$

The equation is satisfied for any value of λ . Thus, the plane $P_1 + \lambda P_2 = 0$ passes through every point on the line of intersection. ■

Example 5.2.5. Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 3$.

Solution. The normal vectors to the planes are $\mathbf{n}_1 = (2, -1, 1)$ and $\mathbf{n}_2 = (1, 1, 2)$. Let θ be the angle between the planes.

$$\cos \theta = \frac{|2(1) + (-1)(1) + 1(2)|}{\sqrt{2^2 + (-1)^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}}$$

$$\cos \theta = \frac{|2 - 1 + 2|}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2}$$

Since $\cos \theta = \frac{1}{2}$, the angle is $\theta = 60^\circ$ (or $\frac{\pi}{3}$). ■

Example 5.2.6. Show that the planes $3x + 4y - 5z = 0$ and $2x + 6y + 6z = 7$ are perpendicular to each other.

Solution. The condition for two planes to be perpendicular is $a_1a_2 + b_1b_2 + c_1c_2 = 0$. Identifying the direction ratios of the normals:

$$a_1 = 3, \quad b_1 = 4, \quad c_1 = -5$$

$$a_2 = 2, \quad b_2 = 6, \quad c_2 = 6$$

Calculating the dot product:

$$a_1a_2 + b_1b_2 + c_1c_2 = (3)(2) + (4)(6) + (-5)(6)$$

$$= 6 + 24 - 30 = 30 - 30 = 0$$

Since the product sum is zero, the planes are perpendicular. ■

Example 5.2.7. Find the length of the perpendicular from the point $(2, 3, -1)$ to the plane $x - 2y + 4z = 10$.

Solution. Rewrite the equation of the plane as $x - 2y + 4z - 10 = 0$. The distance d from the point $(2, 3, -1)$ is:

$$d = \frac{|1(2) - 2(3) + 4(-1) - 10|}{\sqrt{1^2 + (-2)^2 + 4^2}}$$

$$d = \frac{|2 - 6 - 4 - 10|}{\sqrt{1 + 4 + 16}}$$

$$d = \frac{|-18|}{\sqrt{21}} = \frac{18}{\sqrt{21}} = \frac{6\sqrt{21}}{7}$$

Example 5.2.8. Find the distance between the parallel planes $2x - 2y + z + 3 = 0$ and $4x - 4y + 2z + 5 = 0$.

Solution. To use the distance formula for parallel planes, the coefficients of x, y, z must be identical. We divide the second equation by 2:

$$2x - 2y + z + \frac{5}{2} = 0$$

Now, the two equations are:

$$2x - 2y + z + 3 = 0 \quad \text{and} \quad 2x - 2y + z + 2.5 = 0$$

The distance is given by:

$$d = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|3 - 2.5|}{\sqrt{2^2 + (-2)^2 + 1^2}}$$

$$d = \frac{0.5}{\sqrt{4 + 4 + 1}} = \frac{0.5}{\sqrt{9}} = \frac{0.5}{3} = \frac{1}{6}$$

Example 5.2.9. Find the equation of the plane passing through the intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and the point $(1, 1, 1)$.

Solution. The equation of any plane passing through the intersection of the given planes is:

$$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0$$

Since the plane passes through $(1, 1, 1)$, we substitute these coordinates to find λ :

$$(1 + 1 + 1 - 6) + \lambda(2(1) + 3(1) + 4(1) + 5) = 0$$

$$(-3) + \lambda(14) = 0 \implies 14\lambda = 3 \implies \lambda = \frac{3}{14}$$

Substituting $\lambda = \frac{3}{14}$ back into the general equation:

$$(x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$$

Multiplying by 14:

$$14(x + y + z - 6) + 3(2x + 3y + 4z + 5) = 0$$

$$14x + 14y + 14z - 84 + 6x + 9y + 12z + 15 = 0$$

$$20x + 23y + 26z - 69 = 0$$

■

Example 5.2.10. Find the equation of the plane passing through the line of intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ and perpendicular to the plane $5x + 3y + 6z + 8 = 0$.

Solution. The equation of the required plane is:

$$(x + 2y + 3z - 4) + \lambda(2x + y - z + 5) = 0$$

Grouping the coefficients of x, y, z :

$$(1 + 2\lambda)x + (2 + \lambda)y + (3 - \lambda)z + (5\lambda - 4) = 0$$

The normal vector to this plane is $\mathbf{n} = (1 + 2\lambda, 2 + \lambda, 3 - \lambda)$. This plane is perpendicular to the plane $5x + 3y + 6z + 8 = 0$, which has the normal $\mathbf{n}' = (5, 3, 6)$. The dot product of their normals must be zero:

$$5(1 + 2\lambda) + 3(2 + \lambda) + 6(3 - \lambda) = 0$$

$$5 + 10\lambda + 6 + 3\lambda + 18 - 6\lambda = 0$$

$$7\lambda + 29 = 0 \implies \lambda = -\frac{29}{7}$$

Substituting λ back into the grouped equation:

$$\left(1 - \frac{58}{7}\right)x + \left(2 - \frac{29}{7}\right)y + \left(3 + \frac{29}{7}\right)z + \dots = 0$$

Multiplying by 7:

$$(7 - 58)x + (14 - 29)y + (21 + 29)z + 5(-29) - 28 = 0$$

$$-51x - 15y + 50z - 173 = 0$$

Or $51x + 15y - 50z + 173 = 0$.

■

Example 5.2.11. Find the locus of a point whose distance from the origin is three times its distance from the plane $x + y + z = 1$.

Solution. Let $P(x, y, z)$ be the point on the locus. The distance of P from the origin is $\sqrt{x^2 + y^2 + z^2}$. The distance of P from the plane $x + y + z - 1 = 0$ is $\frac{|x + y + z - 1|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{|x + y + z - 1|}{\sqrt{3}}$. According to the problem:

$$\sqrt{x^2 + y^2 + z^2} = 3 \left(\frac{|x + y + z - 1|}{\sqrt{3}} \right)$$

Squaring both sides:

$$x^2 + y^2 + z^2 = 9 \left(\frac{(x + y + z - 1)^2}{3} \right)$$

$$x^2 + y^2 + z^2 = 3(x + y + z - 1)^2$$

$$x^2 + y^2 + z^2 = 3(x^2 + y^2 + z^2 + 2xy + 2yz + 2zx - 2x - 2y - 2z + 1)$$

Rearranging terms:

$$2x^2 + 2y^2 + 2z^2 + 6xy + 6yz + 6zx - 6x - 6y - 6z + 3 = 0$$

This second-degree equation represents the required locus (a sphere-like quadric surface). ■

Example 5.2.12. Find the coordinates of the foot of the perpendicular drawn from the origin to the plane $2x + 3y + 4z - 29 = 0$.

Solution. Let N be the foot of the perpendicular from the origin $O(0, 0, 0)$ to the plane. The line ON is normal to the plane, so its direction ratios are proportional to the coefficients of the plane equation: 2, 3, 4. The equation of line ON passing through the origin is:

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{4} = k$$

Any point on this line is $(2k, 3k, 4k)$. For this point to be the foot of the perpendicular, it must lie on the plane $2x + 3y + 4z = 29$. Substituting the coordinates:

$$2(2k) + 3(3k) + 4(4k) = 29$$

$$4k + 9k + 16k = 29$$

$$29k = 29 \implies k = 1$$

Substituting $k = 1$ back into the coordinates, the foot of the perpendicular is $(2, 3, 4)$. ■

Chapter 6

Advanced Properties of Hyperbola

6.1 The Hyperbola and Tracing of Conics

In this section, we study the hyperbola, its tangent and normal properties, its description relative to its asymptotes, and the general method for tracing conics represented by the general second-degree equation.

Definition 6.1.1 (Hyperbola). A hyperbola is the locus of a point in a plane which moves such that the ratio of its distance from a fixed point (focus) to its distance from a fixed line (directrix) is a constant greater than unity (eccentricity $e > 1$).

Alternatively, it is the locus of a point such that the difference of its distances from two fixed points (foci) is constant and equal to the length of the transverse axis ($2a$).

Theorem 6.1.2 (Standard Equation). *The standard equation of a hyperbola with center at the origin and transverse axis along the x -axis is:*

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $b^2 = a^2(e^2 - 1)$.

Proof. By definition, a hyperbola is the locus of a point whose distance from a fixed point (focus) bears a constant ratio $e > 1$ to its perpendicular distance from a fixed line (directrix).

Let the focus S be at $(ae, 0)$ and the corresponding directrix be the line $x = \frac{a}{e}$. Let $P(x, y)$ be any point on the hyperbola. Let PM be the perpendicular distance from P to the directrix.

By definition:

$$\frac{SP}{PM} = e \implies SP^2 = e^2 PM^2$$

Using the distance formula:

$$(x - ae)^2 + (y - 0)^2 = e^2 \left(x - \frac{a}{e}\right)^2$$

Expanding both sides:

$$\begin{aligned} x^2 - 2aex + a^2e^2 + y^2 &= e^2 \left(x^2 - \frac{2ax}{e} + \frac{a^2}{e^2}\right) \\ x^2 - 2aex + a^2e^2 + y^2 &= e^2x^2 - 2aex + a^2 \end{aligned}$$

Canceling the term $-2aex$ and rearranging the terms:

$$x^2(e^2 - 1) - y^2 = a^2(e^2 - 1)$$

Dividing the entire equation by $a^2(e^2 - 1)$:

$$\frac{x^2}{a^2} - \frac{y^2}{a^2(e^2 - 1)} = 1$$

Since $e > 1$, the term $e^2 - 1$ is positive. We substitute $b^2 = a^2(e^2 - 1)$:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

■

6.1.1 Tangents and Normals

Theorem 6.1.3 (Equation of Tangent). *The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(x_1, y_1)$ is:*

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

In terms of slope m , the tangent lines are $y = mx \pm \sqrt{a^2m^2 - b^2}$.

Proof. Point Form Differentiating the equation of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with respect to x :

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{b^2x}{a^2y}$$

The slope of the tangent at $P(x_1, y_1)$ is $m = \frac{b^2x_1}{a^2y_1}$. The equation of the tangent line passing through $P(x_1, y_1)$ is:

$$y - y_1 = \frac{b^2x_1}{a^2y_1}(x - x_1)$$

Multiplying by $\frac{y_1}{b^2}$ and rearranging:

$$\frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = \frac{xx_1}{a^2} - \frac{x_1^2}{a^2}$$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

Since $P(x_1, y_1)$ lies on the hyperbola, $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$. Thus:

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Slope Form Let the line be $y = mx + c$. For this line to touch the hyperbola, the quadratic equation formed by their intersection must have equal roots. Substituting y into the hyperbola equation:

$$\frac{x^2}{a^2} - \frac{(mx + c)^2}{b^2} = 1$$

$$b^2x^2 - a^2(m^2x^2 + 2mcx + c^2) - a^2b^2 = 0$$

$$x^2(b^2 - a^2m^2) - 2a^2mcx - a^2(c^2 + b^2) = 0$$

For tangency, the discriminant must vanish ($D = 0$):

$$4a^4m^2c^2 - 4(b^2 - a^2m^2)(-a^2(c^2 + b^2)) = 0$$

Dividing by $4a^2$ and simplifying yields:

$$c^2 = a^2m^2 - b^2 \implies c = \pm \sqrt{a^2m^2 - b^2}$$

Substituting c back gives $y = mx \pm \sqrt{a^2m^2 - b^2}$.

■

Theorem 6.1.4 (Equation of Normal). *The equation of the normal to the hyperbola at $P(x_1, y_1)$ is:*

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

Proof. The normal is perpendicular to the tangent. The slope of the tangent at $P(x_1, y_1)$ was found to be $m_T = \frac{b^2x_1}{a^2y_1}$. Therefore, the slope of the normal is:

$$m_N = -\frac{1}{m_T} = -\frac{a^2y_1}{b^2x_1}$$

The equation of the normal passing through $P(x_1, y_1)$ is:

$$y - y_1 = -\frac{a^2y_1}{b^2x_1}(x - x_1)$$

Rearranging to separate the x and y terms:

$$\frac{y - y_1}{a^2y_1} = -\frac{x - x_1}{b^2x_1}$$

Multiplying through by a^2b^2 is one method, but let us cross-multiply directly to match the standard form:

$$b^2x_1(y - y_1) = -a^2y_1(x - x_1)$$

Dividing both sides by x_1y_1 :

$$\begin{aligned} \frac{b^2(y - y_1)}{y_1} &= -\frac{a^2(x - x_1)}{x_1} \\ \frac{b^2y}{y_1} - b^2 &= -\frac{a^2x}{x_1} + a^2 \end{aligned}$$

Transposing the variables to the left and constants to the right:

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

■

6.1.2 Hyperbola Referred to Asymptotes

The asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are the lines $y = \pm \frac{b}{a}x$. These lines pass through the center and touch the hyperbola at infinity.

Theorem 6.1.5 (Equation Referred to Asymptotes as Axes). *If the asymptotes of a hyperbola are taken as the coordinate axes, the equation of the hyperbola takes the form:*

$$xy = c^2$$

$$\text{where } c^2 = \frac{a^2 + b^2}{4}.$$

Proof. Consider the standard equation of the hyperbola:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

The joint equation of its asymptotes is:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \implies \left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = 0$$

Let P be any point on the hyperbola. The lengths of the perpendiculars from P to the two asymptotes are:

$$p_1 = \frac{\frac{x}{a} - \frac{y}{b}}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} \quad \text{and} \quad p_2 = \frac{\frac{x}{a} + \frac{y}{b}}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}$$

Multiplying these lengths:

$$p_1 p_2 = \frac{\frac{x^2}{a^2} - \frac{y^2}{b^2}}{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{1}{\frac{a^2+b^2}{a^2 b^2}} = \frac{a^2 b^2}{a^2 + b^2}$$

Thus, for any point on the hyperbola, the product of the perpendicular distances to the asymptotes is constant.

Now, let us change the coordinate system. Let the asymptotes be the new X and Y axes. Let the coordinates of P in this new system be (X, Y) . Let 2α be the angle between the asymptotes. From the slopes $\pm \frac{b}{a}$, we have $\tan \alpha = \frac{b}{a}$. Then $\sin(2\alpha) = \frac{2 \tan \alpha}{1 + \tan^2 \alpha} = \frac{2(b/a)}{1 + b^2/a^2} = \frac{2ab}{a^2 + b^2}$.

In the oblique coordinate system formed by the asymptotes:

- The perpendicular distance p_1 from $P(X, Y)$ to the X -axis is $Y \sin(2\alpha)$.
- The perpendicular distance p_2 from $P(X, Y)$ to the Y -axis is $X \sin(2\alpha)$.

Substituting these into the product relation derived earlier:

$$(X \sin 2\alpha)(Y \sin 2\alpha) = \frac{a^2 b^2}{a^2 + b^2}$$

$$XY \sin^2 2\alpha = \frac{a^2 b^2}{a^2 + b^2}$$

Substituting $\sin 2\alpha = \frac{2ab}{a^2 + b^2}$:

$$XY \left(\frac{2ab}{a^2 + b^2} \right)^2 = \frac{a^2 b^2}{a^2 + b^2}$$

$$XY \frac{4a^2 b^2}{(a^2 + b^2)^2} = \frac{a^2 b^2}{a^2 + b^2}$$

Simplifying:

$$XY \frac{4}{a^2 + b^2} = 1 \implies XY = \frac{a^2 + b^2}{4}$$

Defining $c^2 = \frac{a^2 + b^2}{4}$, the equation becomes $XY = c^2$. ■

Example 6.1.6. Find the lengths of the transverse and conjugate axes of the hyperbola $xy = 16$.

Solution. The equation $xy = 16$ is in the form $xy = c^2$ where the coordinate axes are the asymptotes. Here $c^2 = 16$. From the theorem, for a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the asymptotic form has $c^2 = \frac{a^2 + b^2}{4}$. However, the form $xy = c^2$ specifically represents a rectangular hyperbola where $a = b$. Substituting $b = a$:

$$c^2 = \frac{a^2 + a^2}{4} = \frac{2a^2}{4} = \frac{a^2}{2}$$

Given $c^2 = 16$:

$$16 = \frac{a^2}{2} \implies a^2 = 32 \implies a = 4\sqrt{2}$$

Since it is a rectangular hyperbola, the transverse axis length is $2a$ and the conjugate axis length is $2b = 2a$.

$$\text{Length of Transverse Axis} = 2(4\sqrt{2}) = 8\sqrt{2}$$

$$\text{Length of Conjugate Axis} = 8\sqrt{2}$$

Example 6.1.7. Show that the area of the triangle formed by the tangent at any point on the hyperbola $xy = c^2$ and the coordinate axes (asymptotes) is constant.

Solution. Let P be any point on the hyperbola $xy = c^2$. We can use parametric coordinates $P(ct, c/t)$. The equation of the tangent at (x_1, y_1) to the curve $xy = c^2$ is given by:

$$\frac{1}{2}(xy_1 + yx_1) = c^2$$

Substituting $x_1 = ct$ and $y_1 = c/t$:

$$\frac{1}{2}\left(x\frac{c}{t} + y(ct)\right) = c^2$$

Dividing by c :

$$\frac{x}{2t} + \frac{yt}{2} = c \implies \frac{x}{t} + yt = 2c$$

This tangent intersects the coordinate axes at points A and B . To find the x -intercept (point A), set $y = 0$:

$$\frac{x}{t} = 2c \implies x = 2ct \implies A(2ct, 0)$$

To find the y -intercept (point B), set $x = 0$:

$$yt = 2c \implies y = \frac{2c}{t} \implies B\left(0, \frac{2c}{t}\right)$$

The area of the triangle formed by the tangent and the axes (OA and OB , where O is origin) is:

$$\begin{aligned}\text{Area} &= \frac{1}{2}|OA| \cdot |OB| \\ \text{Area} &= \frac{1}{2}(2ct) \left(\frac{2c}{t}\right) = \frac{1}{2}(4c^2) = 2c^2\end{aligned}$$

Since c is a constant of the hyperbola, the area $2c^2$ is constant for any tangent. ■

Definition 6.1.8 (Rectangular Hyperbola). If the lengths of the transverse and conjugate axes are equal ($a = b$), the hyperbola is called a rectangular (or equilateral) hyperbola. Its eccentricity is $\sqrt{2}$, and its asymptotes are perpendicular. The standard equation is $x^2 - y^2 = a^2$, and referred to asymptotes, it is $xy = \frac{a^2}{2}$.

6.1.3 Rectangular and Conjugate Diameters

Definition 6.1.9 (Conjugate Diameters). Two diameters of a hyperbola are said to be conjugate if each bisects chords parallel to the other. If $y = m_1x$ and $y = m_2x$ are conjugate diameters of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then:

$$m_1m_2 = -\frac{b^2}{a^2}$$

Theorem 6.1.10 (Properties of Conjugate Diameters). Let CP and CD be two conjugate semi-diameters with lengths a' and b' respectively.

1. **Difference of Squares:** The difference of the squares of two conjugate semi-diameters is constant:

$$CP^2 - CD^2 = a^2 - b^2$$

2. **Area Property:** The area of the parallelogram formed by the tangents at the extremities of conjugate diameters is constant and equal to $4ab$.

Note: In a hyperbola, the extremities of one diameter are real points on the curve, while the extremities of the conjugate diameter are imaginary points (or points on the conjugate hyperbola).

6.1.4 Tracing of Conics

The general equation of the second degree is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.

Theorem 6.1.11 (Conditions for a Hyperbola). *The general equation represents a hyperbola if the discriminant $\Delta \neq 0$ and $h^2 > ab$. It represents a rectangular hyperbola if, in addition, $a + b = 0$.*

Proof. Consider the general equation of the second degree:

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

1. Condition for a Hyperbola: For the equation to represent a conic section (and not a pair of straight lines), the determinant Δ must be non-zero:

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$$

The nature of the conic is determined by the homogeneous part $ax^2 + 2hxy + by^2$, which describes the lines joining the origin to the points at infinity on the curve. Factoring $ax^2 + 2hxy + by^2 = 0$ corresponds to finding the slopes of the asymptotes. Dividing by x^2 :

$$b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$$

This is a quadratic equation in the slope $m = y/x$. For a hyperbola, the curve extends to infinity in two distinct directions (it has two distinct real asymptotes). Therefore, this quadratic must have two distinct real roots. The condition for distinct real roots is that the discriminant of the quadratic is positive:

$$(2h)^2 - 4(b)(a) > 0$$

$$4h^2 - 4ab > 0 \implies h^2 > ab$$

2. Condition for a Rectangular Hyperbola: A hyperbola is rectangular if its asymptotes are perpendicular to each other. The pair of lines represented by $ax^2 + 2hxy + by^2 = 0$ (which are parallel to the asymptotes) are perpendicular if the sum of the coefficients of x^2 and y^2 is zero. Thus, the necessary condition is:

$$a + b = 0$$

■

Steps for Tracing:

- 1. Center:** Find the center (α, β) by solving $\frac{\partial S}{\partial x} = 0$ and $\frac{\partial S}{\partial y} = 0$.
- 2. Shift Origin:** Shift the origin to the center to eliminate linear terms. The new equation is $AX^2 + 2HXY + BY^2 + C' = 0$, where $C' = g\alpha + f\beta + c$.
- 3. Axes:** Determine the lengths and directions of the axes by solving the characteristic equation of the matrix of quadratic terms.
- 4. Asymptotes:** Find the equations of the asymptotes (for hyperbolas).

Example 6.1.12. Determine the nature of the conic represented by the equation $6x^2 - 5xy - 6y^2 + 14x + 5y + 10 = 0$.

Solution. To determine the nature of the conic, we compare the given equation with the general second-degree form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$. The coefficients are:

$$a = 6, \quad b = -6, \quad h = -2.5, \quad g = 7, \quad f = 2.5, \quad c = 10$$

First, we calculate the discriminant Δ to verify that the equation represents a non-degenerate conic.

$$\begin{aligned}\Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 \\ &= (6)(-6)(10) + 2(2.5)(7)(-2.5) - 6(2.5)^2 - (-6)(7)^2 - 10(-2.5)^2 \\ &= -360 - 87.5 - 37.5 + 294 - 62.5 \\ &= -253.5\end{aligned}$$

Since $\Delta \neq 0$, the equation represents a proper conic section, not a pair of straight lines.

Next, we classify the type of conic by evaluating the expression $h^2 - ab$:

$$h^2 - ab = (-2.5)^2 - (6)(-6) = 6.25 + 36 = 42.25$$

As $h^2 > ab$, the conic is a hyperbola.

Finally, we examine the sum of the coefficients of the squared terms:

$$a + b = 6 + (-6) = 0$$

Since $a + b = 0$, the asymptotes of the hyperbola are perpendicular. Therefore, the equation represents a **Rectangular Hyperbola**. ■

Example 6.1.13. Find the equation of the hyperbola whose asymptotes are the lines $x + 2y + 3 = 0$ and $3x + 4y + 5 = 0$, and which passes through the point $(1, -1)$.

Solution. The equation of a hyperbola differs from the joint equation of its asymptotes only by a constant. The joint equation of the asymptotes is:

$$(x + 2y + 3)(3x + 4y + 5) = 0$$

Thus, the equation of the hyperbola is:

$$(x + 2y + 3)(3x + 4y + 5) + k = 0$$

Since the hyperbola passes through the point $(1, -1)$, we substitute these coordinates to find k :

$$(1 + 2(-1) + 3)(3(1) + 4(-1) + 5) + k = 0$$

$$(1 - 2 + 3)(3 - 4 + 5) + k = 0$$

$$(2)(4) + k = 0 \implies 8 + k = 0 \implies k = -8$$

The required equation is:

$$(x + 2y + 3)(3x + 4y + 5) - 8 = 0$$

Expanding this:

$$3x^2 + 4xy + 5x + 6xy + 8y^2 + 10y + 9x + 12y + 15 - 8 = 0$$

$$3x^2 + 10xy + 8y^2 + 14x + 22y + 7 = 0$$

Example 6.1.14. Find the equation of the tangent to the hyperbola $3x^2 - 4y^2 = 12$ which is parallel to the line $y = x - 7$.

Solution. First, write the hyperbola in standard form:

$$\frac{x^2}{4} - \frac{y^2}{3} = 1 \implies a^2 = 4, \quad b^2 = 3$$

The slope of the required tangent is the same as the line $y = x - 7$, so $m = 1$. The equation of a tangent with slope m is given by $y = mx \pm \sqrt{a^2m^2 - b^2}$. Substituting the values:

$$y = 1(x) \pm \sqrt{4(1)^2 - 3}$$

$$y = x \pm \sqrt{1} \implies y = x \pm 1$$

Thus, the two tangents are $x - y + 1 = 0$ and $x - y - 1 = 0$. ■

Example 6.1.15. Prove that the product of the perpendiculars from any point on the hyperbola to its asymptotes is constant.

Solution. Let the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The equations of the asymptotes are:

$$\frac{x}{a} - \frac{y}{b} = 0 \quad \text{and} \quad \frac{x}{a} + \frac{y}{b} = 0$$

Let $P(x_1, y_1)$ be any point on the hyperbola. The perpendicular distances p_1 and p_2 from P to the asymptotes are:

$$p_1 = \frac{\left| \frac{x_1}{a} - \frac{y_1}{b} \right|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} \quad \text{and} \quad p_2 = \frac{\left| \frac{x_1}{a} + \frac{y_1}{b} \right|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}}$$

The product of the perpendiculars is:

$$p_1 p_2 = \frac{\left| \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} \right|}{\frac{1}{a^2} + \frac{1}{b^2}}$$

Since $P(x_1, y_1)$ lies on the hyperbola, $\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1$.

$$p_1 p_2 = \frac{1}{\frac{a^2+b^2}{a^2 b^2}} = \frac{a^2 b^2}{a^2 + b^2}$$

This value is constant for any point on the hyperbola. ■

Example 6.1.16. Trace the conic given by $x^2 - 4y^2 + 6x + 16y - 11 = 0$.

Solution. Comparing with the general equation, $a = 1, b = -4, h = 0$. Since $h^2 - ab = 0 - (1)(-4) = 4 > 0$, the conic is a hyperbola. To find the center, we differentiate partially:

$$\frac{\partial S}{\partial x} = 2x + 6 = 0 \implies x = -3$$

$$\frac{\partial S}{\partial y} = -8y + 16 = 0 \implies y = 2$$

The center is $C(-3, 2)$. A more efficient way to trace is to complete the squares in the original equation:

$$\begin{aligned} (x^2 + 6x) - 4(y^2 - 4y) &= 11 \\ (x^2 + 6x + 9) - 4(y^2 - 4y + 4) &= 11 + 9 - 16 \\ (x + 3)^2 - 4(y - 2)^2 &= 4 \end{aligned}$$

Dividing by 4:

$$\frac{(x + 3)^2}{4} - \frac{(y - 2)^2}{1} = 1$$

This represents a hyperbola with center at $(-3, 2)$, transverse axis parallel to the x -axis, semi-transverse axis $a = 2$, and semi-conjugate axis $b = 1$. ■

Example 6.1.17. If the chord joining the points $P(ct_1, c/t_1)$ and $Q(ct_2, c/t_2)$ on the rectangular hyperbola $xy = c^2$ passes through the center, prove that $t_2 = -t_1$.

Solution. The center of the rectangular hyperbola $xy = c^2$ is the origin $O(0, 0)$. The slope of the line OP is:

$$m_1 = \frac{c/t_1 - 0}{ct_1 - 0} = \frac{1}{t_1^2}$$

The slope of the line OQ is:

$$m_2 = \frac{c/t_2 - 0}{ct_2 - 0} = \frac{1}{t_2^2}$$

Since the chord PQ passes through the center O , the points P, O, Q are collinear. Thus, the slope of OP must equal the slope of OQ .

$$\frac{1}{t_1^2} = \frac{1}{t_2^2} \implies t_1^2 = t_2^2 \implies t_2 = \pm t_1$$

Since P and Q are distinct points forming a chord, $t_1 \neq t_2$. Therefore:

$$t_2 = -t_1$$

This confirms that the parameters of the extremities of any diameter of a rectangular hyperbola are equal in magnitude but opposite in sign. ■

Example 6.1.18. Prove that the product of the lengths of the perpendiculars drawn from the two foci upon any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is constant and equal to b^2 .

Solution. The equation of any tangent to the hyperbola with slope m is:

$$y = mx + \sqrt{a^2m^2 - b^2} \implies mx - y + \sqrt{a^2m^2 - b^2} = 0$$

The coordinates of the two foci are $S(ae, 0)$ and $S'(-ae, 0)$, where $b^2 = a^2(e^2 - 1)$.

Let p_1 and p_2 be the lengths of the perpendiculars from S and S' to the tangent respectively.

$$p_1 = \frac{|m(ae) - 0 + \sqrt{a^2m^2 - b^2}|}{\sqrt{m^2 + 1}}$$

$$p_2 = \frac{|m(-ae) - 0 + \sqrt{a^2m^2 - b^2}|}{\sqrt{m^2 + 1}}$$

Calculating the product p_1p_2 :

$$p_1p_2 = \frac{|(\sqrt{a^2m^2 - b^2} + mae)(\sqrt{a^2m^2 - b^2} - mae)|}{m^2 + 1}$$

Using the identity $(A + B)(A - B) = A^2 - B^2$:

$$p_1p_2 = \frac{|(a^2m^2 - b^2) - (mae)^2|}{m^2 + 1} = \frac{|a^2m^2 - b^2 - m^2a^2e^2|}{m^2 + 1}$$

Grouping terms with m^2 :

$$p_1p_2 = \frac{|a^2m^2(1 - e^2) - b^2|}{m^2 + 1}$$

We know that $b^2 = a^2(e^2 - 1)$, so $-b^2 = a^2(1 - e^2)$. Substituting this:

$$p_1p_2 = \frac{|m^2(-b^2) - b^2|}{m^2 + 1} = \frac{|-b^2(m^2 + 1)|}{m^2 + 1}$$

$$p_1p_2 = b^2$$

Thus, the product of the perpendiculars is constant and equal to the square of the semi-conjugate axis. ■

Example 6.1.19. Show that the area of the triangle formed by any tangent to a hyperbola and its asymptotes is constant and equal to ab .

Solution. Let the equation of the hyperbola referred to its principal axes be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The equations of the asymptotes are $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$. Consider a point $P(a \sec \theta, b \tan \theta)$ on the hyperbola. The equation of the tangent at P is:

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

We find the points of intersection of this tangent with the asymptotes.

1. Intersection with $\frac{x}{a} - \frac{y}{b} = 0$: Substitute $\frac{y}{b} = \frac{x}{a}$ into the tangent equation:

$$\frac{x}{a}(\sec \theta - \tan \theta) = 1 \implies x = \frac{a}{\sec \theta - \tan \theta} = a(\sec \theta + \tan \theta)$$

Then $y = \frac{b}{a}x = b(\sec \theta + \tan \theta)$. Let this point be Q .

2. Intersection with $\frac{x}{a} + \frac{y}{b} = 0$: Substitute $\frac{y}{b} = -\frac{x}{a}$:

$$\frac{x}{a}(\sec \theta + \tan \theta) = 1 \implies x = \frac{a}{\sec \theta + \tan \theta} = a(\sec \theta - \tan \theta)$$

Then $y = -\frac{b}{a}x = -b(\sec \theta - \tan \theta)$. Let this point be R .

The triangle is formed by the vertices $O(0,0)$, Q , and R . The area of $\triangle OQR$ is given by the determinant:

$$\text{Area} = \frac{1}{2} |x_Q y_R - x_R y_Q|$$

$$\text{Area} = \frac{1}{2} |a(\sec \theta + \tan \theta)[-b(\sec \theta - \tan \theta)] - a(\sec \theta - \tan \theta)[b(\sec \theta + \tan \theta)]|$$

Let $K = (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = \sec^2 \theta - \tan^2 \theta = 1$.

$$\text{Area} = \frac{1}{2} |-ab(1) - ab(1)| = \frac{1}{2} |-2ab| = ab$$

Thus, the area is constant. ■

Example 6.1.20. Prove that the locus of the intersection of two perpendicular tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is the circle $x^2 + y^2 = a^2 - b^2$ (Director Circle).

Solution. The equation of any tangent to the hyperbola with slope m is:

$$y = mx + \sqrt{a^2 m^2 - b^2}$$

Let this tangent pass through the point (h, k) . Then:

$$k - mh = \sqrt{a^2 m^2 - b^2}$$

Squaring both sides:

$$(k - mh)^2 = a^2 m^2 - b^2$$

$$k^2 - 2kmh + m^2 h^2 = a^2 m^2 - b^2$$

Rearranging as a quadratic in m :

$$m^2(h^2 - a^2) - 2m(hk) + (k^2 + b^2) = 0$$

The roots of this quadratic, m_1 and m_2 , are the slopes of the two tangents passing through (h, k) . For the tangents to be perpendicular, the product of their slopes must be -1 :

$$m_1 m_2 = -1$$

From the properties of quadratic equations, the product of roots is $\frac{c}{a} = \frac{k^2 + b^2}{h^2 - a^2}$. Thus:

$$\begin{aligned}\frac{k^2 + b^2}{h^2 - a^2} &= -1 \\ k^2 + b^2 &= -(h^2 - a^2) \\ k^2 + b^2 &= -h^2 + a^2 \\ h^2 + k^2 &= a^2 - b^2\end{aligned}$$

Replacing the locus (h, k) with (x, y) , we get the equation of the director circle:

$$x^2 + y^2 = a^2 - b^2$$

■

Example 6.1.21. Prove that the orthocenter of any triangle inscribed in a rectangular hyperbola lies on the hyperbola.

Solution. Let the equation of the rectangular hyperbola be $xy = c^2$. The parametric coordinates of a point on the hyperbola are $(ct, c/t)$. Let the vertices of the triangle be $A(ct_1, c/t_1)$, $B(ct_2, c/t_2)$, and $C(ct_3, c/t_3)$.

First, we find the slope of the side BC :

$$m_{BC} = \frac{c/t_3 - c/t_2}{ct_3 - ct_2} = \frac{c(t_2 - t_3)/(t_2 t_3)}{c(t_3 - t_2)} = -\frac{1}{t_2 t_3}$$

The altitude from A to BC is perpendicular to BC , so its slope is $t_2 t_3$. The equation of the altitude through A is:

$$\begin{aligned}y - \frac{c}{t_1} &= t_2 t_3 (x - ct_1) \\ y &= t_2 t_3 x - ct_1 t_2 t_3 + \frac{c}{t_1}\end{aligned}$$

Similarly, the slope of AC is $-1/(t_1 t_3)$, so the slope of the altitude through B is $t_1 t_3$. The equation of the altitude through B is:

$$y = t_1 t_3 x - ct_1 t_2 t_3 + \frac{c}{t_2}$$

To find the orthocenter, we solve these two equations. Subtracting the second from the first:

$$\begin{aligned}0 &= (t_2 t_3 - t_1 t_3)x + \left(\frac{c}{t_1} - \frac{c}{t_2}\right) \\ 0 &= t_3(t_2 - t_1)x + c\frac{t_2 - t_1}{t_1 t_2}\end{aligned}$$

Since $t_1 \neq t_2$, we divide by $(t_2 - t_1)$:

$$t_3 x + \frac{c}{t_1 t_2} = 0 \implies x = -\frac{c}{t_1 t_2 t_3}$$

Substitute x back into the equation for the altitude through A :

$$\begin{aligned}y &= t_2 t_3 \left(-\frac{c}{t_1 t_2 t_3}\right) - ct_1 t_2 t_3 + \frac{c}{t_1} \\ y &= -\frac{c}{t_1} - ct_1 t_2 t_3 + \frac{c}{t_1} = -ct_1 t_2 t_3\end{aligned}$$

The coordinates of the orthocenter are $\left(-\frac{c}{t_1 t_2 t_3}, -ct_1 t_2 t_3\right)$. Check if this point lies on $xy = c^2$:

$$xy = \left(-\frac{c}{t_1 t_2 t_3}\right)(-ct_1 t_2 t_3) = c^2$$

Since the product is c^2 , the orthocenter lies on the hyperbola. ■

Example 6.1.22. Show that the locus of the midpoint of the segment joining the extremities of two conjugate diameters of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is the pair of asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

Solution. Let P and D be the extremities of two conjugate semi-diameters. For the hyperbola, the extremities of the conjugate diameter are strictly defined on the conjugate hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$. Let the eccentric angle of P be θ . The coordinates are:

$$P(a \sec \theta, b \tan \theta)$$

The corresponding point D on the conjugate hyperbola has coordinates:

$$D(a \tan \theta, b \sec \theta)$$

Let $M(h, k)$ be the midpoint of the segment PD .

$$h = \frac{a \sec \theta + a \tan \theta}{2} = \frac{a}{2}(\sec \theta + \tan \theta)$$

$$k = \frac{b \tan \theta + b \sec \theta}{2} = \frac{b}{2}(\tan \theta + \sec \theta)$$

We observe the relationship between h and k :

$$\frac{h}{a} = \frac{1}{2}(\sec \theta + \tan \theta) \quad \text{and} \quad \frac{k}{b} = \frac{1}{2}(\sec \theta + \tan \theta)$$

Therefore:

$$\frac{h}{a} = \frac{k}{b} \implies \frac{h}{a} - \frac{k}{b} = 0$$

This implies M lies on the line $\frac{x}{a} - \frac{y}{b} = 0$.

If we consider the other extremity of the conjugate diameter $D'(-a \tan \theta, -b \sec \theta)$, the midpoint M' would yield:

$$h' = \frac{a}{2}(\sec \theta - \tan \theta), \quad k' = \frac{b}{2}(\tan \theta - \sec \theta)$$

$$\frac{h'}{a} = -\frac{k'}{b} \implies \frac{h'}{a} + \frac{k'}{b} = 0$$

Combining both cases, the locus is given by:

$$\left(\frac{x}{a} - \frac{y}{b}\right) \left(\frac{x}{a} + \frac{y}{b}\right) = 0 \implies \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

This represents the asymptotes of the hyperbola. ■

Epilogue: A Note to the Student

Congratulations on completing this rigorous course on **Geometry (MMT422J2)**.

The journey from the fundamental properties of the Parabola and Ellipse to the complex three-dimensional structures of Spheres, Cones, and Conicoids is no small feat. Over the course of these units, you have not merely memorized equations; you have developed the ability to visualize and analyze the very fabric of space.

You have transitioned from understanding simple loci in a plane to mastering the general equation of the second degree. You have explored the elegance of the *Radical Plane*, the structural integrity of *Coaxial Systems*, and the intricate symmetries of *Conjugate Diameters*. By mastering the theory of Tangents, Normals, and Asymptotes, you now possess the analytical tools necessary to describe the curvature of the universe.

Remember that the study of Geometry is the study of invariance and symmetry. The analytical skills you have sharpened here—the ability to translate geometric intuition into precise algebraic language—will serve as the bedrock for your future explorations in Multivariable Calculus, Topology, and Theoretical Physics.

As you move forward, carry with you the understanding that every equation tells a story of shape, position, and intersection. You have successfully navigated the dimensions; may your future mathematical endeavors be as structured and elegant as the surfaces you have studied.

“Mighty is geometry; joined with art, resistless.”

— Euripides

Best wishes for your future academic pursuits.